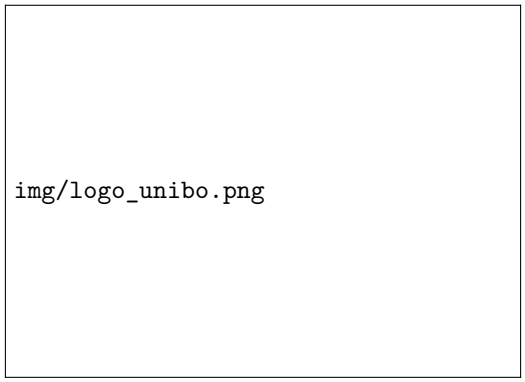


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Quantum Many Body Theory

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Contents

Introduction **1**

1 Weakly Interacting Bose Gas **3**

1.1 The Weak Interaction Assumption 3

1.2 Condensate Depletion and Excitations 13

1.3 Ground State Energy 18

1.4 Spontaneous Symmetry Breaking and Goldstone Theorem 24

1.5 Specific Heat and Condensate Fraction 27

1.6 Landau Criterion for Superfluidity 29

2 BCS Theory of Superconductivity **37**

2.1 Basic Model for Effective Attraction 39

2.2 From Microscopic to Mesoscopic Descriptions 55

2.3 Grand Canonical Hamiltonian and Ground State 60

2.4 Spectrum of the BCS Hamiltonian 72

2.5 Spontaneous Symmetry Breaking and Collective Excitations 89

3 Monte carlo and boh **95**

3.1 The Ising Model 96

3.2 Motivation for Monte Carlo Methods 100

3.3 Markov Chain Monte Carlo 103

Appendices **105**

A Notation and Conventions 105

B Computations and further developments 105

Introduction

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1 | Weakly Interacting Bose Gas

1.1 | The Weak Interaction Assumption

The weakly interacting Bose gas serves as a foundational paradigm in quantum many-body theory. We consider a system of N identical bosons of mass m confined to a box of volume $V = L^3$ with periodic boundary conditions.

The interparticle interaction is characterized by a two-body potential $U(\mathbf{r})$, which we assume to be weak. The constituents of this gas obey Bose-Einstein statistics and are treated as non-relativistic elementary particles, meaning that their internal degrees of freedom are not activated at the relevant energy scales, and the total particle number is strictly fixed (precluding, for example, a gas of photons).

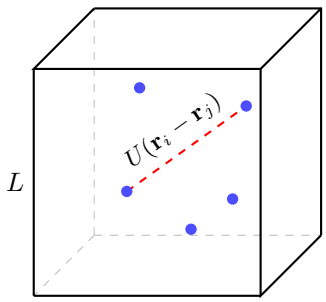


Figure 1.1: Schematic representation of a system of N interacting bosons confined within a cubic box of volume $V = L^3$. The periodic boundary conditions simulate a bulk system. The dashed red line illustrates the two-body interaction potential $U(\mathbf{r}_i - \mathbf{r}_j)$ acting strictly between pairs of particles.

Typical experimental realizations involve dilute gases of bosonic atoms, meaning atoms possessing an even total number of constituent fermions (protons, neutrons, and electrons), such as Sodium (^{23}Na), Rubidium (^{87}Rb), Potassium (^{41}K), Cesium (^{133}Ce), and Lithium (^{7}Li). Historically, this theoretical framework was first motivated by efforts to explain the superfluid behavior of Helium-4 below 2.17K, despite the fact that liquid ^{4}He actually resides in a strongly interacting regime.

The Hamiltonian governing this system of N interacting identical bosons is given in first quantization by

$$\hat{H} = \sum_{i=1}^{N} \frac{\hat{\mathbf{p}}_i^2}{2m} + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j),$$

where $\hat{\mathbf{p}}_i$ is the momentum operator of the i-th particle, and $U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j)$ represents the interaction energy between the i-th and j-th particles.

For a system comprising a macroscopic number of identical bosons, it is highly advantageous to

transition to the second quantization formalism.[1] We introduce bosonic creation and annihilation operators associated with the momentum eigenfunctions of a single particle. For a box with periodic boundary conditions, these single-particle states are quantized plane waves:

$$\phi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}},$$

where the allowed wave vectors $\mathbf{k}$ take the discrete values

$$\mathbf{k} = \frac{\pi}{L}(n_x, n_y, n_z), \quad n_x, n_y, n_z \in \{0, \pm 1, \pm 2, \dots\}.$$

In this momentum basis, the many-body Hamiltonian becomes

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_k \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} U(\mathbf{q}) \hat{a}^\dagger_{\mathbf{k}+\mathbf{q}} \hat{a}^\dagger_{\mathbf{k}'-\mathbf{q}} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{k}}. \tag{1.1.1}$$

Here, ϵ_k denotes the non-interacting energy associated with the wave vector $\mathbf{k}$ (which depends exclusively on its magnitude k), given by the free-particle dispersion

$$\epsilon_k = \frac{\hbar^2 k^2}{2m},$$

and $U(\mathbf{q})$ is the Fourier transform of the real-space two-body potential $U(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2)$. The operators $\hat{a}^\dagger_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}$ represent the creation and destruction operators for the corresponding momentum eigenstates.

In the idealized limit where interactions are completely absent, the ground state of the system is trivial: all N bosons condense into the lowest-energy single-particle state featuring zero momentum ($\mathbf{k} = 0$). This macroscopic formation of a condensate is mathematically expressed by acting N times with the zero-momentum creation operator $\hat{a}^\dagger_0$ on the vacuum state $|0\rangle$ (incorporating the standard $1/\sqrt{N!}$ normalization factor for formal completeness):

$$\left|\psi_0^{(0)}\right\rangle = \frac{(\hat{a}^\dagger_0)^N}{\sqrt{N!}} |0\rangle \,. \tag{1.1.2}$$

Remark. *This clustering phenomenon stands in stark contrast to the behavior of a gas of fermions. Governed by the Pauli exclusion principle, fermions are strictly forbidden from occupying identical quantum states. Consequently, even at zero temperature, non-interacting fermions must sequentially fill all available momentum states up to a maximum threshold known as the Fermi energy, precluding any analogous zero-momentum condensation.*

We now introduce the principal assumption of our approach: the interparticle interaction is sufficiently weak. Given this, we anticipate that the true, interacting ground state remains a modest perturbation of the unperturbed condensate state $\left|\psi_0^{(0)}\right\rangle$. Therefore, even in the presence of interactions, the vast majority of the bosons will remain in the zero-momentum single-particle state. This physical condition is formalized by the inequality

$$\frac{N - N_0}{N} \ll 1, \tag{1.1.3}$$

where $N_0 = \langle \hat{a}^\dagger_0 \hat{a}_0 \rangle$ defines the thermodynamic average number of bosons residing in the condensate. The quantity $N - N_0$ thus counts the number of depleted bosons that have been excited out of the condensate into states with $\mathbf{k} \neq 0$. Due to the weakness of the interaction, these non-zero

[1] For a very clear introduction to the second quantization formalism for many-particle systems, see: H. Tasaki, https://arxiv.org/pdf/1812.10732v5.

momentum states are only microscopically occupied, meaning their occupation numbers scale as $\langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle = \mathcal{O}(1)$. Crucially, the population of these excited states does not scale proportionally with the macroscopic total particle number N.

We leverage this assumption to implement a fundamental approximation on the interaction term of the Hamiltonian (1.1.1). Because operators acting on the condensate ($\hat{a}_0$ and $\hat{a}_0^\dagger$) scale macroscopically with $\sqrt{N_0} \approx \sqrt{N}$, we can safely neglect all interaction terms containing more than two operators equipped with non-zero momentum ($\hat{a}_{\mathbf{k}}$ or $\hat{a}_{\mathbf{k}}^\dagger$ for $\mathbf{k} \neq 0$). (Note that terms containing exactly three zero-momentum operators are inherently forbidden by momentum conservation requirements).

The most substantial and relevant term that survives this truncation is the one containing exclusively operators acting on the condensate ($\mathbf{q} = 0$ and $\mathbf{k} = \mathbf{k}' = 0$):

$$\frac{1}{2V} U(\mathbf{q} = 0) \hat{a}_0^\dagger \hat{a}_0^\dagger \hat{a}_0 \hat{a}_0 = \frac{1}{2V} U(\mathbf{q} = 0)(\hat{N}_0^2 - \hat{N}_0).$$

In the thermodynamic limit, $\hat{N}_0$ diverges linearly with N. Thus, we can completely disregard the trailing $\hat{N}_0$ term in the above expression, as it scales infinitely slower than the dominant quadratic term $\hat{N}_0^2 \sim N^2$.

To systematically express this leading-order contribution in terms of the conserved total particle number, we recognize that the number of bosons in the ground state can be written as the difference between the total number operator $\hat{N}$ and the sum of occupation numbers for all excited states:

$$\hat{N}_0 = \hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}},$$

where the primed summation specifically denotes a sum over all allowed momenta excluding zero. Letting $U_0 \equiv U(\mathbf{q} = 0)$, we substitute this relation back into our dominant interaction term. Expanding the square and purposefully discarding the resulting quartic term (which contains four out-of-condensate operators and is therefore negligible under our approximation), we obtain

$$\frac{1}{2V} U_0 (\hat{N}_0)^2 = \frac{1}{2V} U_0 \left(\hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right)^2 \approx \frac{1}{2V} U_0 \left(\hat{N}^2 - 2\hat{N} \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right).$$

Through this procedure, we successfully reduce the pure zero-momentum interaction into a quadratic form that couples the macroscopic total number N with the single-particle excitations.

Quadratic Interaction Terms and the Short-Range Assumption

Besides the dominant term where all operators are in the condensate state, we must evaluate the terms from the interaction Hamiltonian that contain exactly two operators with momentum different from zero. Before explicitly listing them, we introduce a crucial physical approximation: we replace the momentum-dependent interaction potential $U(\mathbf{q})$ with a simple constant, $U(0)$.

This approximation is mathematically justified if the real-space potential is effectively "short-ranged". For the Fourier transform $U(\mathbf{q}) = \int \mathrm{d}^3\mathbf{r} U(\mathbf{r}) e^{-i\mathbf{q}\cdot\mathbf{r}}$ to be well-behaved and approximately constant for small momenta, its zero-momentum limit $U(0) = \int \mathrm{d}^3\mathbf{r} U(\mathbf{r})$ must be finite. In three spatial dimensions, the spherical integration measure introduces a factor of r^2. Consequently, for the integral $\int dr r^2 U(r)$ to converge at large interparticle distances, the potential $U(r)$ must decay strictly faster than $1/r^3$.

This convergence criterion dictates which physical systems can be described by this simplified model. In dilute atomic gases, van der Waals forces serve as an excellent example of a valid short-ranged potential. In the standard Lennard-Jones model, the long-range attractive tail scales as $1/r^6$. Since $6 > 3$, it comfortably satisfies the convergence requirement. In contrast, dipole-dipole interactions scale exactly as $1/r^3$. This is a tricky marginal case where the spatial integral diverges logarithmically, meaning dipole interactions are mathematically long-ranged in 3D and cannot be treated with this simple approximation. Assuming our potential decays fast enough, we can safely set $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$, making all subsequent calculations significantly simpler.

Remark. *As an analogous example of effective short-range interactions from high-energy physics, the strong nuclear force is also effectively short-ranged (decaying exponentially, much faster than $1/r^3$) because its mediators, pions, are massive.*

With the assumption $U(\mathbf{q}) \approx U_0$ established, we retain the following six terms containing exactly two non-zero momentum operators:

1. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k}' - \mathbf{q} = 0$, we have that $\mathbf{k} = -\mathbf{q}$ and $\mathbf{k}' = \mathbf{q}$, and so we remain with

$$\frac{1}{2V} U_0 a_0^\dagger a_0^\dagger {\sum_{\mathbf{q}}}' a_{\mathbf{q}} a_{-\mathbf{q}},$$

   where the sum $\sum'_{\mathbf{q}}$ is over all possible momenta $\mathbf{q} \neq 0$ because the zero case has been already taken into account in the previous purely macroscopic term.

2. If $\mathbf{k}+\mathbf{q} = 0$ and $\mathbf{k}' = 0$, using again the short-range assumption, the term takes the following form:

$$\frac{1}{2V} U_0 {\sum_{\mathbf{q}}}' a_0^\dagger a_{-\mathbf{q}}^\dagger a_0 a_{-\mathbf{q}} = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{q}}}' a_{-\mathbf{q}}^\dagger a_{-\mathbf{q}}.$$

3. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k} = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 {\sum_{\mathbf{k}'}}' a_0^\dagger a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{k}'}}' a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}.$$

4. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k}' = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 {\sum_{\mathbf{k}}}' a_{\mathbf{k}}^\dagger a_0^\dagger a_0 a_{\mathbf{k}} = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{k}}}' a_{\mathbf{k}}^\dagger a_{\mathbf{k}}.$$

5. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_0^\dagger a_{\mathbf{q}} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_{\mathbf{q}}.$$

6. If $\mathbf{k}' = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger a_0 a_0 = \frac{1}{2V} U_0 a_0 a_0 {\sum_{\mathbf{q}}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger.$$

Condensate Mean-Field Approximation and the Bogoliubov Hamiltonian

A further crucial step is the replacement of the operators a_0 and $a_0^\dagger$ with numbers. Because the non-interacting ground state exhibits macroscopic occupation, its expectation value is $\langle a_0^\dagger a_0 \rangle = N_0 \approx N$. Therefore, we can confidently approximate $a_0 \sim \sqrt{N}$ and $a_0^\dagger \sim \sqrt{N}$ across all the previously derived terms of the approximate Hamiltonian that contain two condensate operators multiplied by two out-of-condensate operators.[2]

This mean-field approximation is formally justified by inspecting the commutator $[a_0, a_0^\dagger] = 1$ and comparing its magnitude to the result of acting with these operators on the non-interacting N-particle ground state $\left|\psi_0^{(0)}(N)\right\rangle$. Specifically, one finds:

$$a_0 \left|\psi_0^{(0)}(N)\right\rangle = \sqrt{N} \left|\psi_0^{(0)}(N-1)\right\rangle \sim \sqrt{N} \left|\psi_0^{(0)}(N)\right\rangle,$$

and

$$a_0^\dagger \left|\psi_0^{(0)}(N)\right\rangle = \sqrt{N+1} \left|\psi_0^{(0)}(N+1)\right\rangle \sim \sqrt{N} \left|\psi_0^{(0)}(N)\right\rangle,$$

where we have safely neglected 1 relative to the macroscopically large number N in these expressions. Because of this scaling, the value of the commutator is entirely negligible, and the operators can be effectively treated as commuting c-numbers.

Furthermore, during this systematic expansion, it is important to recognize why linear terms in $a_{\mathbf{k}}$ or $a_{\mathbf{k}}^\dagger$ do not appear. The terms which cancel out are precisely the ones that contain a single $\hat{a}_0$ or $\hat{a}_0^\dagger$ in the interaction term; if they did not cancel, they would have contributed to an unphysical renormalization of the free term of the Hamiltonian.

By substituting these numerical expectations back into our aggregated interaction terms, the full Hamiltonian becomes:

TODO: expand computations.

$$H = \sum_{\mathbf{k}}{}' (\epsilon_k + nU_0) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{nU_0}{2} \sum_{\mathbf{k}}{}' (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0, \tag{1.1.4}$$

where $n = N/V$ is the particle density.

Note that the second sum in the above equation contains the "anomalous" terms $a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger$ and $a_{\mathbf{k}} a_{-\mathbf{k}}$. Because the Hamiltonian now pairs creation operators together (and annihilation operators together) rather than consisting purely of standard number operators, it is not diagonal. To resolve this and find the true elementary excitations, one must employ the Bogoliubov transformation to diagonalize the Hamiltonian.

As a final remark, we note that if one were to carry out this derivation without invoking the constant potential approximation $U(\mathbf{q}) \simeq U_0$, the resulting Hamiltonian would naturally retain the momentum dependence of the interaction:

$$H = \sum_{\mathbf{k}}{}' (\epsilon_k + nU(k)) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{n}{2} \sum_{\mathbf{k}}{}' U(k) (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0.$$

[2] This can be preatty disturbing, since the expectation value of a single ladder operator is usually zero; but, as we will see later, we are not using eigenstates of number operator, instead superpositions of those (similar to coherent states). Indeed, even if the hamiltonian conserves the number of particle, the ground state description does not really rely on a fixed occupation of bosons. We will stress this point later on.

Summary of Assumptions and Approximations

Before proceeding to diagonalize the Hamiltonian, it is instructive to summarize the hierarchy of physical assumptions and mathematical approximations that have systematically reduced the complex many-body problem to the quadratic Bogoliubov Hamiltonian (1.1.4):

- **Non-Relativistic Elementary Bosons:** We assume the particles are massive, non-relativistic, and constrained to energy scales where internal degrees of freedom are completely frozen. *Consequence:* The total particle number N is strictly conserved (unlike in a photon gas), and the system is well-described by standard Galilean kinetic energy and a static scalar two-body potential.
- **Weak Interaction Hypothesis:** The interparticle potential is assumed to be sufficiently weak such that the true interacting ground state is only a modest perturbation of the ideal Bose-Einstein condensate. *Consequence:* The fraction of depleted particles is vanishingly small ($(N - N_0)/N \ll 1$). This justifies truncating the interaction Hamiltonian to discard all terms containing more than two out-of-condensate operators ($\mathbf{k} \neq 0$).
- **Short-Range Potential:** The two-body interaction $U(\mathbf{r})$ is assumed to decay strictly faster than $1/r^3$ at large interparticle distances. *Consequence:* The Fourier transform of the potential converges and varies slowly at small momenta, allowing us to approximate $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$. This dramatically simplifies the scattering amplitudes in the dilute gas regime.
- **Macroscopic Condensate Occupation (Mean-Field Limit):** The zero-momentum state is macroscopically occupied ($N_0 \approx N \gg 1$), meaning the fundamental commutator $[\hat{a}_0, \hat{a}_0^\dagger] = 1$ is negligible compared to the absolute particle number. *Consequence:* The macroscopic creation and annihilation operators $\hat{a}_0^\dagger$ and $\hat{a}_0$ can be treated as commuting c-numbers and replaced by $\sqrt{N}$. This breaks the $U(1)$ phase symmetry of the Hamiltonian and reduces the complicated quartic interaction term into a tractable quadratic form featuring anomalous pairing terms.

1.1.1 | Bogoliubov Transformations

We wish to diagonalize the Hamiltonian of the weakly interacting Bose gas by introducing new bosonic annihilation and creation operators, $\alpha_{\mathbf{k}}$ and $\alpha_{\mathbf{k}}^\dagger$, known as Bogoliubov operators. We need to construct a Hamiltonian which, in the end, depends only on a diagonal number operator of the form $\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$, yielding a target structure like:

$$H = \sum_{\mathbf{k}} E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \text{const},$$

for some energy dispersion E_k. To eliminate the anomalous terms that mix creation and annihilation operators in our previous Hamiltonian, we can perform a Bogoliubov transformation, which is a linear transformation mapping the original particle operators to the new quasiparticle operators. We start with the following standard ansatz for the Bogoliubov transformation:

$$\begin{cases} \alpha_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, \\ \alpha_{\mathbf{k}}^\dagger = v_k^* a_{\mathbf{k}}^\dagger + u_k^* a_{-\mathbf{k}}, \end{cases} \tag{1.1.5}$$

for $\mathbf{k} \neq 0$, where u_k and v_k are real coefficients. We now assume that these coefficients are real for simplicity, but they could in principle be complex. These coefficients are defined to depend strictly on the momentum magnitude $k = |\mathbf{k}|$, ensuring that rotational invariance is preserved in momentum space.

Because these newly defined operators must describe valid bosonic excitations, they are strictly required to satisfy the standard canonical bosonic commutation relations. We can verify this by evaluating the commutator between two annihilation operators:

$$\begin{aligned}[\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}'}] &= [u_k a_{\mathbf{k}} + v_k a^\dagger_{-\mathbf{k}}, u_{k'} a_{\mathbf{k}'} + v_{k'} a^\dagger_{-\mathbf{k}'}] \\ &= u_k v_{k'} [a_{\mathbf{k}}, a^\dagger_{-\mathbf{k}'}] + v_k u_{k'} [a^\dagger_{-\mathbf{k}}, a_{\mathbf{k}'}] \\ &= u_k v_{k'} \delta_{\mathbf{k},-\mathbf{k}'} - v_k u_{k'} \delta_{-\mathbf{k},\mathbf{k}'} = 0,\end{aligned}$$

which vanishes as expected. Next, we evaluate the critical commutator between a creation and an annihilation operator:

$$\begin{aligned}[\alpha_{\mathbf{k}}, \alpha^\dagger_{\mathbf{k}'}] &= [u_k a_{\mathbf{k}} + v_k a^\dagger_{-\mathbf{k}}, u_{k'} a^\dagger_{\mathbf{k}'} + v_{k'} a_{-\mathbf{k}'}] \\ &= u_k u_{k'} [a_{\mathbf{k}}, a^\dagger_{\mathbf{k}'}] + v_k v_{k'} [a^\dagger_{-\mathbf{k}}, a_{-\mathbf{k}'}] \\ &= (u_k^2 - v_k^2) \delta_{\mathbf{k},\mathbf{k}'}.\end{aligned}$$

To enforce the correct commutation relations, we must impose a hyperbolic normalization condition on the coefficients:

$$|u_k|^2 - |v_k|^2 = 1. \tag{1.1.6}$$

Provided this condition holds, the transformation linking the old and new operators can be easily inverted. We can express the linear mapping and its Hermitian conjugate in matrix notation:

$$\begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha^\dagger_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ v_k & u_k \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ a^\dagger_{-\mathbf{k}} \end{pmatrix}.$$

Because the determinant of the transformation matrix is exactly $u_k^2 - v_k^2 = 1$, the matrix is trivially invertible:

$$\begin{pmatrix} a_{\mathbf{k}} \\ a^\dagger_{-\mathbf{k}} \end{pmatrix} = \begin{pmatrix} u_k & -v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha^\dagger_{-\mathbf{k}} \end{pmatrix},$$

which explicitly yields the inverted transformation:

$$\begin{cases} a_{\mathbf{k}} = u_k \alpha_{\mathbf{k}} - v_k \alpha^\dagger_{-\mathbf{k}}, \\ a^\dagger_{\mathbf{k}} = u_k \alpha^\dagger_{\mathbf{k}} - v_k \alpha_{-\mathbf{k}}. \end{cases}$$

We now proceed with the formal diagonalization of the Hamiltonian by expressing it entirely in terms of the new Bogoliubov operators:

$$\begin{aligned}H = &\sum_{\mathbf{k}}{}' \left[(\epsilon_k + nU_0) v_k^2 - nU_0 u_k v_k \right] + \frac{1}{2} V n^2 U_0 \\ &+ \sum_{\mathbf{k}}{}' \left[(\epsilon_k + nU_0)(u_k^2 + v_k^2) - 2nU_0 u_k v_k \right] \alpha^\dagger_{\mathbf{k}} \alpha_{\mathbf{k}} \\ &+ \sum_{\mathbf{k}}{}' \left[\frac{nU_0}{2}(u_k^2 + v_k^2) - u_k v_k (\epsilon_k + nU_0) \right] (\alpha^\dagger_{\mathbf{k}} \alpha^\dagger_{-\mathbf{k}} + \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}}).\end{aligned}$$

It may initially appear that we have not made significant progress, as the final line still contains non-diagonal anomalous terms. However, we now have the mathematical freedom to choose arbitrary coefficients that eliminate these specific terms. By selecting an appropriate v_k (which

simultaneously determines u_k via the constraint $u_k^2 = 1 + v_k^2$), we can force the bracketed prefactor of the off-diagonal terms to vanish identically:

$$\frac{nU_0}{2}(u_k^2 + v_k^2) - u_k v_k(\epsilon_k + nU_0) = 0.$$

Through simple algebraic manipulation and applying our normalization condition, this requirement simplifies to a second-order polynomial equation for v_k^2:

$$4E_k^2 v_k^4 + 4E_k^2 v_k^2 - (nU_0)^2 = 0,$$

where we have suggestively defined the factor E_k^2 as:

$$E_k^2 = (\epsilon_k + nU_0)^2 - (nU_0)^2. \tag{1.1.7}$$

Solving this equation for v_k^2 directly yields the functional form of the transformation coefficients:

$$v_k^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU_0}{E_k} - 1\right),$$

and consequently,

$$u_k^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU_0}{E_k} + 1\right),$$

which implies for the cross-term:

$$u_k v_k = \frac{nU_0}{2E_k}.$$

By inserting these rigorously derived coefficients back into our expanded operator expression, the off-diagonal terms strictly cancel, yielding the final form of the diagonalized Hamiltonian:

$$H = \sum_{\mathbf{k}}{}' E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \mathcal{E}_0, \tag{1.1.8}$$

where $\mathcal{E}_0$ is a macroscopic constant energy shift defined by:

$$\mathcal{E}_0 = \frac{1}{2}Vn^2U_0 - \frac{1}{2}\sum_{\mathbf{k}}{}'(\epsilon_k + nU_0 - E_k).$$

The quantity $E_k = [(\epsilon_k + nU_0)^2 - (nU_0)^2]^{1/2} > 0$ represents the fundamental quasiparticle excitation spectrum of the system.

The Non-Interacting Limit

In the non-interacting limit, characterized by a vanishing interaction strength $U_0 \to 0$, it is instructive to verify that our diagonalized theory smoothly recovers the foundational physics of the ideal Bose gas.

First, we examine the Bogoliubov quasiparticle excitation spectrum E_k. By taking the limit of the dispersion relation, we find:

$$E_k = \lim_{U_0 \to 0} \sqrt{(\epsilon_k + nU_0)^2 - (nU_0)^2} = \sqrt{\epsilon_k^2} = \epsilon_k.$$

As physically expected, the quasiparticle spectrum trivially reduces to the free-particle kinetic energy $\epsilon_k = \hbar^2 k^2/(2m)$.

Next, we evaluate the asymptotic behavior of the Bogoliubov transformation coefficients v_k^2 and u_k^2. Substituting $E_k \to \epsilon_k$ and $U_0 \to 0$, we obtain:

$$v_k^2 = \lim_{U_0 \to 0} \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right) = \frac{1}{2}(1-1) = 0,$$

and correspondingly,

$$u_k^2 = \lim_{U_0 \to 0} \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} + 1 \right) = \frac{1}{2}(1+1) = 1.$$

This implies that $u_k \to 1$ and $v_k \to 0$. Consequently, the Bogoliubov quasiparticle operators $\alpha_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a^\dagger_{-\mathbf{k}}$ reduce identically to the bare particle operators $a_{\mathbf{k}}$. In the absence of interactions, the elementary excitations of the system are simply the original non-interacting bosons themselves, and the quantum depletion of the condensate drops to zero.

Finally, we analyze the macroscopic constant energy shift $\mathcal{E}_0$. The purely classical mean-field contribution $\frac{1}{2}Vn^2U_0$ obviously vanishes. For the quantum fluctuation sum, we note that $E_k \approx \epsilon_k + nU_0 - \frac{(nU_0)^2}{2\epsilon_k}$ for small U_0, meaning the summand $(\epsilon_k + nU_0 - E_k) \sim \mathcal{O}(U_0^2)$. Therefore, the entire shift vanishes:

$$\lim_{U_0 \to 0} \mathcal{E}_0 = 0.$$

The ground state energy of the system correctly defaults to zero, recovering the trivial result for an ideal Bose-Einstein condensate where all particles occupy the lowest available energy level with zero kinetic energy.

Ground State

With the Hamiltonian successfully diagonalized, we first seek its ground state, denoted $|\psi_0\rangle$. This state is mathematically defined as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$ for all non-zero momenta:

$$\alpha_{\mathbf{k}}|\psi_0\rangle = 0, \quad \forall \mathbf{k} \neq 0,$$

since any mode added to the vacuum would necessarily increase the energy of the system.

The corresponding energy eigenvalue of this ground state is simply the zero-point constant $\mathcal{E}_0$:

$$H|\psi_0\rangle = \mathcal{E}_0|\psi_0\rangle.$$

By acting with the creation operator $\alpha^\dagger_{\mathbf{k}}$ on this vacuum state, we can construct the entire spectrum of excited states:

$$\frac{(\alpha^\dagger_{\mathbf{k}_1})^{n_{\mathbf{k}_1}}}{\sqrt{n_{\mathbf{k}_1}!}} \frac{(\alpha^\dagger_{\mathbf{k}_2})^{n_{\mathbf{k}_2}}}{\sqrt{n_{\mathbf{k}_2}!}} \cdots \frac{(\alpha^\dagger_{\mathbf{k}_N})^{n_{\mathbf{k}_N}}}{\sqrt{n_{\mathbf{k}_N}!}} |\psi_0\rangle,$$

whose corresponding energy eigenvalues are given by:

$$\mathcal{E}_0 + \sum_{i=1}^{N} n_{\mathbf{k}_i} E_{\mathbf{k}_i},$$

where $n_{\mathbf{k}_i} = \alpha^\dagger_{\mathbf{k}_i} \alpha_{\mathbf{k}_i}$ is the occupation number of the i-th state in the *Bogoliubov quasiparticle basis*.

It is critical to note that the vacuum of these newly defined operators, $|\psi_0\rangle$, is distinct from the vacuum of the original bare particle operators, $a_{\mathbf{k}}$. This distinction becomes physically apparent

when we evaluate the expectation value of the bare particle number operator in the interacting ground state:

$$\begin{aligned}\langle\psi_0|a^\dagger_{\mathbf{k}}a_{\mathbf{k}}|\psi_0\rangle &= \langle\psi_0|(u_k\alpha^\dagger_{\mathbf{k}} - v_k\alpha_{-\mathbf{k}})(u_k\alpha_{\mathbf{k}} - v_k\alpha^\dagger_{-\mathbf{k}})|\psi_0\rangle \\ &= \langle\psi_0|\left[u_k^2\alpha^\dagger_{\mathbf{k}}\alpha_{\mathbf{k}} + v_k^2\alpha_{-\mathbf{k}}\alpha^\dagger_{-\mathbf{k}} - u_kv_k(\alpha^\dagger_{\mathbf{k}}\alpha^\dagger_{-\mathbf{k}} + \alpha_{-\mathbf{k}}\alpha_{\mathbf{k}})\right]|\psi_0\rangle \\ &= v_k^2 \neq 0,\end{aligned}$$

where the final evaluation leverages the commutation relation $[\alpha_{-\mathbf{k}}, \alpha^\dagger_{-\mathbf{k}}] = 1$ and the fact that annihilation operators destroy the vacuum. This previous calculation elegantly demonstrates a profound physical consequence of the theory: *even in the absolute zero-temperature ground state* $|\psi_0\rangle$*, the mean occupation number of states with non-zero momentum is strictly positive.* In terms of the original operators, we would state that $n_{\mathbf{k}} \neq 0$ (where $n_{\mathbf{k}} = a^\dagger_{\mathbf{k}}a_{\mathbf{k}}$) even when $\mathbf{k} \neq 0$. Because of the interactions, a finite fraction of bosons is fundamentally pushed out of the zero-momentum state, leading to a phenomenon known as quantum depletion, where the condensate is only partially occupied even at zero temperature.

1.2 | Condensate Depletion and Excitations

In the previous section, we established that the approximate Hamiltonian of a weakly interacting Bose gas can be rewritten in a diagonal form using the Bogoliubov transformation. The ground state $|\psi_0\rangle$ of this diagonalized Hamiltonian is defined precisely as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$, meaning $\alpha_{\mathbf{k}}|\psi_0\rangle = 0$ for all $\mathbf{k} \neq 0$.

In terms of the original bare particle creation and annihilation operators, we can compute the momentum distribution function in this interacting ground state for both the zero-momentum ($\mathbf{k} = 0$) and non-zero-momentum ($\mathbf{k} \neq 0$) states:

$$\langle\psi_0|a_0^\dagger a_0|\psi_0\rangle = N_0 \sim N,$$

$$\langle\psi_0|a_{\mathbf{k}}^\dagger a_{\mathbf{k}}|\psi_0\rangle = v_k^2 \neq 0, \quad \mathbf{k} \neq 0.$$

While in the non-interacting case all N particles condense into the single-particle state with $\mathbf{k} = 0$, the presence of interactions induces a partial quantum depletion of the condensate even at absolute zero.

We now want to calculate the total number of these non-condensed bosons, denoted as N'. This non-condensate fraction is evaluated by summing the expectation value of the bare number operator over all non-zero momentum states:

$$N' = N - N_0 = \sum_{\mathbf{k}}{}' \langle\psi_0|a_{\mathbf{k}}^\dagger a_{\mathbf{k}}|\psi_0\rangle = \sum_{\mathbf{k}}{}' v_k^2,$$

where the primed sum indicates summation over all $\mathbf{k} \neq 0$. As a consistency check, we recall the explicit form of v_k^2 derived earlier; in the strictly non-interacting limit, this excited occupation correctly vanishes:

$$v_k^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU_0}{E_k} - 1\right) \xrightarrow{U_0 \to 0} 0.$$

However, for interacting bosons, there is a fundamental partial depletion of the condensate even at zero temperature. Let us calculate the total number of non-condensed bosons N'. To evaluate this macroscopic quantity, we transition to the thermodynamic limit—taking $V \to \infty$ and $N \to \infty$ while keeping the density $n = N/V$ constant—which allows us to replace the discrete momentum sum with a continuous integral over k-space:

$$N' = \sum_{\mathbf{k}} v_k^2 \xrightarrow[\text{TD limit}]{N,V\to\infty,\ n \text{ const.}} V \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} v_k^2.$$

Dividing both sides by the volume V, the density of non-condensed bosons n' is given by:

$$n' = \frac{N'}{V} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} v_k^2. \tag{1.2.1}$$

Before explicitly calculating this integral, we will verify its mathematical convergence in both the infrared (long-wavelength, $k \to 0$) and ultraviolet (short-wavelength, $k \to \infty$) limits. When $k \to 0$, the Bogoliubov excitation energy $E_k \to 0$, which simplifies the occupation number to:

$$v_k^2 \sim \frac{nU_0}{2E_k} \quad (k \to 0).$$

Infrared limit. To analyze exactly how E_k scales towards zero, we substitute the free particle dispersion $\epsilon_k = \hbar^2 k^2/2m$:

$$E_k \sim \sqrt{2nU_0\epsilon_k} = c\hbar k \quad (k \to 0),$$

where we have conveniently defined the parameter c as:

$$c = \sqrt{\frac{nU_0}{m}}. \tag{1.2.2}$$

This quantity c possesses the dimensions of a velocity and physically corresponds to the *speed of sound* propagating through the weakly interacting Bose gas. Combining these small-k expressions, the ground state occupation number diverges near the origin as:

$$v_k^2 \sim \frac{mc}{2\hbar k} \quad (k \to 0).$$

Despite this $1/k$ divergence in the integrand, the integration measure $\mathrm{d}^3\mathbf{k}$ in spherical coordinates introduces a geometric factor of k^2, ensuring that the integral (1.2.1) is safely convergent for $k \to 0$.

Ultraviolet limit. Let us now examine the ultraviolet behavior as $k \to \infty$. By Taylor expanding the square root definition of E_k in the limit where kinetic energy dominates the interaction ($\epsilon_k \gg nU_0$), we have:

$$E_k = \epsilon_k\sqrt{1 + \frac{2nU_0}{\epsilon_k}} \sim \epsilon_k\left[1 + \frac{nU_0}{\epsilon_k} - \frac{1}{2}\left(\frac{nU_0}{\epsilon_k}\right)^2\right].$$

Substituting this expansion back into the full expression for v_k^2, the leading-order terms cancel, leaving a rapid decay:

$$v_k^2 \sim \frac{1}{2}\left[\frac{\epsilon_k + nU_0}{\epsilon_k\left(1 + \frac{nU_0}{\epsilon_k} - \frac{1}{2}\left(\frac{nU_0}{\epsilon_k}\right)^2\right)} - 1\right] \sim \frac{(nU_0)^2}{4\epsilon_k^2} = \left(\frac{mc}{\hbar k}\right)^4.$$

Because the occupation number decays as $1/k^4$ at large momenta, the integral converges absolutely at the $k \to \infty$ upper bound as well.

To formally evaluate the integral, we introduce a natural dimensionless momentum variable $\tilde{k}$ defined as:

$$\tilde{k} = \frac{\hbar k}{mc},$$

such that the occupation number can be elegantly rewritten as a universal function of $\tilde{k}$:

$$v_k^2 = v_{\tilde{k}}^2 = \frac{1}{2}\left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1\right).$$

This final expression is considered universal in the sense that all system-specific dependencies, such as the particle density n or the interaction potential strength U_0, have been completely absorbed into the definition of the dimensionless coordinate $\tilde{k}$.

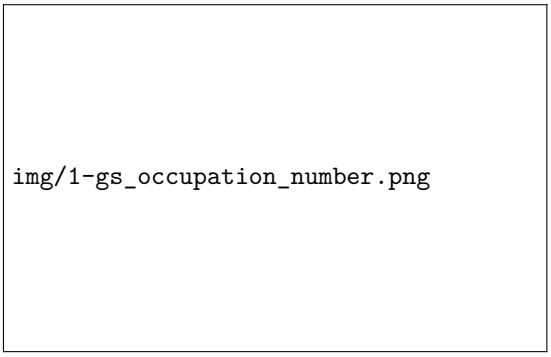


Figure 1.2: Occupation numbers of the ground state $v_{\tilde{k}}^2$. Notice that for small $\tilde{k}$ one has $v_{\tilde{k}}^2 \sim 1/(2\tilde{k})$, while for large $\tilde{k}$ one has $v_{\tilde{k}}^2 \sim 1/\tilde{k}^4$.

Having established its convergence and universality, we can explicitly compute the non-condensed density n' by executing the change of variables from k to $\tilde{k}$ inside the integral:

$$n' = \left(\frac{mc}{\hbar}\right)^3 \int \frac{\mathrm{d}^3\tilde{k}}{(2\pi)^3} \frac{1}{2} \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) = \left(\frac{mc}{\hbar}\right)^3 \frac{1}{4\pi^2} \int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) \mathrm{d}\tilde{k}.$$

This remaining one-dimensional integral over $\tilde{k}$ can be solved exactly, yielding a factor of $4/3$:

$$\int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) \mathrm{d}\tilde{k} = \frac{4}{3}.$$

Applying this result, the density of depleted bosons elegantly simplifies to:

$$n' = \frac{1}{3\pi^2} \left(\frac{mc}{\hbar}\right)^3.$$

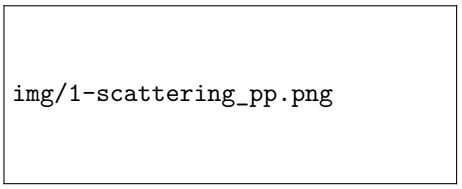
The fundamental fractional depletion of the condensate, given by the ratio $N'/N = n'/n$, is therefore:

$$\frac{N'}{N} = \frac{1}{3\pi^2 n} \left(\frac{mc}{\hbar}\right)^3. \tag{1.2.3}$$

We can thus interpret and manipulate the condensate depletion by operating strictly on the macroscopic density n and the characteristic sound velocity $c = \sqrt{nU_0/m}$. Because c entirely encapsulates the underlying interaction strength U_0, it effectively acts as the single defining free parameter for the weakly interacting system's thermodynamic properties.

Example (Scattering Length). The scattering length a is a measure of the strength of the interaction between two particles. It is defined as the distance at which the wave function of two interacting particles goes to zero. In the context of Bose-Einstein condensates, the scattering length plays a crucial role in determining the properties of the condensate, including its stability and the nature of its excitations.

Figure 1.3: Scattering process with incoming momentum $\mathbf{p}$ and outgoing momentum $\mathbf{p}'$. If the scattering is elastic, momentum conservation ensures that $|\mathbf{p}'| = |\mathbf{p}|$.



To rigorously justify this parameterization, recall that within the Born approximation—which is formally valid for weak interactions—the scattering amplitude $f(\mathbf{p}' \leftarrow \mathbf{p})$ is directly related

to the Fourier transform of the interaction potential $U(\mathbf{q})$ evaluated at the momentum transfer $\mathbf{q} = \mathbf{p}' - \mathbf{p}$:

$$f(\mathbf{p}' \leftarrow \mathbf{p}) = -\frac{m_r}{2\pi\hbar^2} U(\mathbf{q}),$$

where m_r is the reduced mass of the two-body collision. Because we are considering identical bosons of mass m, the reduced mass simplifies to $m_r = m/2$.

At very low collision energies, the momentum transfer approaches zero, and the scattering becomes completely isotropic, entirely losing its angular dependence. In this fundamental low-energy limit, the scattering length a is mathematically defined as the negative asymptotic limit of the scattering amplitude:

$$a = -\lim_{\mathbf{p},\mathbf{p}' \to 0} f(\mathbf{p}' \leftarrow \mathbf{p}).$$

Because the scattering is isotropic, this single parameter is sufficient to completely control both the differential cross-section, $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |a|^2$, and the total cross-section, $\sigma_{\text{tot}} = 4\pi a^2$.

By substituting the zero-momentum limit of the Born approximation into this definition, we obtain $a = \frac{m}{4\pi\hbar^2} U_0$. Consequently, for weakly interacting Bose gases, the scattering length can be inverted to define the effective interaction potential U_0 through the relation:

$$U_0 = \frac{4\pi\hbar^2 a}{m}, \tag{1.2.4}$$

where m is the mass of the particles. This relation allows us to express various physical quantities in terms of the scattering length, providing a more intuitive understanding of how interactions affect the condensate.

1.2.1 | Condensate Depletion in Terms of the Gas Parameter

With the effective interaction parameterized strictly by the geometric scattering length a, we can recast our thermodynamic results into a universal form. We first substitute the expression for U_0 into our previous definition of the sound velocity, $c = \sqrt{nU_0/m}$:

$$c = \frac{1}{m}\sqrt{4\pi\hbar^2 n a}.$$

Rearranging these terms to isolate the specific velocity-momentum ratio used in our depletion integral yields:

$$\frac{mc}{\hbar} = 2\sqrt{\pi n a}.$$

We can now insert this expression back into the previously derived equation for the fraction of non-condensed bosons, $\frac{N'}{N} = \frac{1}{3\pi^2 n}\left(\frac{mc}{\hbar}\right)^3$. Performing this substitution directly provides the final analytic result for the condensate depletion:

$$\frac{N'}{N} = \frac{8}{3\sqrt{\pi}}\sqrt{n a^3}. \tag{1.2.5}$$

A profound observation regarding Equation 1.2.5 is the complete absence of the reduced Planck constant, $\hbar$. Because the final result does not contain any $\hbar$ term, there is no mathematical pathway to expand the expression or take a limit to recover a classical or semiclassical analog. This is a direct consequence of the underlying physics: quantum depletion—and Bose-Einstein condensation in general—are purely quantum mechanical effects, fundamentally arising from the zero-point energy and non-commuting nature of the interactions between particles in the condensate.

The quantity N'/N physically represents the fraction of bosons that are pushed out of the condensate ground state. Crucially, Equation 1.2.5 reveals that this depletion depends entirely on a dimensionless quantity called the **gas parameter**, na^3, which scales as the density multiplied by the characteristic scattering volume. Alternatively, we can interpret this parameter as the cube of the ratio between the scattering length a and the average inter-particle distance $n^{-1/3}$.

For the initial Bogoliubov mean-field treatment to be mathematically valid, we explicitly required the depleted fraction N'/N to be small. Consequently, our theory requires the gas parameter to satisfy the strict condition $na^3 \ll 1$. The real physical meaning of this approximation is that it can be satisfied in two distinct ways: the parameter can be small either because the scattering length a is intrinsically small (representing a genuinely weak interaction), or because the particle density n is exceedingly low (representing a dilute gas).

Because of this density dependence, it can be proven that the Bogoliubov theory remains highly accurate even for infinitely strong interactions, provided the gas is sufficiently dilute. Consider, for instance, a hard-sphere core potential parameterized by a radius r_0:

$$V(r) = \begin{cases} +\infty & r < r_0, \\ 0 & r > r_0. \end{cases}$$

For such an impenetrable hard-sphere potential, the scattering length is simply equal to the core radius, $a = r_0$. Even in this highly non-perturbative scenario, the established theory remains perfectly valid as long as $nr_0^3 \ll 1$. This specific condition of ultra-low density is universally realized in experimental systems of ultracold atomic gases.

1.3 | Ground State Energy

Let us now study the macroscopic ground state energy $\mathcal{E}_0$ of the interacting system. Recall that for an ideal, non-interacting Bose gas, the ground state energy is trivially zero because all particles occupy the zero-momentum state without any kinetic or potential cost. However, the introduction of a weak interparticle potential fundamentally alters this scenario. By taking the thermodynamic limit, the discrete sum over the momentum modes transforms into a continuous integral over all possible wave vectors, yielding the energy density:

$$\frac{\mathcal{E}_0}{V} = \frac{1}{2}U_0 n^2 - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}(\epsilon_k + nU_0 - E_k).$$

An immediate problem arises when we inspect the behavior of this integral at large momenta (the ultraviolet limit). Based on our earlier asymptotic expansion, the Bogoliubov spectrum E_k approaches the free-particle energy with a leading-order correction, meaning the term in the integrand, $\epsilon_k + nU_0 - E_k$, decays asymptotically as $1/k^2$. Because the three-dimensional integration measure d^3k introduces a k^2 factor, the integral diverges linearly.

The physical origin of this unphysical ultraviolet divergence lies in the strict short-range approximation we previously adopted, namely replacing the momentum-dependent potential $U(\mathbf{k})$ with a constant $U(\mathbf{k} = 0) = U_0$ for all $\mathbf{k}$. In reality, only a mathematically pathological Dirac delta potential possesses a Fourier transform that remains constant for arbitrarily large momenta. For any realistic physical potential, the Fourier transform must eventually decay to zero at large wave vectors.

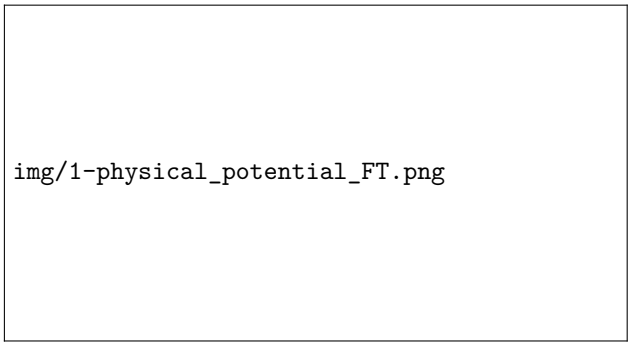


Figure 1.4: Fourier transform of a generic physical potential $U(\mathbf{r})$ in momentum space, $U(\mathbf{k})$. The Fourier transform is a smooth function that decays to zero at large wave vectors. In contrast, the delta-function approximation for the interaction potential is represented by a constant horizontal line, which is valid only for low-momentum scattering processes.

1.3.1 | Renormalization and the Lee-Huang-Yang Correction

Fortunately, the assumption of a strictly constant potential is not strictly necessary for the theory's structural integrity. If one repeats the rigorous diagonalization procedure while retaining the full momentum dependence of the potential, the Bogoliubov transformation coefficients become:

$$v_{\mathbf{k}}^2 = \frac{1}{2}\left(\frac{\epsilon_k + nU(\mathbf{k})}{E_{\mathbf{k}}} - 1\right),$$

and the quasiparticle dispersion takes the generalized form:

$$E_{\mathbf{k}} = \left[(\epsilon_k + nU(\mathbf{k}))^2 - (nU(\mathbf{k}))^2\right]^{1/2}.$$

With these generalized expressions, the corrected ground state energy density becomes:

$$\frac{\mathcal{E}_0}{V} = \frac{1}{2}U_0 n^2 - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}(\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}}). \tag{1.3.1}$$

By evaluating the large momentum limit of this refined integrand, we find:

$$\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} \sim \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}.$$

Therefore, to guarantee the convergence of the energy integral, it is entirely sufficient that the Fourier transform of the potential, $U(\mathbf{k})$, vanishes faster than $1/\sqrt{k}$ at large k. This is a remarkably mild assumption, naturally verified by any real-space potential $U(r)$ that does not diverge faster than $1/r^{5/2}$ as $r \to 0$.

Although this resolves the mathematical divergence, it introduces an interpretational challenge: the ground state energy now explicitly depends on the high-energy microscopic details of the interaction potential $U(\mathbf{k})$ rather than on a single, universally measurable macroscopic parameter like the scattering length a. To eliminate this unpleasant feature and express the energy purely in terms of physical observables, we employ a standard procedure analogous to classical renormalization. We add and subtract the problematic asymptotic ultraviolet behavior from our integral expression:

$$\frac{\mathcal{E}_0}{V} = \frac{U_0 n^2}{2} - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left[\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k} + \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}\right].$$

By splitting the integral and factoring out the density, we obtain:

$$\frac{\mathcal{E}_0}{V} = \frac{n^2}{2}\left(U_0 - \int \frac{d^3k}{(2\pi)^3}\frac{U(\mathbf{k})^2}{2\epsilon_k}\right) - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left[\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}\right].$$

The bracketed prefactor of the first term can be elegantly linked to the physically observable scattering length a by moving beyond the first-order Born approximation and utilizing the second-order Born approximation for the scattering amplitude $f(\mathbf{p}' \leftarrow \mathbf{p})$:

$$f(\mathbf{p}' \leftarrow \mathbf{p}) = -\frac{m_r}{2\pi\hbar^2}\left[U(\mathbf{p}' - \mathbf{p}) + \int \frac{d^3p''}{(2\pi)^3}\frac{U(\mathbf{p}' - \mathbf{p}'')U(\mathbf{p}'' - \mathbf{p})}{\frac{p^2}{2m_r} - \frac{p''^2}{2m_r} + i0^+}\right],$$

where 0^+ denotes an infinitesimal positive regulator. Recalling the definition of the scattering length in the low-energy limit, the expression simplifies to:

$$a = \frac{m}{4\pi\hbar^2}\left(U_0 - \int \frac{d^3p''}{(2\pi)^3}\frac{U(\mathbf{p}'')^2}{\frac{p''^2}{m} - i0^+}\right).$$

Because the integrand naturally vanishes for $p'' = 0$, the $i0^+$ regulator becomes irrelevant. By changing the integration variable to the wave vector $\mathbf{k}$ and identifying $\epsilon_k = \hbar^2k^2/2m$, we uncover the crucial identity:

$$a = \frac{m}{4\pi\hbar^2}\left(U_0 - \int \frac{d^3k}{(2\pi)^3}\frac{U(\mathbf{k})^2}{2\epsilon_k}\right).$$

By successfully replacing the bare coupling constant U_0 with the physical scattering length a, the ground state energy is renormalized:

$$\frac{\mathcal{E}_0}{V} = \frac{2\pi\hbar^2 n^2 a}{m} - \frac{1}{2}\int \frac{d^3k}{(2\pi)^3}\left(\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}\right).$$

Because the remaining integral is now manifestly convergent in the ultraviolet regime, we can safely reapply the low-energy approximation $U(\mathbf{k}) \approx U_0$ inside the integral without generating any infinities. The integral then becomes:

$$\int \frac{d^3k}{(2\pi)^3} \left(\epsilon_k + nU_0 - E_k - \frac{(nU_0)^2}{2\epsilon_k} \right).$$

By rewriting this in terms of our previously defined dimensionless momentum variable $\tilde{k}$, the integral scales out as:

$$\left(\frac{mc}{\hbar}\right)^3 mc^2 \frac{1}{2\pi^2} \int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2}{2} + 1 - \sqrt{\frac{\tilde{k}^4}{4} + \tilde{k}^2} - \frac{1}{\tilde{k}^2} \right) d\tilde{k}.$$

This dimensionless integral can be evaluated analytically and yields an exact value of $-32/15$. Inserting this numerical result alongside the definition of the sound velocity c, we arrive at the celebrated final expression for the ground state energy of a dilute Bose gas:

$$\frac{\mathcal{E}_0}{V} = \frac{2\pi\hbar^2 n^2 a}{m} \left[1 + \frac{128}{15\sqrt{\pi}} \sqrt{na^3} \right]. \tag{1.3.2}$$

The leading term in Equation 1.3.2 represents the standard mean-field energy, $\frac{1}{2}n^2U_0$, which precisely matches the expectation value of the Hamiltonian if all quantum field operators were simply replaced by their classical expectation value $\sqrt{n}$. The subsequent correction, proportional to the gas parameter $\sqrt{na^3}$, is the renowned Lee-Huang-Yang (LHY) term. This pivotal term specifically accounts for the zero-point quantum fluctuations of the interacting field around its classical mean-field value.

The macroscopic stability of the gas is dictated by these energy contributions. For a repulsive interaction ($a > 0$), the energy increases quadratically with density, meaning the macroscopic compressibility is positive and the gas will homogenously fill the confining volume to minimize its energy. Conversely, if the effective potential is attractive ($a < 0$), the scattering length is negative, and the energy is minimized by allowing the density to approach infinity. Because bosons are not protected by the Pauli exclusion principle, an attractive Bose gas is mechanically unstable and will spontaneously undergo spatial collapse.

Remark. *The interplay between the mean-field energy and the LHY quantum fluctuations can yield fascinating new phases of matter. In 2015, D. S. Petrov theoretically demonstrated the stabilization of a collapsing binary Bose-Bose mixture. Consider a mixture of two bosonic species with intra-species repulsion ($a_{11}, a_{22} > 0$) but a sufficiently strong inter-species attraction ($a_{12} < 0$) such that the total mean-field compressibility becomes negative, triggering a collapse. Because the attractive mean-field term scales as $\propto n^2$ while the repulsive LHY fluctuation term scales as $\propto n^{5/2}$, the repulsive quantum fluctuations dominate at higher densities. This delicate competition creates an optimal finite density that minimizes the energy, halting the collapse. The resulting state is a self-bound quantum droplet that remains in equilibrium with the vacuum even without external trapping potentials, a phenomenon successfully observed in experiments in 2018.*

Pressure and Speed of Sound

Returning to a single-component Bose gas, we observe that the macroscopic pressure at absolute zero, $T = 0$, is derived from the standard thermodynamic energy-volume relation:

$$P = -\left(\frac{\partial E}{\partial V}\right)_N = -\frac{E}{V} + n\frac{\partial (E/V)}{\partial n}.$$

By substituting the Lee-Huang-Yang expression for the ground state energy density, we obtain the corrected pressure:

$$P = \frac{2\pi\hbar^2 n^2 a}{m}\left[1 + \frac{64}{5\sqrt{\pi}}\sqrt{na^3}\right].$$

From this thermodynamic equation of state, the macroscopic speed of sound, c_s, in the interacting gas is given by the density derivative of the pressure:

$$c_s^2 = \frac{1}{m}\frac{\partial P}{\partial n} = \frac{4\pi\hbar^2 na}{m^2}\left[1 + \frac{16}{\sqrt{\pi}}\sqrt{na^3}\right].$$

We immediately see that to leading order in the gas parameter, the thermodynamic speed of sound c_s agrees exactly with the microscopic velocity c extracted from the linear slope of the single-particle Bogoliubov dispersion E_k in the long-wavelength limit ($k \to 0$) introduced previously. It can be formally proven that the fundamental identity $c = c_s$ should hold exactly in any fully consistent many-body theory[3][4]. However, maintaining strict consistency with the perturbative result for the speed of sound to next-to-leading order requires extending calculations beyond the present Bogoliubov mean-field approximation when computing the quasiparticle spectrum E_k[5].

1.3.2 | Ground State Wave Function

We established that the exact ground state $|\psi_0\rangle$ of the diagonalized effective Hamiltonian must be the vacuum state for the Bogoliubov quasiparticle excitations. Mathematically, this mandates that for all non-zero wave vectors $\mathbf{k} \neq 0$, the annihilation operator yields $\alpha_{\mathbf{k}}|\psi_0\rangle = 0$. If we substitute the defining Bogoliubov transformation—which expresses the quasiparticle operator $\alpha_{\mathbf{k}}$ as a linear combination of the original bare bosonic creation and destruction operators—into this vacuum condition, we obtain:

$$u_k a_{\mathbf{k}}|\psi_0\rangle = -v_k a^\dagger_{-\mathbf{k}}|\psi_0\rangle, \tag{1.3.3}$$

for $\mathbf{k} \neq 0$.

The physical implications of this constraint are profound. On the left-hand side of Equation 1.3.3, the bare annihilation operator $a_{\mathbf{k}}$ acts on the ground state, explicitly removing a boson of momentum $\mathbf{k}$. Simultaneously, on the right-hand side, a creation operator $a^\dagger_{-\mathbf{k}}$ acts on the same ground state, creating a particle with the exact opposite momentum $-\mathbf{k}$. This dynamic equality makes it abundantly clear that the true interacting ground state $|\psi_0\rangle$ cannot be a simple eigenvector of the total particle number operator $\hat{N}$. If it were an eigenstate, the left-hand side of the equation would project onto a subspace with $N - 1$ particles, whereas the right-hand side would project onto a subspace with $N + 1$ particles, creating an irreconcilable contradiction.

[3] C.R. Cabrera, L. Tanzi, J. Sanz, B. Naylor, P. Thomas, P. Cheiney, and L. Tarruell, *Quantum liquid droplets in a mixture of Bose-Einstein condensates*, Science **359**, 301 (2018).
[4] P. C. Hohenberg and P. C. Martin, Annals of Physics **34**, 291 (1965).
[5] S.T. Beliaev, Soviet Phys. JETP **7**, 299 (1958).

Instead, the structural property demanded by Equation 1.3.3 is satisfied by a state heavily featuring coherent superpositions of correlated particle pairs. We propose the following ansatz for the non-condensate portion of the state:

$$\exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle.$$

To rigorously verify that this state satisfies our requirement, we act upon it with the operator $u_k a_{\mathbf{k}}$:

$$u_k a_{\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle = u_k \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right) a_{\mathbf{k}}|0\rangle$$
$$+u_k \left[a_{\mathbf{k}}, \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)\right]|0\rangle.$$

The first term trivially vanishes because the bare destruction operator $a_{\mathbf{k}}$ acts directly on the absolute vacuum $|0\rangle$. The second term can be explicitly evaluated by expanding the exponential function as a Taylor series. Recalling the fundamental commutator identity $[A, BC] = [A, B]C + B[A, C]$, we evaluate the commutator of a single annihilation operator with a pair creation operator:

$$[a_{\mathbf{k}}, a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}] = [a_{\mathbf{k}}, a^\dagger_{\mathbf{p}}]a^\dagger_{-\mathbf{p}} + a^\dagger_{\mathbf{p}}[a_{\mathbf{k}}, a^\dagger_{-\mathbf{p}}]$$
$$= a^\dagger_{-\mathbf{p}}\delta_{\mathbf{k},\mathbf{p}} + a^\dagger_{\mathbf{p}}\delta_{\mathbf{k},-\mathbf{p}}.$$

More generally, this can be easily verified by mathematical induction for arbitrary powers of the pair operator:

$$[a_{\mathbf{k}}, (a^\dagger_{\mathbf{p}})^n (a^\dagger_{-\mathbf{p}})^n] = [a_{\mathbf{k}}, (a^\dagger_{\mathbf{p}})^n](a^\dagger_{-\mathbf{p}})^n + (a^\dagger_{\mathbf{p}})^n[a_{\mathbf{k}}, (a^\dagger_{-\mathbf{p}})^n]$$
$$= n(a^\dagger_{\mathbf{p}})^{n-1}\delta_{\mathbf{k},\mathbf{p}}(a^\dagger_{-\mathbf{p}})^n + (a^\dagger_{\mathbf{p}})^n n(a^\dagger_{-\mathbf{p}})^{n-1}\delta_{\mathbf{k},-\mathbf{p}}$$
$$= \frac{\partial}{\partial a^\dagger_{\mathbf{k}}}\left((a^\dagger_{\mathbf{p}})^n (a^\dagger_{-\mathbf{p}})^n\right).$$

In this way, one can see that the operational action of the commutator is completely mathematically analogous to a formal differentiation with respect to the conjugate creation operator $a^\dagger_{\mathbf{k}}$. Leveraging this powerful result, we evaluate the commutator with the full exponential:

$$\left[a_{\mathbf{k}}, \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)\right] = \frac{\partial}{\partial a^\dagger_{\mathbf{k}}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)$$
$$= -\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p}(\delta_{\mathbf{k},\mathbf{p}} a^\dagger_{-\mathbf{p}} + \delta_{\mathbf{k},-\mathbf{p}} a^\dagger_{\mathbf{p}}) \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)$$
$$= -\frac{v_k}{u_k} a^\dagger_{-\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right).$$

By substituting this evaluated commutator back into our initial expression, we obtain the exact relation:

$$u_k a_{\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle = -v_k a^\dagger_{-\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}{}' \frac{v_p}{u_p} a^\dagger_{\mathbf{p}} a^\dagger_{-\mathbf{p}}\right)|0\rangle,$$

which rigorously proves that our proposed exponential pair-state possesses exactly the property demanded by Equation 1.3.3. Note that the exponential operator acting on the vacuum creates an infinite superposition of bosonic pairs with equal and opposite non-zero momenta, serving as the formal quantum mechanical description of the condensate depletion.

Finally, let us address the macroscopic component corresponding to $\mathbf{k} = 0$. Recalling the mean-field substitution utilized to diagonalize the Hamiltonian ($a_0 \to \sqrt{N_0}$ and $a_0^\dagger \to \sqrt{N_0}$), we physically require that the expectation values of the macroscopic creation and annihilation operators over the true ground state equal this condensed value:

$$\langle \psi_0 | a_0 | \psi_0 \rangle = \sqrt{N_0},$$

$$\langle \psi_0 | a_0^\dagger | \psi_0 \rangle = \sqrt{N_0}.$$

These two expectation value equations are simultaneously satisfied by enforcing the eigenvalue relation:

$$a_0 | \psi_0 \rangle = \sqrt{N_0} | \psi_0 \rangle.$$

This defining equation is universally satisfied by Glauber coherent states, which are mathematically structured as:

$$e^{-N_0/2} e^{\sqrt{N_0} a_0^\dagger} | 0 \rangle.$$

Because the zero-momentum operators completely commute with the pair-creation operators at finite momenta ($\mathbf{p} \neq 0$), both the mean-field condensate requirement and the Bogoliubov pair-correlation requirement can be satisfied simultaneously. Collecting these distinct physical contributions, we arrive at the complete, unified expression for the interacting many-body ground state:

$$| \psi_0 \rangle = \frac{1}{\mathcal{N}} e^{\sqrt{N_0} a_0^\dagger} \exp \left(-\frac{1}{2} {\sum_{\mathbf{p}}}' \frac{v_p}{u_p} a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger \right) | 0 \rangle, \tag{1.3.4}$$

where $\mathcal{N}$ is an overall normalization constant that ensures $\langle \psi_0 | \psi_0 \rangle = 1$. By explicitly computing the norm of this state, this constant can be analytically proven to be:

$$\mathcal{N} = e^{N_0/2} \left(\prod_{\mathbf{k}} u_k^2 \right)^{1/2}.$$

1.4 | Spontaneous Symmetry Breaking and Goldstone Theorem

It is straightforward to verify that the original unperturbed Hamiltonian commutes with the total number operator $\hat{N} = \sum_{\mathbf{k}} a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}}$. This commutation property is intimately connected with the fundamental invariance of the Hamiltonian under global U(1) phase transformations:

$$a_{\mathbf{k}} \to e^{i\phi} a_{\mathbf{k}},$$
$$a_{\mathbf{k}}^{\dagger} \to e^{-i\phi} a_{\mathbf{k}}^{\dagger}.$$

These phase rotations are formally realized by the application of the unitary continuous symmetry operator $U(\phi)$:

$$U(\phi) = e^{i\phi \hat{N}}.$$

Indeed, it is easy to explicitly check the generator action on the annihilation operator:

$$U^{\dagger}(\phi) a_{\mathbf{k}} U(\phi) = e^{i\phi} a_{\mathbf{k}}.$$

Because the Hamiltonian is invariant under this specific transformation—meaning $U^{\dagger}(\phi) H U(\phi) = H$—Noether's theorem guarantees there should be a conserved macroscopic charge. This conserved charge is nothing else than the mathematical generator of the transformation itself, namely, the total number operator $\hat{N}$.

Crucially, even if the Hamiltonian governing the system is perfectly invariant under the above U(1) transformation, the interacting ground state $|\psi_0\rangle$ is not. This profound physical phenomenon is known as spontaneous symmetry breaking. There are multiple analytical ways to see this symmetry breaking. Consider the macroscopic expectation value of the zero-momentum annihilation operator a_0. We previously established that over the coherent ground state, $\langle \psi_0 | a_0 | \psi_0 \rangle \neq 0$. Applying the unitary transformation, we have:

$$\langle \psi_0 | (U^{\dagger}(\phi) a_0 U(\phi)) | \psi_0 \rangle = \langle \psi_0 | (e^{i\phi} a_0) | \psi_0 \rangle.$$

However, if the ground state $|\psi_0\rangle$ were truly invariant under the symmetry group, then by definition $U(\phi)|\psi_0\rangle = |\psi_0\rangle$. This invariant assumption would require that the equation evaluates to:

$$\langle \psi_0 | a_0 | \psi_0 \rangle = e^{i\phi} \langle \psi_0 | a_0 | \psi_0 \rangle,$$

which can logically be satisfied for an arbitrary phase ϕ if and only if $\langle \psi_0 | a_0 | \psi_0 \rangle = 0$. This immediately leads to a direct contradiction with our foundational result that the condensate is macroscopically occupied. The exact same mathematical proof can be repeated for the pair correlation expectation value at finite momenta $\mathbf{k} \neq 0$. In that case, we notice that

$$\langle \psi_0 | a_{\mathbf{k}} a_{-\mathbf{k}} | \psi_0 \rangle = -u_k v_k \neq 0,$$

and applying the operator $U(\phi)$ to both bare operators and falsely imposing the invariance of the ground state $|\psi_0\rangle$ once again yields a fatal contradiction.

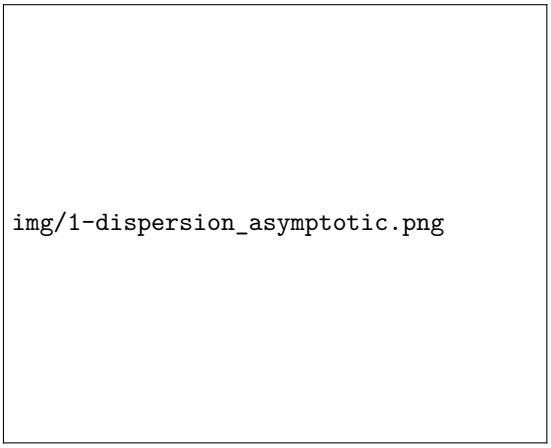
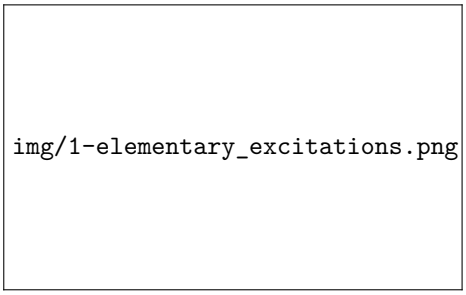


Figure 1.5: Plot of the dispersion $E_k = \sqrt{\epsilon_k^2 + 2nU_0\epsilon_k}$ (blue line) with its two asymptotic behaviours: for $k \to 0$ we have that the dispersion is linear with $E_k \sim c\hbar k$, while for $k \to \infty$ we have that the dispersion is quadratic with $E_k \sim \epsilon_k + nU_0$.

Once we have rigorously verified that the chosen ground state explicitly breaks the continuous U(1) symmetry, we know by Goldstone's theorem that there must emerge massless Goldstone bosons. These excitations must exhibit an energy dispersion that vanishes continuously (specifically, linearly) with the momentum p, yielding $E \sim cp$ for $p \to 0$. Recall that after executing the Bogoliubov transformation, the interacting Hamiltonian becomes diagonalized, featuring the quasiparticle energy spectrum given by E_k. The asymptotic limit $k \to 0$ for E_k has been already evaluated, successfully reproducing exactly the linear phononic dispersion that Goldstone's theorem demands: $E_k \sim c\hbar k$ (as visualized in Figure 1.5), with the characteristic sound velocity c scaling with the interaction strength.

Figure 1.6: Elementary excitations are modes that allow us to describe low energy states. If we excite the ground state GS, increasing the energy by $+E_k$, it's like creating an elementary excitation (a quasi-particle).



The derived energy values E_k structurally describe the dispersion of these elementary excitations, which are collective modes that allow us to describe effectively low-energy many-body states as a simple ideal gas of quasi-particles (see Figure 1.6). This quasiparticle framework is particularly relevant at very low temperatures T, when only these low-energy excited states are thermally activated, since random thermal fluctuations are small and strictly of order $k_B T$.

Consequently, complex interacting states at low energy can be cleanly described as a dilute gas of elementary excitations. It is critical to recognize that these independent excitations are not the original bare bosons created by the standard operators $a^\dagger_{\mathbf{k}}$, but rather new hybridized bosonic modes created by the Bogoliubov operators $\alpha^\dagger_{\mathbf{k}}$ for small k. These elementary excitations are governed by an effective interacting dispersion E_k that is structurally distinct from the purely quadratic free-particle dispersion ϵ_k of the original bare bosons.

Remarkably, even for a strongly interacting bare Bose gas, provided the temperature remains sufficiently low, very few of these hybridized modes are activated. Because the thermal density of quasi-particles is low, these elementary excitations are highly dilute and can thus be treated to an excellent approximation as a non-interacting ideal gas. Lastly, we note a fundamental statistical

constraint: for a system comprised exclusively of underlying bosons, like in the present model, only bosonic elementary excitations are physically possible. In contrast, for a many-body system of underlying fermions, the collective excitations can manifest as either fermionic (quasiparticles) or bosonic (e.g., Cooper pairs, plasmons) elementary excitations depending on the specific pairing mechanics.

1.5 | Specific Heat and Condensate Fraction

Let us proceed with the calculation of the specific heat of a dilute Bose gas. From the diagonalized Hamiltonian, we established that at absolute zero temperature, $T = 0$, the system resides in the quasiparticle vacuum with no elementary excitations. At a finite temperature $T > 0$, the thermal mean number of elementary excitations, $\langle \alpha_{\mathbf{k}}^{\dagger} \alpha_{\mathbf{k}} \rangle$, is governed by the standard Bose-Einstein distribution—since these Bogoliubov quasiparticles are fundamentally bosons:

$$\langle \alpha_{\mathbf{k}}^{\dagger} \alpha_{\mathbf{k}} \rangle = \frac{1}{e^{\beta E_k} - 1},$$

where $\beta = (k_B T)^{-1}$. In this expression, the chemical potential μ associated with the elementary excitations is strictly zero. This physically arises because the total number of quasiparticle excitations is not a conserved quantity; it is determined dynamically by the thermal equilibrium at temperature T (entirely analogous to phonons in a crystalline solid or photons in black-body radiation). Consequently, the total internal energy $E(T)$ of the gas at temperature T is given by the sum of the zero-point ground state energy and the thermal excitation energy:

$$E(T) = \mathcal{E}_0 + \sum_{\mathbf{k}} \frac{E_k}{e^{\beta E_k} - 1}.$$

Taking the thermodynamic limit, the discrete summation over wave vectors transitions into a continuous integral, yielding the total energy density:

$$\frac{E(T)}{V} = \frac{\mathcal{E}_0}{V} + \int \frac{d^3k}{(2\pi)^3} \frac{E_k}{e^{\beta E_k} - 1}.$$

For high-energy modes where $E_k \gtrsim k_B T$, the integrand is exponentially suppressed by the Bose distribution. Therefore, the dominant contribution to the integral comes strictly from the low-energy domain where $E_k \lesssim k_B T$. At very low temperatures, this restricts our focus to the extreme long-wavelength limit of the dispersion relation.

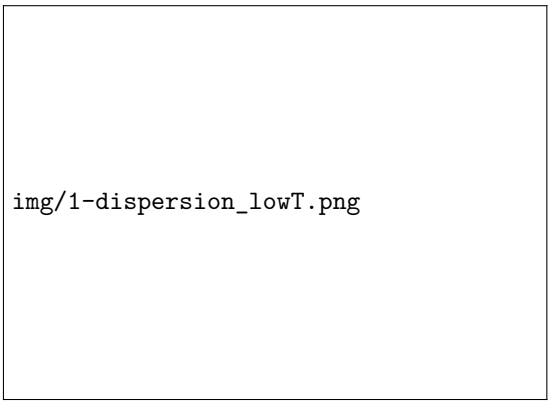


Figure 1.7: At low temperature T, if we are interested just in the region $E_k \lesssim k_B T$, we can see that the dispersion E_k goes linearly with the momentum k.

In this low-momentum regime, we recall that the Bogoliubov dispersion is strictly linear, $E_k \approx c\hbar k$. Thus, in the asymptotic limit $T \to 0$, we can safely replace the full dispersion relation with its phononic approximation inside the integral:

$$\frac{E(T)}{V} = \frac{\mathcal{E}_0}{V} + \int \frac{d^3k}{(2\pi)^3} \frac{c\hbar k}{e^{\beta c\hbar k} - 1}.$$

Evaluating this integral by switching to spherical coordinates and introducing the dimensionless integration variable $x = \beta c\hbar k$, we obtain:

$$\begin{aligned}\frac{E(T)}{V} &= \frac{\mathcal{E}_0}{V} + \left(\frac{k_B T}{c\hbar}\right)^3 k_B T \frac{4\pi}{(2\pi)^3} \int_0^\infty \frac{x^3}{e^x - 1} dx \\ &= \frac{\mathcal{E}_0}{V} + \frac{\pi^2}{30} \frac{(k_B T)^4}{(c\hbar)^3}.\end{aligned}$$

From this energy density, we directly compute the isochoric heat capacity, C_V:

$$C_V = \left(\frac{\partial E(T)}{\partial T}\right)_V = \frac{2\pi^2}{15} V k_B \left(\frac{k_B T}{c\hbar}\right)^3.$$



Figure 1.8: Plot of the behavior of the specific heat C_V at low temperature for a bosonic system. The blue line represents the specific heat of an interacting Bose gas, which scales proportionally to the cube of the temperature. The red line represents the specific heat of a non-interacting Bose gas, scaling proportionally to the temperature to the power of three halves.

This final expression reveals that the low-temperature specific heat of the weakly interacting Bose gas scales as T^3. This characteristic cubic temperature dependence mirrors the behavior of acoustic phonons in a Debye solid or photons in a cavity, fundamentally stemming from the linear phononic dispersion relation. We can starkly contrast this interacting result with the ideal, non-interacting Bose gas, which exhibits a $T^{3/2}$ dependence due to its purely quadratic free-particle dispersion. In the interacting system, the linear dispersion creates a lower density of states at low energies compared to the non-interacting case, making the elementary excitations significantly harder to thermally excite at temperatures near absolute zero.

1.6 | Landau Criterion for Superfluidity

In a normal fluid flowing through a pipe, friction between the fluid and the pipe boundaries leads to a continuous dissipation of energy and a corresponding macroscopic pressure drop along the direction of flow. Conversely, superfluidity (a remarkable quantum phenomenon famously discovered in liquid helium by Pyotr Kapitza) is characterized by the ability of certain fluids to flow without any friction. In a perfectly superfluid state, even at a finite macroscopic flow velocity $\mathbf{v} > 0$, the pressure remains exactly constant along the fluid, such that $\nabla p = 0$.

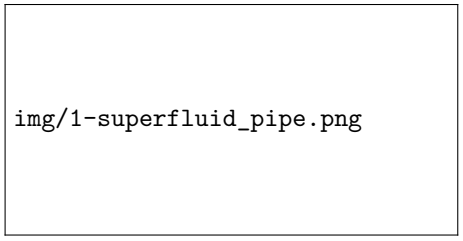


Figure 1.9: A fluid flows in a pipe at a velocity $\mathbf{v}$ and along the flow the pressure decreases due to viscosity, generating a gradient of pressure ∇p.

Our primary goal is to leverage the spectral properties of the many-body Hamiltonian and the associated elementary excitation dispersion relation to rigorously derive a fundamental condition known as the Landau criterion for superfluidity.

To understand the microscopic origin of friction, consider a classical many-body fluid, such as water, flowing through a pipe. If we assume the system is initially prepared in a state devoid of internal excitations, physical friction with the pipe boundaries will eventually induce their formation. Consequently, the total internal energy of the macroscopic system increases by the cumulative energy of these newly generated thermal excitations. For instance, if the initial state follows a classical Maxwell-Boltzmann momentum distribution, the dissipation of macroscopic forward momentum into random thermal excitations will eventually cause the distribution to narrow and shift toward lower net momenta. This physical picture implies that by examining the energy spectrum and the quasiparticle dispersion relation, we can determine precisely when adding energy to the system through such thermal or frictional processes is kinematically allowed.

1.6.1 | Kinematic Framework and Galilean Transformations

Lev Landau proposed an elegant kinematic criterion to establish whether a zero-temperature ($T = 0$) system behaves as a superfluid, specifically allowing one to predict the critical velocity at which the frictionless superfluid flow breaks down. Landau's derivation relies on comparing the energetics of the system in two distinct inertial reference frames: the laboratory frame K, where the fluid moves with a macroscopic velocity $\mathbf{v}$ and the pipe is perfectly at rest, and the co-moving frame K', where the fluid is strictly at rest and the pipe moves backwards with a velocity $-\mathbf{v}$. The frame K' thus translates with a velocity $\mathbf{v}$ with respect to the laboratory frame K.

In the co-moving frame K', the fluid is statically at rest. If the system experiences absolutely no friction with the pipe walls, it remains in its pristine many-body ground state, characterized by the complete absence of thermal excitations ($\langle \alpha^\dagger_{\mathbf{k}} \alpha_{\mathbf{k}} \rangle = 0$ for all $\mathbf{k}$). In this unperturbed scenario, the total energy of the fluid is simply its zero-point ground state energy, $E'_{(0)} = \mathcal{E}_0$.

If, however, friction were to occur, the interaction with the moving pipe walls would perturb the fluid, doing work to create elementary excitations. The system's internal energy would consequently

increase by $\sum_{\mathbf{k}} E_k \langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle$, where the expectation values $\langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle$ are positive integers (or strictly non-zero for finite-temperature statistical ensembles). To analyze the threshold of this dissipative process, let us consider the creation of a single elementary excitation with momentum $\mathbf{p}$. The total energy of the perturbed fluid in frame K' then becomes $E'_{(\mathrm{exc})} = \mathcal{E}_0 + E_p$.

We now transform this energy back to the laboratory frame K using the standard **Galilean transformation** for the total energy of a many-particle system:

$$E = E' + \frac{MV^2}{2} + \mathbf{P}' \cdot \mathbf{V},$$

where E is the energy in frame K, E' and $\mathbf{P}'$ are the total internal energy and total internal momentum in frame K', M is the total macroscopic mass of the fluid, and $\mathbf{V}$ is the relative velocity of K' with respect to K. In our setup, this relative frame velocity is exactly $\mathbf{V} = \mathbf{v}$.

Remark. *When performing this Galilean boost, one must be exceedingly careful regarding the definition of momentum if the system is coupled to an external gauge field (such as an electromagnetic field for charged superfluids like superconductors). In such cases, the canonical momentum is shifted by the vector potential and does not simply correspond to the inertial mass times the velocity. For our neutral bosonic superfluid, however, this complication does not arise.*

For the fluid hosting a single elementary excitation, its internal energy in K' is $E' = E'_{(\mathrm{exc})} = \mathcal{E}_0 + E_p$. Because the macroscopic fluid is at rest in K', the only contribution to the total internal momentum comes from the newly created excitation itself, so $\mathbf{P}' = \mathbf{p}$. Substituting these into the transformation yields the total energy of the excited fluid in the laboratory frame K:

$$E_{(\mathrm{exc})} = \mathcal{E}_0 + E_p + \frac{Mv^2}{2} + \mathbf{p} \cdot \mathbf{v}.$$

Conversely, if the fluid remains in the unperturbed ground state, its internal energy in K' is merely $E' = E'_{(0)} = \mathcal{E}_0$, and its total internal momentum is identically zero ($\mathbf{P}' = 0$). Substituting these into the Galilean transformation, the energy of the frictionless flowing fluid in frame K is simply:

$$E_{(0)} = \mathcal{E}_0 + \frac{Mv^2}{2}.$$

Combining the last two expressions, we determine the energy cost to create the elementary excitation in the laboratory frame:

$$E_{(\mathrm{exc})} - E_{(0)} = E_p + \mathbf{p} \cdot \mathbf{v}.$$

To definitively establish whether the system behaves as a superfluid, we must examine the sign of this energy difference. Friction fundamentally occurs when the creation of an excitation is energetically favorable, which happens when the perturbed state has a lower total energy than the unperturbed flow ($E_{(\mathrm{exc})} < E_{(0)}$). In contrast, robust superfluidity is maintained whenever the spontaneous creation of excitations is kinematically forbidden because it costs energy ($E_{(\mathrm{exc})} > E_{(0)}$).

The critical velocity v_c is defined as the minimum macroscopic flow velocity at which the system's total energy can be spontaneously lowered (or kept equal) by the addition of elementary excitations. This threshold dictates exactly when it becomes thermodynamically convenient to dissipate kinetic energy. By imposing the boundary equality $E_{(\mathrm{exc})} = E_{(0)}$, we will obtain the precise conditions for the critical velocity v_c, which sharply separates the dissipative friction regime from the stable frictionless regime, thereby marking a dynamical phase transition to superfluidity.

1.6.2 | The Critical Velocity Threshold

In order for the spontaneous creation of an excitation to be kinematically possible in the laboratory reference frame K, the energy difference $E_{(\text{exc})} - E_{(0)}$ must be negative. This is because friction physically transfers kinetic energy from the moving fluid to the stationary pipe; consequently, the macroscopic fluid must decrease its total energy in this dissipative process. Friction will thus occur only if, for some values of the excitation momentum $\mathbf{p}$, the following inequality holds:

$$E_p + \mathbf{p} \cdot \mathbf{v} < 0.$$

The above condition is most easily satisfied if the created momentum $\mathbf{p}$ is aligned exactly anti-parallel (opposite) to the flow velocity $\mathbf{v}$. In this optimal dissipative configuration, the vector dot product becomes $-pv$, resulting in the simplified scalar condition:

$$\frac{E_p}{p} < v.$$

So, for a given macroscopic flow velocity v, friction will be physically possible if and only if the above condition is satisfied by some value of the excitation momentum p.

To find the absolute threshold for dissipation, we must consider the global minimum value of the ratio E_p/p as the momentum p is varied over all possible accessible states:

$$v_c \equiv \min_p \frac{E_p}{p}. \tag{1.6.1}$$

According to our kinematic inequality, only if the velocity v of the fluid strictly exceeds this critical threshold v_c is the creation of excitations (and thus friction) possible. We therefore conclude the fundamental Landau criterion: if v_c is strictly greater than zero, then the fluid will flow without any friction for all velocities $v < v_c$, behaving as a true superfluid. Mathematically, at a local minimum of the ratio E_p/p, simple calculus dictates that its derivative must vanish:

$$\frac{d}{dp}\left(\frac{E_p}{p}\right) = 0,$$

yielding the geometric condition:

$$\frac{dE_p}{dp} = \frac{E_p}{p}.$$

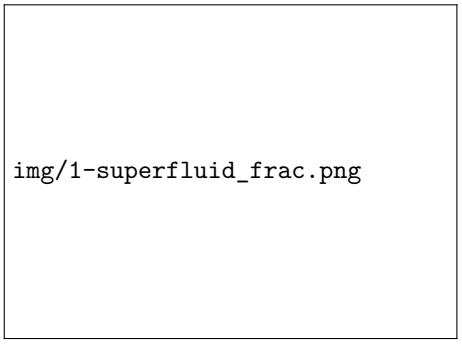


Figure 1.10: Plot of the dispersion E_p as a function of the momentum p (the blue line). For a given generic point P of coordinates (p, E_p), the two red lines represent respectively the tangent to the curve in P (with slope $\frac{dE_p}{dp}$) and the straight line connecting the origin to P, with slope E_p/p.

Note that this derivative equation has a highly intuitive geometric interpretation: if one plots the excitation energy E_p as a function of the momentum p, the derivative dE_p/dp represents the local slope of the tangent to the curve at the point (p, E_p), while the ratio E_p/p represents the overall

slope of the secant line joining the origin directly to that same point (p, E_p) (see Figure 1.10). When these two distinct slopes perfectly coincide, the secant line from the origin becomes tangent to the dispersion curve. It is exactly at this point of tangency that a minimum of E_p/p occurs, and the numerical value of this tangent slope identically defines the critical velocity v_c.

1.6.3 | Application to Physical Bose Systems

It is now time to apply the Landau criterion to various physical Bose systems, including both ideal and weakly interacting dilute gases, as well as strongly interacting systems like liquid helium. By analyzing the dispersion relations of elementary excitations in these systems, we can determine their critical velocities and assess their superfluid properties.

Ideal and Weakly Interacting Dilute Gases

The ideal Bose gas is a particularly instructive example. In this case, the energy dispersion is purely kinetic, given by $E_p = \frac{p^2}{2m}$. Let us analyze this system through the following example.

Example (No Superfluidity for Ideal BEC). Let us apply the Landau criterion to an ideal, non-interacting Bose-Einstein condensate. For a free particle, the energy dispersion is purely kinetic, meaning $E_p = \frac{p^2}{2m} \sim p^2$. When evaluating the critical velocity ratio, we find $\frac{E_p}{p} \propto \frac{p^2}{p} = p$. Taking the limit to find the minimum over all accessible momenta, $\min_p \left(\frac{E_p}{p}\right) \xrightarrow{p\to 0} 0$. Because $v_c = 0$, any arbitrarily small flow velocity $v > 0$ satisfies the condition for dissipation. It is therefore always kinematically convenient to create excitations, meaning an ideal BEC strictly exhibits no superfluidity.

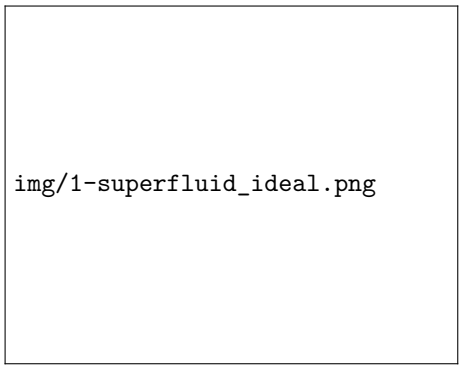


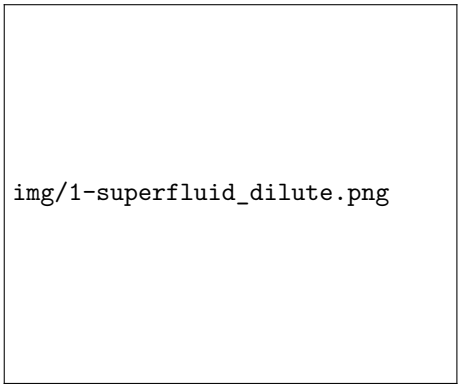
Figure 1.11: Dispersion E_p of elementary excitations in an ideal Bose gas as a function of the momentum p (blue line). The point in which the Landau criterion is satisfied is exactly at the origin, and the slope of the tangent in this point (horizontal red line) gives the critical velocity $v_c = 0$.

For the weakly interacting dilute Bose gas analyzed previously via Bogoliubov theory, recalling the behavior of E_p for small p, the dispersion is linear: $E_p \sim cp$. Consequently, the ratio E_p/p is a constant c near the origin. The global minimum condition is realized precisely as $p \to 0$ (see Figure 1.12):

$$v_c = \min_p \frac{E_p}{p} = c.$$

This profound result means that the weakly interacting Bose gas is a true superfluid for any flow velocity $v < c$. Only for supersonic flows $v > c$ are Bogoliubov quasiparticle excitations created, and macroscopic dissipation becomes possible.

Figure 1.12: Dispersion E_p of elementary excitations in a dilute Bose gas as a function of the momentum p (blue line). The point in which the Landau criterion is satisfied is the origin, and the slope of the tangent in this point (red line) gives the critical velocity: $v_c = \min_p E_p/p$. For $v \geq v_c$ there is friction, while for $v < v_c$ the gas is superfluid.



Superfluidity is not a universal guarantee; there will always be some critical parameter (such as a threshold velocity or a critical temperature) beyond which the system will spontaneously start to generate excitations and subsequently lose its superfluid properties.

Strongly Correlated Regimes: Liquid Helium-4 and Rotons

In strongly interacting Bose systems, such as liquid Helium-4, the situation becomes more complex. The measured spectrum of elementary excitations in liquid ^{4}He reveals a non-trivial dispersion relation that includes a local minimum at finite momentum, known as the "roton minimum". This feature significantly influences the critical velocity for superfluidity.

Example (Helium-4). Even in strongly interacting Bose systems that defy simple perturbative treatment, like liquid Helium-4, robust superfluid behavior emerges at sufficiently low temperatures. Finally, let us consider the measured spectrum of elementary excitations for liquid ^{4}He (as determined directly from neutron scattering experiments). We see that in this complex strongly-correlated case, there are actually two distinct geometric solutions (points of tangency) satisfying the Landau condition.

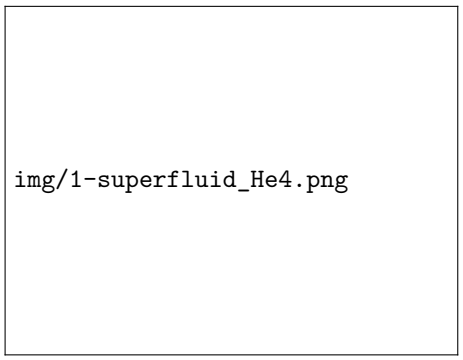


Figure 1.13: Dispersion E_p of elementary excitations in ^{4}He as a function of the momentum p (blue line). The points in which the Landau criterion is satisfied are two (straight red lines tangent to the curve). The one with the lowest slope determines the true critical velocity $v_c \approx 60$ m/s (while the initial sound velocity is much higher at $c = 240$ m/s). The gas is a superfluid for $v < v_c$.

The tangent line with the absolute lowest slope geometrically corresponds to the true global minimum of E_p/p. In Helium-4, this absolute minimum is reached not at the origin, but at a finite momentum close to the famous "roton minimum" of the dispersion curve E_p. The slope of this tangent gives the corresponding critical velocity for Helium-4: $v_c \approx 60$ m/s. Notice that this critical threshold for creating rotons is significantly lower than the speed of sound determined by the initial linear slope at the origin, which is $c = 240$ m/s.

The energy gap at this local minimum is called the roton gap, denoted Δ, and is experimentally

measured such that $\Delta/k_B = 8.7$ K. The elementary excitations grouped close to this finite-momentum minimum acquire a distinctly quadratic, particle-like effective form:

$$E_p = \Delta + \frac{(p - p_0)^2}{2m^*}$$

with an effective mass $m^* = 0.16m$ (where m is the bare mass of a ^{4}He atom) and a characteristic momentum $p_0/\hbar = 1.9$ Å^{-1}. These specific high-momentum elementary excitations are called rotons (while the low-momentum excitations corresponding to the initial linear part of the spectrum are standard acoustic phonons). We thus see that when the macroscopic superfluid velocity v exceeds $v_c \approx 60$ m/s, the spontaneous kinematic excitation of rotons completely destroys the superfluid behavior, even though the flow remains highly subsonic relative to c.

1.6.4 | The Macroscopic Probe and the Superfluid Phase Boundary

To deeply understand the physical mechanism underpinning the Landau argument, it is essential to clarify the asymmetric roles of the macroscopic boundary (e.g. the pipe or a moving capillary) and the quantum fluid. *Friction* in this context arises exclusively when the macroscopic motion of the object relative to the fluid creates elementary quasiparticle excitations within the fluid. At absolute zero temperature ($T = 0$), the superfluid resides entirely in its many-body ground state. Consequently, it cannot dissipate energy by transitioning to a lower energy state: it can only act as an **energy sink**, absorbing energy from the moving object by becoming excited.

In this framework, the pipe is treated strictly as an inert, classical macroscopic body moving at a well-defined velocity **v**. Its internal quantum spectrum, structure, or thermal kinetic energy are completely irrelevant to the onset of the dispersion process. The generation of friction is strictly a consequence of the fluid's capacity to generate excitations. As dictated by the Landau criterion, this unidirectional transfer of kinetic energy from the moving macroscopic probe to the quantum fluid becomes kinematically permitted only when the object's velocity exceeds the critical threshold v_c.

This critical boundary can be visualized geometrically in a parameter space plotting the temperature T against the flow velocity parameter v. The phase boundary cleanly separates the parameter space into a normal dissipative regime and a protected superfluid phase.

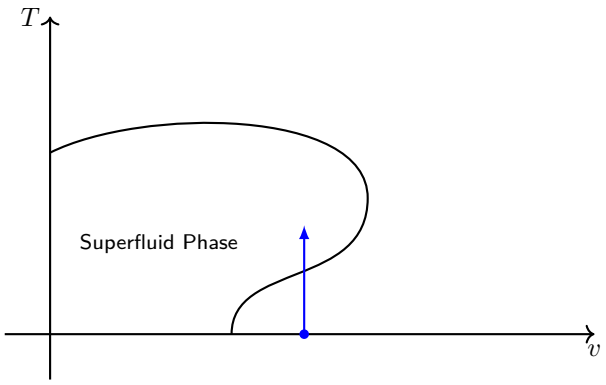


Figure 1.14: Schematic phase diagram illustrating the superfluid boundary as a function of the flow velocity parameter v and the temperature T. The solid curve defines the onset of excitation creation (friction). The blue arrow depicts a specific constant-velocity trajectory traversing the parameter space.

1.6.5 | Finite Temperature Effects and the Two-Fluid Model

It is crucial to note that the entire Landau derivation presented above assumes absolute zero temperature. If we consider a finite temperature $T > 0$, we must take into account the pre-existing gas of thermal excitations that are statistically present in the system. Indeed, even if the kinematic line pv sits safely below the zero-temperature critical dispersion threshold, we must consider the energy contributions and collisions from these thermal excitations, which can effectively transfer momentum and turn off the macroscopic superfluidity by traversing the dispersion gap. Consequently, the true critical velocity will be strictly lower than at zero temperature, and it will exhibit a strong dependence on the ambient temperature of the system.

Furthermore, at finite temperatures, there will always be a "normal" fraction of the fluid comprised of these thermal excitations that experiences ordinary friction. However, there concurrently exists a remaining fraction of the fluid that remains perfectly superfluid. According to the two-fluid model, the fraction of the fluid that is superfluid scales directly with the total number of particles, ensuring it remains a macroscopically sizable and observable portion of the system up until the critical transition temperature is reached.

Concluding Remarks: The Physics of Emergence

Through the mathematical machinery of the Bogoliubov transformation, we systematically mapped the bare bosonic degrees of freedom, $a_{\mathbf{k}}$, to a new set of operators, $\alpha_{\mathbf{k}}$. This was not merely a convenient mathematical change of basis, but a profound physical restructuring of the system. The new operators create and destroy collective, hybridized modes that intermix particles and holes. These emergent entities are defined as **quasiparticles**, and they—rather than the original bare bosons—constitute the true elementary excitations of our interacting many-body description.

This conceptual shift perfectly illustrates the celebrated paradigm articulated by P. W. Anderson: "*More is Different*". Quantum Many-Body Theory (QMBT) demonstrates that macroscopic physics is fundamentally distinct from single-particle quantum mechanics. Merely sticking individual constituent pieces together is entirely insufficient to understand the whole. As the system grows to the thermodynamic limit, non-trivial interactions generate entirely new emergent phenomena that simply do not exist at the level of isolated single particles.

A quintessential example of this macroscopic emergence is **spontaneous symmetry breaking**, which serves as a foundational pillar of modern QMBT. As we have seen, while the original Hamiltonian inherently respects a continuous global U(1) phase symmetry, the macroscopic interacting ground state spontaneously breaks it by developing a definite coherent phase. According to Goldstone's theorem, this broken continuous symmetry mathematically necessitates the emergence of massless Goldstone bosons. In the context of the interacting Bose gas, these Goldstone modes physically manifest as the acoustic phonons characterized by the linear dispersion relation, $E_k \sim c\hbar k$, at low momenta.

Ultimately, by closely analyzing the experimental observables and the low-energy spectrum of this system, we arrive at a powerful concluding lesson: **weak interactions can do a lot**. It is tempting to assume that exotic phenomena require strong, non-perturbative couplings. However, we have demonstrated that even an arbitrarily weak interaction is sufficient to completely rewire the vacuum state of the system. A profoundly rich phenomenology (encompassing quantum condensate depletion at absolute zero, the formation of a well-defined sound velocity, spontaneous symmetry breaking, and the onset of macroscopic superfluidity governed by the Landau criterion) emerges not from strong forces, but from the collective many-body nature of weakly interacting quasiparticles.

2 | BCS Theory of Superconductivity

Superconductivity was experimentally observed well before the formal discovery of quantum mechanics, presenting a profound physical mystery that eventually necessitated the development of a comprehensive quantum many-body description. The definitive microscopic explanation of this phenomenon, known as the BCS theory, was developed in 1957 by John Bardeen, Leon Cooper, and John Robert Schrieffer. At its core, the theory explains the superconducting state through the formation of bound states of electrons, known as Cooper pairs.

To formally develop this, we will consider a general system composed of spin-1/2 fermions experiencing a mutual *attractive* interaction between particles of opposite spin. Our primary theoretical interest is to determine the low-temperature phase of this many-body system. The central idea is that this mutual attraction can induce the formation of Cooper pairs—pairs of fermions with opposite spin. Because these composite pairs possess an integer total spin, they behave effectively as bosons and can undergo macroscopic condensation at sufficiently low temperatures.

The original motivation for studying such an interacting fermionic system was the microscopic explanation of metallic superconductivity. Superconductivity is a physical phenomenon occurring in certain metals and compounds whereby an electric current flows with absolutely zero electrical resistance below a specific critical temperature T_c. This phenomenon is deeply analogous to superfluidity; whereas a superfluid involves the macroscopic flow of a fluid with zero viscosity (the mechanical analog of electrical resistance) below a critical temperature, a superconductor involves the frictionless flow of charged carriers, namely electrons.

A crucial thermodynamic clue pointing toward the existence of these bound pairs of fermions in a superconductor is provided by the electronic specific heat, c_v. For an ideal Fermi gas at low temperatures T—which serves as a highly effective model for the conduction electron gas in a normal solid—the specific heat scales linearly with T. However, when a material transitions into a superconductor, the specific heat is observed to vanish exponentially with decreasing temperature. This exponential suppression is the hallmark thermodynamic behavior expected when there is a finite energy gap in the excitation spectrum separating the macroscopic ground state from the first available excited states. The magnitude of this energy gap can be extracted directly from specific heat measurements, revealing it to be extremely small—typically several orders of magnitude smaller—compared to the Fermi energy E_F of the electrons. It is natural to guess that this energy gap is directly associated with the finite binding energy of the fermion pairs. (The microscopic mechanism by which normally repulsive electrons can experience an effective mutual attraction in a metal lattice will be discussed in detail in the subsequent section).

Beyond its foundational success in condensed matter systems, the BCS theoretical framework we are going to develop is remarkably universal. The collection of phenomena that can be governed by these same equations spans an astonishing range of more than ten orders of magnitude in temperature. At the lower extreme, in the nanokelvin regime, the theory applies directly to ultracold Fermi gases trapped in optical lattices. These are dilute, neutral systems of fermionic atoms where an effective mutual attraction—arising fundamentally from van der Waals forces—can be accurately modeled as a simple contact interaction due to the extreme diluteness of the gas. These highly tunable systems have been studied very actively over recent years because experimentalists can artificially tune the strength of the interparticle interaction at will. The theory is equally relevant to other macroscopic neutral systems, such as liquid ^{3}He, where fermionic neutral atoms also experience a mutual attraction. At the absolute opposite extreme of the temperature and density scale, the very same BCS formalism can be extended to describe the pairing of quarks inside the ultra-dense, extremely high-temperature cores of neutron stars (a phenomenon known as color superconductivity).

2.1 | Basic Model for Effective Attraction

Let us consider a simplified model of a solid consisting of a cubic lattice of positively charged ions with a lattice spacing a, permeated by a gas of itinerant electrons. We assume the system is isolated, without any external electric or magnetic fields. The electrons are subject to a periodic potential generated by the ionic lattice; consequently, the potential minima are located in the immediate vicinity of the positive ions, marking the regions where the electrons are most likely to be found.

The electrons interact not only with this periodic ionic potential but also with one another via the standard Coulomb repulsion, $V_{e-e} \sim e^2/r$. However, the presence of the mobile electron gas embedded in the positively charged background activates the many-body phenomenon of screening. In a metal, the spatial redistribution of conduction electrons around any single negative charge effectively screens that negative charge at a distance from it. This screening effect drastically modifies the long-range Coulomb repulsion into a short-range effective interaction, which decays exponentially as $\sim e^{-r/\xi}$, where the screening length ξ is typically on the order of a few lattice spacings a. Although this screened interaction is significantly weaker at macroscopic distances, it remains fundamentally repulsive. To explain the formation of Cooper pairs, we must identify a physical mechanism capable of overcoming this residual repulsion to produce a net effective attraction between the electrons.

The Electron-Phonon Interaction and Fröhlich Mechanism

This requisite attraction arises from the interaction between the electrons and the collective vibrations of the ionic lattice, known as phonons. Because the typical Debye temperature characterizing these macroscopic phonon modes is on the order of $\omega_D \sim 100\,\mathrm{K}$, while superconducting phenomena typically occur at cryogenic temperatures around $1\,\mathrm{K}$, the thermal phonons are effectively frozen out. Consequently, we can safely treat the quantized phonon field as a perturbation to the interacting electron system.

As famously demonstrated by H. Fröhlich, in certain metals, the combination of the screening of the Coulomb repulsion and the virtual exchange of phonons between electrons may lead to an overall effective attraction between them. Specifically, the dynamic interaction between electrons and phonons produces a retarded attractive force. When an electron moves through the crystal, it locally modifies the ionic lattice by pulling the positive ions slightly toward its trajectory, creating a localized cloud of positive charges, and then rapidly moves away. Subsequently, a second electron may be electrostatically attracted by this localized accumulation of positive charge.

A key physical ingredient of this retarded mechanism is the stark difference in dynamical time scales: the heavy distortions of the ionic lattice fade away rather slowly in comparison with the extremely fast time scale over which one electron is replaced by a second one at a certain position due to the rapid motion of the electrons. Ultimately, these two combined effects—electronic screening and virtual phonon exchange—may lead in certain materials to an overall effective attraction between the electrons, provided those electrons belong to a tiny energy shell situated immediately around the Fermi energy E_F.

2.1.1 | Effective Interaction

We aim to describe the behavior of electrons in a solid and understand how they interact with the lattice vibrations. A solid can be modeled as a static lattice of ions, with electrons occupying localized orbitals and tunneling through them. The classical positions of the ions are perturbed by dynamical excitations, which we describe as **phonons**.

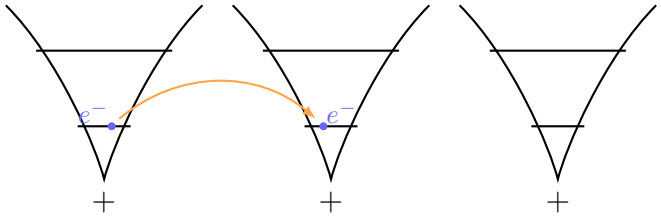


Figure 2.1: Schematic representation of a solid modeled as a static lattice of positive ions. The electrons occupy localized orbitals within the ionic Coulomb potential wells and can tunnel (hop) to adjacent sites. This baseline periodic structure is subsequently perturbed by dynamical lattice vibrations (phonons), which modulate the effective electron-electron interactions.

In the harmonic approximation, the phonon Hamiltonian is constructed as a sum over independent harmonic oscillators:

$$H_{\text{ion}} = \frac{1}{2}\sum_{\mathbf{q}} \hbar\omega_{\mathbf{q}} a^\dagger_{\mathbf{q}} a_{\mathbf{q}},$$

where $a^\dagger_{\mathbf{q}}$ and $a_{\mathbf{q}}$ are the bosonic creation and annihilation operators for a phonon with momentum $\mathbf{q}$, and $\omega_{\mathbf{q}}$ is the phonon frequency.

The Hamiltonian for the electrons is composed of an interaction part and a kinetic part:

$$H_{\text{el}} = H_{\text{int}} + H_{\text{kin}}.$$

Interaction Hamiltonian. Naively, the interaction between electrons is the bare Coulomb repulsion. However, the presence of a background of positive charges screens this interaction. The effective screened potential is often described by the Thomas-Fermi model, which introduces an exponential decay over the distance $|\mathbf{r}|$:

$$V(\mathbf{r}) \propto \frac{e^{-|\mathbf{r}|/\xi}}{|\mathbf{r}|} \sim e^{-|\mathbf{r}|/\xi},$$

where ξ is the Thomas-Fermi screening length.

While the kinetic energy can be formulated on a real-space lattice (yielding the Hubbard model), it is much more convenient to diagonalize it directly in momentum space. Assuming that the kinetic energy is independent of the electron spin σ, we have:

$$H_{\text{kin}} = \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma},$$

where $c^\dagger_{\mathbf{k},\sigma}$ and $c_{\mathbf{k},\sigma}$ are the fermionic creation and annihilation operators for an electron with momentum $\mathbf{k}$ and spin σ, and $\epsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m}$ is the bare electron dispersion.

Total Hamiltonian. Combining the lattice and electronic components, the total Hamiltonian is:

$$H = H_{\text{ion}} + H_{\text{int}} + H_{\text{kin}}.$$

Our ultimate goal is to use perturbation theory to trace out the phonon degrees of freedom. This procedure will yield an effective Hamiltonian that only contains the fermionic operators $(c, c^\dagger)$, effectively incorporating the phonons as a mediated interaction between the electrons:

$$H \to H_{\text{eff}}(c, c^\dagger).$$

2.1.2 | Single Particle Problem

Electron-Phonon Interaction. How can an electron interact with a phonon? By traveling through the solid, an electron can kick the ions and excite their physical modes. Based on the principle of Occam's razor, we consider only the simplest fundamental processes: the emission and the absorption of a single phonon.

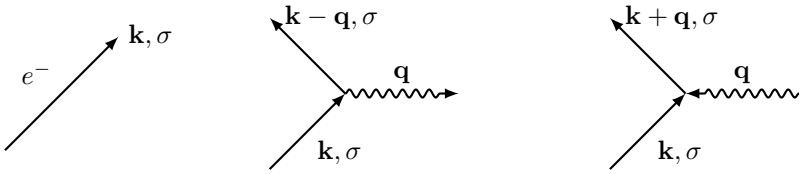


Figure 2.2: Fundamental electron-phonon interaction processes. Left: A free electron propagating with momentum $\mathbf{k}$ and spin σ. Middle: Emission of a single phonon with momentum $\mathbf{q}$, scattering the electron to a new momentum state $\mathbf{k}-\mathbf{q}$. Right: Absorption of a single phonon with momentum $\mathbf{q}$, scattering the electron to momentum $\mathbf{k}+\mathbf{q}$. In both interactions, the spin σ is conserved.

The phenomenological interaction potential V_1 describing these scattering processes must contain a sum over all possible electron momenta $\mathbf{k}$, spins σ, and phonon momenta $\mathbf{q}$:

$$V_1 = \sum_{\mathbf{k},\sigma} \sum_{\mathbf{q}} \left(M_{\mathbf{q},\sigma} a_{\mathbf{q}} c^\dagger_{\mathbf{k}+\mathbf{q},\sigma} c_{\mathbf{k},\sigma} + M_{-\mathbf{q},\sigma} a^\dagger_{\mathbf{q}} c^\dagger_{\mathbf{k}-\mathbf{q},\sigma} c_{\mathbf{k},\sigma} \right).$$

The amplitude of this interaction is governed by the matrix element $M_{\mathbf{q},\sigma}$, which takes the form:

$$M_{\mathbf{q},\sigma} \simeq \frac{|\mathbf{q}|}{\sqrt{M_{\text{ion}} \omega_{\mathbf{q}}}}.$$

Physically, $M_{\mathbf{q},\sigma}$ is inversely proportional to the square root of the ion mass M_{ion}. If the lattice is composed of heavy ions, it is extremely challenging for the electrons to excite the phononic modes, causing the matrix element to vanish rapidly. Conversely, lighter ions allow for an easier transfer of energy from the electrons to the ion chain.

Perturbation Theory in V_1. Our full system is described by:

$$H = H_{\text{kin}} + H_{\text{ph}} + V_1.$$

We proceed by treating V_1 as a perturbation. To do so, we must first define the unperturbed vacuum (ground state) for both fermions and bosons.

Unperturbed Vacuum. We evaluate the vacuum in the presence of some initial particle excitations. This state, $|\phi_{n_e,n_{ph}}\rangle$, is formed by the tensor product of the electron and phonon states:

$$|\phi_{n_e,n_{ph}}\rangle = \left(\prod_{\substack{\mathbf{k}_j=\mathbf{k}_1,\ldots,\mathbf{k}_{n_e} \\ \sigma_j=\sigma_1,\ldots,\sigma_{n_e}}} c^\dagger_{\mathbf{k}_j,\sigma_j} \right) \left(\prod_{\mathbf{q}_j=\mathbf{q}_1,\ldots,\mathbf{q}_{n_{ph}}} a^\dagger_{\mathbf{q}_j} \right) |0,0\rangle.$$

Here, the absolute vacuum is defined as $|0,0\rangle = |n_e = 0, n_{ph} = 0\rangle$. The phonon vacuum itself is a tensor product over all possible momentum states:

$$|0,0\rangle = |0\rangle_{\mathbf{q}_1} \otimes |0\rangle_{\mathbf{q}_2} \otimes \ldots$$

Because the system is confined in a box/lattice, the allowed momenta are discrete and defined within the Brillouin zone. For example, along the x-direction:

$$q_x^{(J)} = \frac{2\pi J}{L_x}, \quad J \in \mathbb{Z}.$$

For the electrons, this highly populated unperturbed ground state is commonly referred to as the **Fermi Sea**.

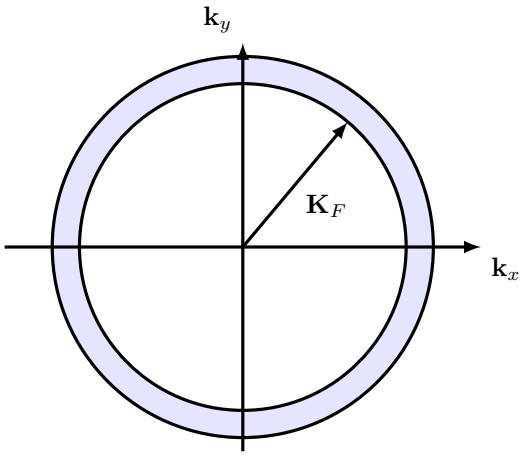


Figure 2.3: Two-dimensional representation of the momentum space for the unperturbed electron gas. The highly populated ground state forms a Fermi sea, bounded by the Fermi circle of radius K_F. The shaded narrow shell highlights the tiny energy window around the Fermi surface where the effective electron-electron interaction, mediated by phonons, is non-zero and attractive.

First Order Perturbation Theory

Let us compute the first-order correction to the energy, which fundamentally requires evaluating the expectation value of the perturbation V_1 over the unperturbed vacuum state $|\phi_0\rangle$:

$$E_1 = \langle\phi_0|V_1|\phi_0\rangle.$$

If we define our initial state explicitly by its definite number of electrons and phonons, $|\phi_0\rangle = |n_e, n_{ph}\rangle$, applying the interaction potential V_1 physically yields a new state where the total number of phonons has changed by exactly one (due to the fundamental processes of either a single emission or a single absorption):

$$V_1|\phi_0\rangle = |n_e, n_{ph} \pm 1\rangle.$$

Because the number of phonons strictly changes under the action of V_1, the resulting state is mathematically orthogonal to the initial state. Under the physically motivated assumption that the phonons do not macroscopically condense (which would give them a finite vacuum expectation value), the first-order energy correction strictly vanishes:

$$\implies \langle n_e, n_{ph}|n_e, n_{ph} \pm 1\rangle = 0.$$

Second Order Perturbation Theory

Since the leading first-order term is identically zero, we must evaluate the perturbation to second order to capture the leading interaction effects. The second-order energy correction E_2 is formally given by summing over all possible intermediate states $|\phi_{n,m}\rangle$:

$$E_2 = \sum_{n,m} \frac{\langle\phi_0|V_1|\phi_{n,m}\rangle\langle\phi_{n,m}|V_1|\phi_0\rangle}{E_0^{(\text{unp.})} - E_{n,m}^{(\text{unp.})}}.$$

Out of this seemingly infinite sum over all basis states, only highly specific terms survive the projection. The number of electrons n in the intermediate state must strictly remain equal to n_e, and the number of phonons m must be exactly $n_{ph}+1$ or $n_{ph}-1$. This selection rule allows us to simplify the numerator structure using Kronecker deltas:

$$E_2 = \sum_{n,m} \frac{\delta_{n,n_e}\left(\delta_{m,n_{ph}+1} + \delta_{m,n_{ph}-1}\right)}{E_0^{(\text{unp.})} - E_{n,m}^{(\text{unp.})}}.$$

Example (Single electron application). To intimately understand the underlying physics, let us perform the explicit calculation for the simplest possible case: a single electron with initial momentum $\mathbf{p}$ and spin σ, and exactly zero initial phonons ($n_e = 1$, $n_{ph} = 0$). The initial state is constructed as $|\phi_0\rangle = c^\dagger_{\mathbf{p},\sigma}|0\rangle$. Applying the interaction V_1 to this specific state and carefully enforcing both momentum and spin conservation through the resulting delta functions, we obtain:

$$\begin{aligned} V_1|\phi_0\rangle &= \sum_{\mathbf{q},\sigma'}\sum_{\mathbf{k}'} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{k}'+\mathbf{q},\sigma'}c_{\mathbf{k}',\sigma'}c^\dagger_{\mathbf{p},\sigma}|0\rangle \\ &= \sum_{\mathbf{q},\sigma'}\sum_{\mathbf{k}'} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{k}'+\mathbf{q},\sigma'}\delta_{\sigma,\sigma'}\delta_{\mathbf{k}',\mathbf{p}}|0\rangle \\ &= \sum_{\mathbf{q}} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{p}+\mathbf{q},\sigma}|0\rangle. \end{aligned}$$

This rigorous evaluation explicitly shows the initial electron's momentum being shifted by $\mathbf{q}$ due to the interaction with the phonon field.

Now, we compute the quantum overlap of this resulting state with a generic intermediate state $|\phi_{n,m}\rangle$. For a non-zero overlap to exist, the intermediate state must contain exactly 1 electron and 1 phonon ($n = 1$, $m = 1$), meaning its bra form is constructed as $\langle\phi_{n,m}| = \langle 0|c_{\mathbf{p}+\mathbf{q}',\sigma'}a_{\mathbf{q}'}$. Taking the inner product systematically isolates the exact phonon mode that was emitted:

$$\begin{aligned} \langle\phi_{n,m}|V_1|\phi_0\rangle &= \left(\sum_{\mathbf{q}',\sigma'}\langle 0|c_{\mathbf{p}+\mathbf{q}',\sigma'}a_{\mathbf{q}'}\right)\left(\sum_{\mathbf{q}} M_{\mathbf{q}}(a^\dagger_{\mathbf{q}} + a_{-\mathbf{q}})c^\dagger_{\mathbf{p}+\mathbf{q},\sigma}|0\rangle\right) \\ &= \sum_{\sigma',\mathbf{q}'} \delta_{\sigma,\sigma'}\delta_{\mathbf{q},\mathbf{q}'}M_{\mathbf{q}} = M_{\mathbf{q}}. \end{aligned}$$

Finally, substituting this squared matrix element back into the formal second-order energy formula, we must evaluate the corresponding energy denominator. The initial unperturbed energy is simply the kinetic energy of the single electron, $E_0^{(\text{unp.})} = \epsilon_{\mathbf{p}}$, and the intermediate unperturbed energy includes both the shifted electron and the newly emitted phonon, $E_{n,m}^{(\text{unp.})} = \epsilon_{\mathbf{p}+\mathbf{q}} + \hbar\omega_{\mathbf{q}}$. Thus, the second-order correction is:

$$E_2 = \frac{M_{\mathbf{q}}^2}{-\hbar\omega_{\mathbf{q}} + (\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}})}. \tag{2.1.1}$$

This crucial formula explicitly defines the effective energy shift that the virtual phonon cloud induces on a single isolated fermion.

2.1.3 | Two Body Problem

To rigorously study the effective electron-electron interactions mediated by phonons, we must transition from a single-particle picture to an initial unperturbed state $|\phi_0\rangle$ containing at least two particles. We define this two-fermion ground state as:

$$|\phi_0\rangle = c^\dagger_{\mathbf{p},\alpha} c^\dagger_{\mathbf{p}',\alpha'} |0\rangle.$$

As established previously, the first-order perturbation theory correction trivially vanishes because the interaction Hamiltonian strictly creates or destroys a single phonon, ensuring orthogonality with the initial phonon-free state. We therefore proceed directly to evaluate the second-order energy correction, which requires computing the transition matrix elements connecting our initial two-particle state to the virtual intermediate states:

$$\langle \phi_{n,m} | V_1 | \phi_0 \rangle \quad \text{and} \quad \langle \phi_{2,1} | V_1 | \phi_0 \rangle.$$

Applying the electron-phonon interaction operator V_1 to our initial state allows the fermionic destruction operator $c_{\mathbf{k},\sigma}$ within V_1 to contract with either of the two initial electrons (possessing momenta $\mathbf{p}$ or $\mathbf{p}'$):

$$\begin{aligned} V_1|\phi_0\rangle &= \sum_{\mathbf{q}} M_{\mathbf{q}} (a^\dagger_{-\mathbf{q}} + a_{\mathbf{q}}) \sum_{\mathbf{k},\sigma} c^\dagger_{\mathbf{k}+\mathbf{q},\sigma} c_{\mathbf{k},\sigma} c^\dagger_{\mathbf{p},\alpha} c^\dagger_{\mathbf{p}',\alpha'} |0\rangle \\ &= \sum_{\mathbf{q}} M_{\mathbf{q}} (a^\dagger_{-\mathbf{q}} + a_{\mathbf{q}}) \left(c^\dagger_{\mathbf{p}+\mathbf{q},\alpha} c^\dagger_{\mathbf{p}',\alpha'} + c^\dagger_{\mathbf{p},\alpha} c^\dagger_{\mathbf{p}'+\mathbf{q},\alpha'} \right) |0\rangle. \end{aligned}$$

This branching leads to two distinct physical scattering processes:

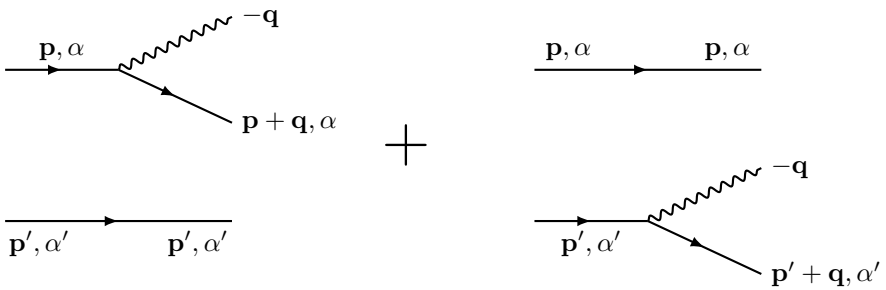


Figure 2.4: Visual representation of the intermediate state created by the interaction operator V_1 acting on the two-electron vacuum, with time flowing horizontally from left to right. The state is a coherent quantum superposition (denoted by the + sign) of two distinct physical processes. Left: The first electron scatters by emitting a phonon $-\mathbf{q}$, while the second electron propagates freely as a spectator. Right: The first electron acts as the free spectator, while the second electron scatters by emitting a phonon $\mathbf{q}$.

(?) Coherent quantum superposition?

These two diagrams correspond precisely to the two possible intermediate states containing exactly one virtual phonon, which can be emitted by either the first electron or the second electron. The bra forms of these specific intermediate states that yield a non-zero overlap are given by:

$$\langle \phi_{2,1}| = \langle 0 | a_{-\mathbf{q}} c_{\mathbf{p}+\mathbf{q},\alpha} c_{\mathbf{p}',\alpha'},$$

$$\langle \tilde{\phi}_{2,1}| = \langle 0 | a_{\mathbf{q}} c_{\mathbf{p},\alpha} c_{\mathbf{p}'-\mathbf{q},\alpha'}.$$

We now substitute these transition matrix elements into the standard second-order perturbation formula. The expectation values of the paired creation and annihilation operators simply trace out to unity, allowing us to factor out the squared electron-phonon matrix element $M_{\mathbf{q}}^2$:

$$E_2' = \left[\frac{\langle \phi_0 | V_1 | \phi_{2,1} \rangle \langle \phi_{2,1} | V_1 | \phi_0 \rangle}{E_0 - E_{2,1}^{(\text{unp.})}} + \frac{\langle \phi_0 | V_1 | \tilde{\phi}_{2,1} \rangle \langle \tilde{\phi}_{2,1} | V_1 | \phi_0 \rangle}{E_0 - \tilde{E}_{2,1}^{(\text{unp.})}} \right]$$
$$= M_{\mathbf{q}}^2 \left(\frac{1}{E_0 - E_{2,1}^{(\text{unp.})}} + \frac{1}{E_0 - \tilde{E}_{2,1}^{(\text{unp.})}} \right).$$

By evaluating this algebraic sum, we can determine whether the resulting energy shift represents a repulsive or an attractive effective interaction between the electrons, a property completely dictated by the sign of the unperturbed energy denominators.

Defining the initial unperturbed energy and the intermediate unperturbed energies for these two distinct exchange processes, we have:

$$E_0 = \epsilon_{\mathbf{p}} + \epsilon_{\mathbf{p}'},$$
$$E_{2,1}^{(\text{unp.})} = \epsilon_{\mathbf{p}+\mathbf{q}} + \epsilon_{\mathbf{p}'} + \hbar\omega_{-\mathbf{q}},$$
$$\tilde{E}_{2,1}^{(\text{unp.})} = \epsilon_{\mathbf{p}'-\mathbf{q}} + \epsilon_{\mathbf{p}} + \hbar\omega_{\mathbf{q}}.$$

Substituting these explicit energies into our second-order expression, the energy shift E_2' becomes:

$$E_2' = M_{\mathbf{q}}^2 \left(\frac{1}{\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}} - \hbar\omega_{\mathbf{q}}} + \frac{1}{\epsilon_{\mathbf{p}'} - \epsilon_{\mathbf{p}'-\mathbf{q}} - \hbar\omega_{\mathbf{q}}} \right).$$

To simplify this result, we apply the physical limit of a small momentum exchange $\mathbf{q}$. We assume the relevant scattering processes keep the electrons tightly confined close to the Fermi surface (within the heavily populated Fermi sea), which imposes the approximate kinematic relation:

$$\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}} \simeq \epsilon_{\mathbf{p}'-\mathbf{q}} - \epsilon_{\mathbf{p}'}.$$

Applying this small-$\mathbf{q}$ approximation, the effective energy shift neatly reduces to the Fröhlich interaction form:

$$E_2' \simeq M_{\mathbf{q}}^2 \left(\frac{2\hbar\omega_{\mathbf{q}}}{(\epsilon_{\mathbf{p}+\mathbf{q}} - \epsilon_{\mathbf{p}})^2 - (\hbar\omega_{\mathbf{q}})^2} \right). \tag{2.1.2}$$

This pivotal result reveals that the effective phonon-mediated interaction can dynamically manifest as either attractive or repulsive, depending entirely on the sign of the energy denominator.

Attractive Interaction and Isotope Effect

We now restrict our focus to the low-temperature regime, specifically when $T \ll T_D$, where $T_D \sim \omega_D$ is the characteristic Debye temperature of the crystal lattice. In this regime, the electrons lack the thermal energy necessary to scatter far from the Fermi surface. Consequently, the kinetic energy difference between the initial and scattered electron states is constrained to be much smaller than the typical phonon energy (the Debye energy):

$$|\epsilon_{\mathbf{k}+\mathbf{q}} - \epsilon_{\mathbf{k}}| \ll \hbar\omega_D.$$

Under this stringent condition, the denominator in Equation 2.1.2 is overwhelmingly dominated by the phonon energy squared term, $-(\hbar\omega_{\mathbf{q}})^2$, ensuring it naturally becomes negative. The effective interaction therefore simplifies to a constant, weakly attractive potential:

$$E_2' \simeq \frac{-|M_{\mathbf{q}}|^2}{\hbar\omega_D}. \tag{2.1.3}$$

Recall from our fundamental coupling definitions that the squared electron-phonon matrix element, $|M_{\mathbf{q}}|^2$, is inversely proportional to the mass of the lattice ions (M_{ion}). Therefore, the overall strength of the attractive interaction scales directly as:

$$E_2' \propto \frac{1}{M_{\text{ion}}}.$$

This simple inverse proportionality provides a profound microscopic explanation for the experimentally observed **isotope effect** in conventional superconductors, where the critical transition temperature is found to vary predictably with the specific mass of the isotopes that comprise the crystal lattice.

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Remark. *While our derivation treats V_1 as a small perturbation within a simplified cubic lattice, this is not always universally applicable. Because the matrix element $M_{\mathbf{q}}$ scales inversely with the ion mass, the effective electron-phonon interaction can become exceedingly strong in materials composed of very light ions. In such strong-coupling regimes, the perturbative approach breaks down, necessitating more advanced non-perturbative methods (such as Eliashberg theory). Furthermore, real materials often possess complex crystal geometries far beyond a simple cubic structure; these specific geometric details fundamentally alter the phonon spectrum and, consequently, the precise structure of the effective electron-electron interaction.*

The BCS Effective Potential

For the specific purposes of successfully developing the macroscopic BCS theory (Step 1 of the paradigm), we treat this phonon-mediated interaction as a separable, strictly attractive potential. We completely drop its explicit momentum dependence in favor of a constant effective coupling strength, phenomenologically denoted as $-|g_{\text{eff}}|^2$.

Furthermore, while the complete interaction Hamiltonian realistically includes the residual screened Coulomb repulsion (the Thomas-Fermi term), the fundamental assumption of BCS theory is that the attractive phonon-mediated part robustly dominates for electrons near the Fermi surface. Thus, we neglect the residual electrostatic repulsion and approximate the total effective electron-electron potential as purely attractive:

$$V_{e-e,\mathbf{k},\mathbf{k}'} \simeq -|g_{\text{eff}}|^2 + \cancel{V_{\text{Thomas-Fermi}}} \simeq -|g_{\text{eff}}|^2.$$

It is crucial to emphasize that while this attractive energy shift $\Delta E = -|g_{\text{eff}}|^2$ (derived by discarding scattering processes with energy exchanges far beyond the Debye cutoff ω_D) is exceedingly small, it is fundamentally sufficient to destabilize the unperturbed Fermi sea and induce the formation of a bound state between two electrons. Having established the scalar amplitude of this

attractive potential, the immediate next step is to rigorously construct the full operatorial part of the interaction Hamiltonian to be utilized in solving the macroscopic many-body problem.

We are assuming weak interactions, but that is not always the case: the coefficient $M(\mathbf{q})$ is inversely proportional to the mass of the ions in the lattice, and thus the interaction can be very strong in the case of light ions. In this case, we need to use a non-perturbative approach.

One has to pay attention also to the geometry of the lattice, since our treatment was based on a cubic lattice, but in general the lattice can be more complicated. The geometry of the lattice can affect the phonon spectrum and thus the effective interaction between electrons.

We got $\Delta E = -|g_{eff}|^2$ for phonons with $\omega < \omega_D$ (discarding processes going to far away from the Fermi surface). This is a very small energy, but it is enough to create a bound state of two electrons. Now we have to find the operatorial part of the Hamiltonian, which is the part that will be used in the many-body problem.

2.1.4 | Effective Hamiltonian and Two-Body Ground State

Our primary objective is to trace out the phononic degrees of freedom completely and construct an effective electronic Hamiltonian $H^{(2)}$ valid up to second order in perturbation theory:

$$H^{(2)} = H_0 + H_2.$$

Rigorously, this systematic decoupling of the bosonic modes is achieved through a canonical **Schrieffer-Wolff transformation**. However, we can intuitively reconstruct the operatorial form of the effective interaction H_2 (often denoted as H_{int}) directly from the second-order transition matrix elements evaluated previously:

$$\langle \phi_0 | V_1 | \phi_{mn} \rangle \langle \phi_{mn} | V_1 | \phi_0 \rangle.$$

Physically, this second-order process corresponds to two incoming fermions exchanging a virtual phonon of momentum $\mathbf{q}$, as depicted in the following Feynman diagram:

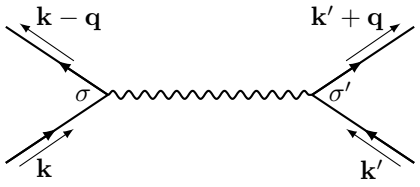


Figure 2.5: Feynman diagram representing the effective electron-electron interaction mediated by the virtual exchange of a phonon. Two incoming fermions with momenta $\mathbf{k}$ and $\mathbf{k}'$ scatter into two outgoing fermions with momenta $\mathbf{k}-\mathbf{q}$ and $\mathbf{k}'+\mathbf{q}$, exchanging a momentum $\mathbf{q}$ and preserving their respective spins σ and σ'.

In the second quantization formalism, the general interacting part of the many-body Hamiltonian, H_{int}, capturing this fundamental scattering process between pairs of fermions (parameterized in terms of center-of-mass and relative momenta), can be modeled as follows:

$$H_{\text{int}} = \frac{1}{2V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} \sum_{\sigma,\sigma'} V(\mathbf{k}',\mathbf{k}) c^\dagger_{\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma} c^\dagger_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma'} c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\sigma'} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\sigma}, \tag{2.1.4}$$

where V is the macroscopic volume of the system, and $c_{\mathbf{k},\sigma}$ and $c^\dagger_{\mathbf{k},\sigma}$ are respectively the fermionic annihilation and creation operators for an electron possessing momentum $\mathbf{k}$ and spin projection σ. The scattering amplitude is governed by the matrix element $V(\mathbf{k}', \mathbf{k})$. We sum over all possible incoming and outgoing relative momenta (as well as over all feasible values of the center-of-mass momentum $\mathbf{q}$), and over the spin indices σ and σ'. The prefactor of $1/2$ is explicitly included to avoid double counting of identical two-body scattering processes.

Translating our specific phonon-exchange diagram into second quantization operators using the constant attractive BCS coupling $-|g_{\text{eff}}|^2$, the effective two-body interaction term H_2 explicitly takes the form:

$$H_2 = -|g_{\text{eff}}|^2 \sum_{\substack{\sigma,\sigma' \\ \mathbf{p},\mathbf{p}' \\ \mathbf{q}\in\text{shell}}} \left[\left(c^\dagger_{\mathbf{p}+\mathbf{q},\sigma} c^\dagger_{\mathbf{p}'-\mathbf{q},\sigma'} \right) \left(c_{\mathbf{p}',\sigma'} c_{\mathbf{p},\sigma} \right) + \left(c^\dagger_{\mathbf{p},\sigma} c^\dagger_{\mathbf{p}',\sigma'} \right) \left(c_{\mathbf{p}-\mathbf{q},\sigma} c_{\mathbf{p}'+\mathbf{q},\sigma'} \right) \right].$$

The summation constraint $\mathbf{q} \in$ shell mathematically enforces the crucial physical requirement that the attractive interaction only operates for small energy exchanges bounded by the Debye frequency ω_D.

Notice how the phonon operators have completely disappeared from the final expression for the interaction Hamiltonian (2.1.4). By casting the Hamiltonian in this form, we have successfully integrated out the bosonic degrees of freedom, reducing the complex coupled electron-phonon problem to a purely fermionic model governed by an effective attractive potential. The phonons are still physically present in the crystal, but they are now treated purely as virtual excitations that mediate the interaction between the electrons.

Our goal is now to solve the many-body problem of interacting fermions governed by $H = H_0 + H_2$, which remains a highly non-trivial task. Thus, we will initially limit our analysis to the simplest possible model that still captures the essential physics of the instability: finding the ground state of a system containing just two electrons interacting via the effective potential $V(\mathbf{k}', \mathbf{k})$ on top of an inert, filled Fermi sea of non-interacting electrons (which blocks states below k_F via Pauli exclusion). This is the celebrated **Cooper problem**, and it serves as the foundational starting point for the full macroscopic BCS theory.

Solving the Two-Electron Ground State: Wavefunction Factorization

To solve the effective Hamiltonian $H = H_0 + H_2$ for the two-electron ground state, we define the general two-particle wavefunction as a function of both the spatial positions and spin projections of the electrons:

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2).$$

Assuming spatial translational invariance across the crystal, we can cleanly separate the center-of-mass (CM) motion from the relative motion. The center-of-mass motion is described by a simple plane wave $e^{i\mathbf{K}_{\text{CM}}\cdot\mathbf{R}_{\text{CM}}}$. Furthermore, because our starting Hamiltonian does not contain terms that explicitly couple spin and momentum (i.e., there is no entanglement between the external positional degrees of freedom and the internal spin degrees of freedom), we can factorize the wavefunction into a spatial part and a spin part:

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2) = e^{i\mathbf{K}_{\text{CM}}\cdot\mathbf{R}_{\text{CM}}} \varphi(\mathbf{r}_1 - \mathbf{r}_2) \phi^{\text{spin}}(\sigma_1, \sigma_2), \tag{2.1.5}$$

where $\varphi(\mathbf{r}_1 - \mathbf{r}_2)$ is the external spatial wavefunction governing the relative motion, and $\phi^{\text{spin}}(\sigma_1, \sigma_2)$ is the internal spin wavefunction. (Note that to maximize the phase space available for scattering

near the Fermi surface and achieve the lowest possible ground state energy, one typically sets the center-of-mass momentum to zero, $\mathbf{K}_{\mathrm{CM}} = 0$).

Remark. *The clean factorization of the wavefunction into spatial and spin components is strictly valid because the Hamiltonian can be separated into operators acting solely on position and spin, respectively:* $H = \hat{\Lambda}_1^{\mathbf{r}} \hat{\Lambda}_2^{\sigma}$. *This separability is a general feature of models that neglect relativistic spin-orbit coupling. A prominent example where local interactions are treated explicitly alongside spin is the* ***Hubbard Model****:*

$$H_{Hubbard} = U n_{i\uparrow} n_{i\downarrow},$$

where $U > 0$ *represents an on-site Coulomb repulsion energy penalty, which is fundamental for studying strongly correlated phenomena such as the* ***Mott Transition****. However, in our current derivation for conventional superconductivity, we utilize the phonon-mediated attractive potential* H_2 *and assume spin-orbit effects are negligible.*

Spin Wavefunction and Fermionic Antisymmetry

For a system composed of two spin-1/2 particles, the composite spin state can be configured either as a symmetric triplet ($S = 1$) or an antisymmetric singlet ($S = 0$):

- **Triplet ($S = 1$, symmetric):** $|\uparrow_1\uparrow_2\rangle, \quad |\downarrow_1\downarrow_2\rangle, \quad \frac{1}{\sqrt{2}}(|\uparrow_1\downarrow_2\rangle + |\downarrow_1\uparrow_2\rangle)$.
- **Singlet ($S = 0$, antisymmetric):** $\frac{1}{\sqrt{2}}(|\uparrow_1\downarrow_2\rangle - |\downarrow_1\uparrow_2\rangle)$.

The triplet states are energetically unfavorable in this simplified model. To maximally benefit from the localized attractive potential mediated by the phonons, the two electrons must have a finite probability of occupying the exact same local spatial site (i.e., $\mathbf{r}_1 = \mathbf{r}_2$). The Pauli exclusion principle strictly forbids two electrons with parallel spins (triplets) from occupying the same spatial position, forcing their spatial wavefunction to vanish at zero separation. Similarly, a simple magnetic interaction model $H \propto \mathbf{S}_1 \cdot \mathbf{S}_2$ naturally selects the singlet as the lowest energy ground state. Therefore, we assume the Cooper pair forms in a spin singlet state:

$$\phi^{\mathrm{spin}}(\sigma_1, \sigma_2) = \frac{1}{\sqrt{2}} \left(|\uparrow_1\downarrow_2\rangle - |\downarrow_1\uparrow_2\rangle\right).$$

Because electrons are fundamental fermions, the total many-body wavefunction must be strictly antisymmetric under the simultaneous exchange of both particles' spatial and spin coordinates:

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2) = -\Psi(\mathbf{r}_2, \sigma_2; \mathbf{r}_1, \sigma_1).$$

Substituting our factorized ansatz (2.1.5) into this condition (and noting that the center-of-mass coordinate $\mathbf{R}_{\mathrm{CM}} = (\mathbf{r}_1 + \mathbf{r}_2)/2$ remains invariant upon swapping particles 1 and 2), we obtain:

$$\varphi(\mathbf{r}_1 - \mathbf{r}_2)\phi^{\mathrm{spin}}(\sigma_1, \sigma_2) = -\varphi(\mathbf{r}_2 - \mathbf{r}_1)\phi^{\mathrm{spin}}(\sigma_2, \sigma_1).$$

Since we established that the spin component is a singlet, it is explicitly antisymmetric under spin exchange:

$$\phi^{\mathrm{spin}}(\sigma_2, \sigma_1) = -\phi^{\mathrm{spin}}(\sigma_1, \sigma_2).$$

The negative signs cancel out on the right-hand side, which imposes a strict parity symmetry condition on the spatial part of the wavefunction:

$$\varphi(\mathbf{r}_1 - \mathbf{r}_2) = +\varphi(\mathbf{r}_2 - \mathbf{r}_1).$$

Defining the relative distance vector as $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, this simplifies to:

$$\varphi(\mathbf{r}) = +\varphi(-\mathbf{r}).$$

We conclude that for a spin-singlet pair, the relative spatial wavefunction $\varphi(\mathbf{r})$ must be an **even function** of the relative separation (corresponding to s-wave, d-wave, etc., orbital symmetries).

Fourier Expansion and the Eigenvalue Equation

Our goal is now to determine the explicit functional form of $\varphi(\mathbf{r})$. Because the problem depends solely on the relative distance, it is vastly simpler to treat in momentum space. We expand the relative spatial wavefunction into its Fourier modes $\varphi_{\mathbf{k}}$:

$$\varphi(\mathbf{r}_1 - \mathbf{r}_2) = \sum_{\mathbf{k}} \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}.$$

Applying the spatial even-parity symmetry condition $\varphi(\mathbf{r}_1 - \mathbf{r}_2) = +\varphi(\mathbf{r}_2 - \mathbf{r}_1)$, we can manipulate the expansion by relabeling the dummy summation index $\mathbf{k} \to -\mathbf{k}$:

$$\sum_{\mathbf{k}} \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} = \varphi(\mathbf{r}_1-\mathbf{r}_2) = +\varphi(\mathbf{r}_2-\mathbf{r}_1) = \sum_{\mathbf{k}} \varphi_{\mathbf{k}} e^{-i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} = \sum_{-\mathbf{k}} \varphi_{-\mathbf{k}} e^{+i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} = \sum_{\mathbf{k}} \varphi_{-\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)}.$$

Comparing the first and last expressions, since the sum over all $\mathbf{k}$ and $-\mathbf{k}$ spans the exact same momentum space, the Fourier coefficients themselves must be symmetric:

$$\varphi_{\mathbf{k}} = \varphi_{-\mathbf{k}}.$$

We now wish to solve the time-independent Schrödinger eigenvalue equation for the two-body ground state energy E in the center-of-mass frame ($\mathbf{K}_{\mathrm{CM}} = 0$):

$$E\Psi = H\Psi = (H_0 + H_{\mathrm{int}})\Psi, \quad \text{with} \quad \Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2) = \sum_{\mathbf{k}} \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \phi^{\mathrm{spin}}_{S=0}(\sigma_1, \sigma_2).$$

To find the closed equation governing the coefficients $\varphi_{\mathbf{k}}$, we transition fully to the second quantization representation and project the Schrödinger equation onto a specific basis state $\langle\Psi_{\mathbf{k}}|$:

$$\langle\Psi_{\mathbf{k}}|E|\Psi\rangle = \langle\Psi_{\mathbf{k}}|H_0 + H_{\mathrm{int}}|\Psi\rangle,$$

where the total two-particle state is expanded as $|\Psi\rangle = \sum'_{\mathbf{k}'} \varphi_{\mathbf{k}'}|\Psi_{\mathbf{k}'}\rangle$ (the prime on the summation indicates that the momenta are restricted to the narrow attractive energy shell above the filled Fermi sea), and the normalized basis state for a spin-singlet pair with opposite momenta $(\mathbf{k}, -\mathbf{k})$ is defined in second quantization as:

$$|\Psi_{\mathbf{k}}\rangle = \frac{1}{\sqrt{2}} \left(c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} - c^\dagger_{-\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\downarrow} \right) |0\rangle. \tag{2.1.6}$$

We now split the evaluation of $H|\Psi\rangle$ into two parts: the free non-interacting kinetic part and the interaction part. The action of the kinetic Hamiltonian H_0 is straightforward:

$$H_0|\Psi\rangle = \left(\frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} \right) \left[\sum_{\mathbf{k}}' \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \phi^{\mathrm{spin}}_{S=0} \right] = (\epsilon_{\mathbf{k}} + \epsilon_{-\mathbf{k}}) \left[\sum_{\mathbf{k}}' \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \phi^{\mathrm{spin}}_{S=0} \right] = 2 \sum_{\mathbf{k}}' \epsilon_{\mathbf{k}} \varphi_{\mathbf{k}} |\Psi_{\mathbf{k}}\rangle,$$

where we used the isotropic dispersion symmetry $\epsilon_{\mathbf{k}} = \epsilon_{-\mathbf{k}} = \hbar^2 k^2/(2m)$.

Next, we evaluate the action of the attractive interaction term H_{int} (or H_2) on the state $|\Psi\rangle$:

$$H_{\text{int}}|\Psi\rangle = -|g_{\text{eff}}|^2 \sum_{\sigma_1,\sigma_2} \sum_{\mathbf{k}_1,\mathbf{k}_2} {\sum_{\mathbf{q}}}' c^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma_1} c^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma_2} c_{\mathbf{k}_1,\sigma_1} c_{\mathbf{k}_2,\sigma_2} \left({\sum_{\mathbf{k}}}' \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \right) \left(c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} - c^\dagger_{-\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\downarrow} \right) |0\rangle.$$

Contracting the annihilation operators $c_{\mathbf{k}_1,\sigma_1} c_{\mathbf{k}_2,\sigma_2}$ with the creation operators of the singlet state resolves the internal summations over $\mathbf{k}_1, \mathbf{k}_2$ and spins (specifically selecting the anti-parallel configurations such as $\sigma_1 = \uparrow, \sigma_2 = \downarrow$) via Kronecker deltas. After performing this operator algebra, the scattered state simplifies to:

$$H_{\text{int}}|\Psi\rangle = -|g_{\text{eff}}|^2 {\sum_{\mathbf{q}}}' {\sum_{\mathbf{k}}}' \varphi_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \left(c^\dagger_{\mathbf{k}+\mathbf{q},\uparrow} c^\dagger_{-\mathbf{k}-\mathbf{q},\downarrow} - c^\dagger_{-\mathbf{k}+\mathbf{q},\uparrow} c^\dagger_{\mathbf{k}-\mathbf{q},\downarrow} \right) |0\rangle.$$

Now we compute the complete matrix elements by projecting onto the bra $\langle\Psi_{\mathbf{k}}|$. For the left-hand side of the Schrödinger equation, the orthogonality of the basis states immediately yields:

$$\langle\Psi_{\mathbf{k}}|E|\Psi\rangle = E\varphi_{\mathbf{k}}.$$

For the right-hand side, we bracket the kinetic and interaction results with $\langle\Psi_{\mathbf{k}}|$:

$$\begin{aligned}
\langle\Psi_{\mathbf{k}}|H_0 + H_{\text{int}}|\Psi\rangle &= 2\epsilon_{\mathbf{k}}\varphi_{\mathbf{k}} + \langle\Psi_{\mathbf{k}}|H_{\text{int}}|\Psi\rangle \\
&= 2\epsilon_{\mathbf{k}}\varphi_{\mathbf{k}} + \langle 0| \left(c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} - c_{\mathbf{k},\downarrow} c_{-\mathbf{k},\uparrow} \right) {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'} e^{-i\mathbf{k}'\cdot(\mathbf{r}_1-\mathbf{r}_2)} \\
&\quad \times {\sum_{\mathbf{q}}}' (-|g_{\text{eff}}|^2) \left(c^\dagger_{\mathbf{k}'+\mathbf{q},\uparrow} c^\dagger_{-\mathbf{k}'-\mathbf{q},\downarrow} - c^\dagger_{-\mathbf{k}'+\mathbf{q},\uparrow} c^\dagger_{\mathbf{k}'-\mathbf{q},\downarrow} \right) |0\rangle.
\end{aligned}$$

To obtain a non-zero vacuum overlap in the interaction bracket, the momenta of the fermions created by the ket must precisely match those destroyed by the bra. This conservation requirement establishes strict algebraic constraints on the allowed momentum transfer vectors $\mathbf{q}$ (for instance, matching the indices in the cross-terms imposes simultaneous relations such as $\mathbf{q} = -\mathbf{k} - \mathbf{k}'$ and $\mathbf{q} = +\mathbf{k} - \mathbf{k}'$, or simply forces a direct transition from pair state $(\mathbf{k}', -\mathbf{k}')$ to pair state $(\mathbf{k}, -\mathbf{k})$ via the momentum shift $\mathbf{q} = \mathbf{k} - \mathbf{k}'$). Consequently, the sum over $\mathbf{q}$ collapses, leaving only the single scattering channel that connects $\mathbf{k}'$ to $\mathbf{k}$. Evaluating the remaining fermionic contractions reduces the total matrix element to a remarkably simple form:

$$\langle\Psi_{\mathbf{k}}|H|\Psi\rangle = 2\epsilon_{\mathbf{k}}\varphi_{\mathbf{k}} - |g_{\text{eff}}|^2 {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'}.$$

The Consistency Condition and the Cooper Bound State

Equating the left and right sides of our projected Schrödinger equation, we obtain the explicit eigenvalue equation for the Fourier coefficients $\varphi_{\mathbf{k}}$:

$$E\varphi_{\mathbf{k}} = 2\epsilon_{\mathbf{k}}\varphi_{\mathbf{k}} - |g_{\text{eff}}|^2 {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'}.$$

Rearranging the terms to isolate $\varphi_{\mathbf{k}}$ yields:

$$(E - 2\epsilon_{\mathbf{k}})\varphi_{\mathbf{k}} = -|g_{\text{eff}}|^2 {\sum_{\mathbf{k}'}}' \varphi_{\mathbf{k}'}. \tag{2.1.7}$$

Notice that the summation over $\mathbf{k}'$ on the right-hand side is completely independent of the free index $\mathbf{k}$; it is simply a constant scalar, which we define as C:

$$C = \sum_{\mathbf{k}'}{}' \varphi_{\mathbf{k}'}.$$

Substituting C back into the rearranged eigenvalue equation (2.1.7), we can explicitly solve for the functional form of the Fourier components $\varphi_{\mathbf{k}}$:

$$\varphi_{\mathbf{k}} = \frac{-|g_{\text{eff}}|^2 C}{E - 2\epsilon_{\mathbf{k}}}.$$

We now insert this expression for $\varphi_{\mathbf{k}}$ back into the definition of C by summing both sides over the allowed states $\mathbf{k}$ to obtain a closed self-consistency equation:

$$C = \sum_{\mathbf{k}}{}' \varphi_{\mathbf{k}} = \sum_{\mathbf{k}}{}' \left(\frac{-|g_{\text{eff}}|^2 C}{E - 2\epsilon_{\mathbf{k}}} \right) = -|g_{\text{eff}}|^2 C \sum_{\mathbf{k}}{}' \frac{1}{E - 2\epsilon_{\mathbf{k}}}.$$

Assuming $C \neq 0$ (which physically corresponds to finding a non-trivial bound state solution), we divide both sides by C to arrive at the fundamental Cooper consistency condition:

$$1 = -|g_{\text{eff}}|^2 \sum_{\mathbf{k}}{}' \frac{1}{E - 2\epsilon_{\mathbf{k}}}. \tag{2.1.8}$$

To evaluate this self-consistency sum analytically, we transition to the continuum limit, replacing the discrete sum over momenta $\mathbf{k}$ with an integral over the single-particle energy ϵ weighted by the density of states $g(\epsilon)$:

$$\sum_{\mathbf{k}}{}' \cdots \xrightarrow{\text{cont. lim.}} \int_0^{\hbar\omega_D} d\epsilon \, g(\epsilon) \cdots$$

Here, $g(\epsilon)$ represents the number of available electronic states per unit energy window $[\epsilon, \epsilon + d\epsilon]$. We Measure the single-particle energies relative to the Fermi level (setting $\epsilon_F = 0$) and integrate strictly from 0 (the Fermi surface, since states below it are Pauli-blocked by the inert Fermi sea) up to the Debye cutoff energy $\hbar\omega_D$. This upper limit is enforced because, beyond the Debye frequency, there are no phonon modes available in the crystal lattice to mediate the attractive scattering. Applying the continuum limit to Eq. (2.1.8) gives:

$$1 = -|g_{\text{eff}}|^2 \int_0^{\hbar\omega_D} d\epsilon \, \frac{g(\epsilon)}{E - 2\epsilon}.$$

Because the relevant phonon integration energy scale ($\hbar\omega_D$) is typically orders of magnitude smaller than the electronic Fermi energy (E_F), the density of states varies negligibly over this narrow shell. We can therefore safely approximate $g(\epsilon)$ as a constant evaluated precisely at the Fermi surface, $g(\epsilon) \simeq g(E_F)$:

$$1 \simeq -|g_{\text{eff}}|^2 g(E_F) \int_0^{\hbar\omega_D} d\epsilon \, \frac{1}{E - 2\epsilon}.$$

Evaluating this elementary logarithmic integral yields our implicit algebraic equation for the two-body bound state energy E[1]:

$$1 = -|g_{\text{eff}}|^2 g(E_F) \left(-\frac{1}{2} \right) \ln\left(\frac{E - 2\hbar\omega_D}{E} \right).$$

[1]Note on the integral prefactor: Mathematically, evaluating $\int_0^{\hbar\omega_D} \frac{d\epsilon}{E-2\epsilon}$ strictly yields a factor of $-\frac{1}{2}\ln\left(\frac{E-2\hbar\omega_D}{E}\right)$, whereas lecture board notes occasionally reflect a coefficient of -2 (as in $1 \simeq -2|g_{\text{eff}}|^2 g(E_F)\ln(\ldots)$). During the lecture, this discrepancy was addressed verbally as an arbitrary algebraic scaling factor (such as spin-degeneracy factors or density-of-states conventions) that is ultimately absorbed directly into the definition of the dimensionless interaction parameter λ. Below, we show the exact $1/2$ integration alongside the lecture's λ parameterization.

Following the standard parameterization, we define the dimensionless electron-phonon coupling constant λ (absorbing the numerical prefactors into its definition, e.g., $\frac{1}{\lambda} = -\frac{1}{2|g_{\text{eff}}|^2 g(E_F)}$ or $\lambda \equiv |g_{\text{eff}}|^2 g(E_F)$ depending on the chosen convention):

$$-\frac{1}{\lambda} \simeq \ln\left(\frac{E - 2\hbar\omega_D}{E}\right) \implies e^{-\frac{1}{\lambda}} = 1 - \frac{2\hbar\omega_D}{E},$$

(or equivalently, keeping the negative sign explicit in the exponent definition as written in the notes, $1 - e^{\frac{1}{\lambda}} = \frac{2\hbar\omega_D}{E}$). Solving explicitly for the energy E, we obtain:

$$E = \frac{2\hbar\omega_D}{1 - e^{\frac{1}{\lambda}}} = 2\hbar\omega_D \underbrace{\left(\frac{1}{1 - e^{\frac{1}{|g_{\text{eff}}|^2 g(E_F)}}}\right)}_{<0} < 0. \tag{2.1.9}$$

(Note: In the lecture notes, the algebraic rearrangement is also briefly sketched in the simplified product form $E = 2\hbar\omega_D(1 - e^{1/\lambda}) < 0$ to immediately highlight the negative sign of the quantity in parentheses).

Physical Analysis: Weak-Coupling Limit and Temperature Scales

Let us carefully analyze the physical consequences of Eq. (2.1.9). The dimensionless parameter $\lambda = |g_{\text{eff}}|^2 g(E_F)$ acts as a strictly positive constant measuring the strength of the effective attraction. Since $\lambda > 0$, the exponential term $e^{1/\lambda}$ is strictly greater than 1, which guarantees that the denominator $(1 - e^{1/\lambda})$ is definitively negative. Because the Debye energy $\hbar\omega_D$ is positive, this mathematically guarantees that $E < 0$ (relative to the Fermi surface $2\epsilon_F = 0$). This is a profound theoretical result: we have rigorously proven that the two-fermion system lowers its energy below the Fermi sea to form a stable **bound state**, the **Cooper pair**.

In the physically realistic weak-coupling limit, where the coupling $\lambda \ll 1$ and the Debye energy is much larger than the resulting binding energy ($\hbar\omega_D \gg |E|$), we can expand the logarithmic relation for $|E| = -E$:

$$\frac{1}{\lambda} \simeq \ln\left(1 - \frac{2\hbar\omega_D}{E}\right) = \ln\left(1 + \frac{2\hbar\omega_D}{|E|}\right) \simeq \ln\left(\frac{2\hbar\omega_D}{|E|}\right).$$

Inverting this asymptotic relation yields the celebrated Cooper formula for the absolute magnitude of the pair binding energy:

$$|E| = \frac{2\hbar\omega_D}{e^{\frac{1}{\lambda}}} = 2\hbar\omega_D e^{-\frac{1}{\lambda}} = 2\hbar\omega_D \exp\left(-\frac{1}{|g_{\text{eff}}|^2 g(E_F)}\right). \tag{2.1.10}$$

(Depending on whether the factor of 1/2 from the integral is kept explicit rather than absorbed into λ, this is equivalently written as $E = -2\hbar\omega_D e^{-2/\lambda}$).

This final analytical expression carries two critical physical messages:

1. **Cooper Instability** $E < 0$: We have successfully formed a stable two-electron bound pair. Crucially, unlike in three-dimensional vacuum where an attractive potential must exceed a finite threshold depth to support a bound state, the presence of the Pauli-blocking Fermi sea reduces the effective dimensionality of the density of states to become constant at the Fermi surface. Consequently, *even an arbitrarily weak attractive interaction* ($\lambda \to 0^+$) *produces a bound state*!

2. **Non-Perturbative Singularity** $E \propto e^{-\frac{1}{|g_{\text{eff}}|^2 g(E_F)}}$: The binding energy depends exponentially on the *inverse* of the coupling strength. Mathematically, the function $e^{-1/\lambda}$ possesses an essential singularity at $\lambda = 0$; all of its Taylor series derivatives vanish at zero coupling. From a theoretical standpoint, one might initially be unhappy with this feature because it proves that the superconducting bound state cannot be reached at any finite order in standard perturbation theory starting from the normal metal. However, it also naturally explains why the superconducting energy scale is so vastly smaller than the bare electronic scales.

Let us conclude by reasoning about the characteristic temperature and energy scales implied by this hierarchy. The exponential dependence implies that the energy gap created by these bound states is extremely small, meaning the pairs are fragile and can only resist thermal destruction up to very low temperatures. We can roughly estimate the typical order-of-magnitude scales for a conventional metal as follows:

- **Fermi Temperature ($T_F \sim 10^4$ K):** This sets the bare kinetic energy scale of the rapid electron motion in the metal.
- **Effective Phonon Scale ($g_{\text{eff}} \sim \frac{10^4}{10^2} \sim 10^2$ K):** Because the effective attraction arises at second order in electron-phonon perturbation theory (and is bounded by the Debye temperature $T_D = \hbar\omega_D/k_B$), the characteristic interaction energy scale drops by roughly two orders of magnitude compared to T_F, landing around ~ 100 K.
- **Critical Temperature ($T_c \sim E/k_B \sim \frac{10^2}{10^2} \sim 1$ K):** Finally, the non-perturbative exponential suppression $e^{-1/\lambda}$ reduces the energy scale by another one to two orders of magnitude.

Thus, this simple two-body Cooper model provides a completely natural microscopic explanation for why the critical temperatures T_c observed in conventional metallic superconductors are on the order of $\sim$ 1–10 Kelvin, four orders of magnitude smaller than the Fermi temperature. (Note that while there may in principle exist other exotic bound states or excitations at higher energies, our low-energy effective perturbative expansion around the Fermi surface does not allow us to access them).

2.2 | From Microscopic to Mesoscopic Descriptions

We have previously established that starting from the complete electron-phonon Hamiltonian and applying the weak-coupling approximation allows us to systematically integrate out the phononic degrees of freedom. This procedure yields an effective Hamiltonian governed purely by an attractive electron-electron interaction, $V_{e^-e^-} \propto -|g_{\text{eff}}|^2$. By analyzing this restricted two-body problem within a narrow momentum annulus surrounding the Fermi surface, we demonstrated that this effective attraction supports a stable bound state ($E_2 < 0$). This is a profoundly non-trivial theoretical consequence, given the strong inherent Coulomb repulsion between bare electrons in a vacuum.

> **Remark.** *This intellectual paradigm of integrating out collective bosonic modes to generate effective interactions between fundamental fermions is not exclusive to condensed matter physics. An elegant analogy exists in the realm of Cavity Quantum Electrodynamics (Cavity QED), a field significantly advanced by Serge Haroche (awarded the 2012 Nobel Prize in Physics). In both disciplines, the core physical phenomena emerge from the intricate interplay between microscopic elements (such as electrons mapping to micro degrees of freedom) and collective bosonic modes (such as phonons or cavity photons mapping to collective degrees of freedom).*

Moving forward, our current objective is to formally bridge the microscopic and mesoscopic descriptions of these interacting gases. At the two-body level, we will employ the quantum **T-matrix** to connect microscopic parameters (specifically the universal atomic interaction defined by the scattering length a) to the macroscopic properties of the effective interaction potential. Subsequently, we will tackle the full many-body grand-canonical problem by deploying Mean-Field approximations and the Bogoliubov transformation, methodologies mathematically homologous to those utilized for studying weakly interacting Bose gases.

2.2.1 | The Effective Attractive Potential

To capture the essential physics of the Fröhlich mechanism, specifically the crucial detail that the phonon-mediated attraction only overcomes Coulomb repulsion for electrons energetically adjacent to the Fermi surface, the effective attractive potential is explicitly modeled by a piecewise function:

$$V(\mathbf{k}', \mathbf{k}) = \begin{cases} -g_{\text{e,ph}}^2 & \text{if } |\epsilon_k - \mu| < \hbar\omega_D \text{ and } |\epsilon_{k'} - \mu| < \hbar\omega_D, \\ 0 & \text{elsewhere,} \end{cases} \tag{2.2.1}$$

where $g_{\text{e,ph}}$ represents the effective coupling constant between the electrons and phonons, μ is the chemical potential of the electron gas (recalling that in the low-temperature limit $T \to 0$, the chemical potential intimately approaches the Fermi energy, $\mu(T) \xrightarrow{T\to 0} E_F$), ϵ_k is the non-interacting kinetic energy of the fermion, and ω_D is the characteristic Debye frequency of the ionic lattice.

By analyzing the explicit energy bounds in the expression for the potential (2.2.1), we clearly see a profound physical restriction: only those electrons located within a remarkably narrow energy shell surrounding the Fermi sphere are permitted to interact attractively with each other.

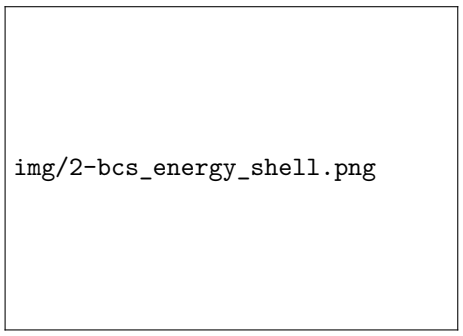


Figure 2.6: The model of the potential $V(\mathbf{k}', \mathbf{k})$ for superconductors is given by (2.2.1) and it involves just the fermions with energies ϵ_k and $\epsilon_{k'}$ residing strictly within a tiny energy shell of width $2\hbar\omega_D$ centered around the Fermi energy.

To express this restricted interaction more compactly, it is highly advantageous to introduce a dimensionless cut-off function $w_k = \Theta(\hbar\omega_D - |\epsilon_k - \mu|)$, where Θ denotes the standard Heaviside step function. Utilizing this notation, the piecewise potential (2.2.1) can be elegantly rewritten in a strictly factorized form:

$$V(\mathbf{k}', \mathbf{k}) = -g_{\mathrm{e,ph}}^2 w_k w_{k'}. \tag{2.2.2}$$

This mathematically represents the localized attractive potential acting exclusively between fermions scattered around the Fermi energy in a superconductor. Note that this specific theoretical form of the potential is entirely *separable*; the dependence on the incoming momentum $\mathbf{k}$ and the outgoing momentum $\mathbf{k}'$ factors into independent terms, a property that will dramatically simplify the subsequent many-body calculations.

The theoretical framework developed here for electrons in solid-state crystal lattices can also be seamlessly applied to model ultracold Fermi gases. In these synthetic quantum systems, the microscopic interactions between two distinct particles occurring at different momenta are analogously modeled using a separable potential:

$$V_{\mathbf{k}',\mathbf{k}} = v_0 w_{\mathbf{k}} w_{\mathbf{k}'}.$$

This structure is mathematically identical to the effective electron-electron potential derived previously. However, for ultracold atomic Fermi gases, the interaction is dictated directly by nature, and the fundamental physical parameter used to describe it is the scattering length a. Typically, for these highly tunable atomic interactions, this characteristic value is on the order of $a \simeq 100a_0$, where a_0 represents the Bohr radius.

2.2.2 | Ultracold Fermi Gases

As previously established, the macroscopic theoretical framework developed for BCS superconductivity can be seamlessly adapted to study synthetic quantum systems of ultracold Fermi atoms. In these highly controlled atomic systems, the effective interparticle interaction possesses a very short range compared to the average interparticle distance. Because of this extreme separation of scales, it is highly convenient to model the interaction using a separable potential in momentum space:

$$V(\mathbf{k}', \mathbf{k}) = v_0 w_k w_{k'},$$

where $w_k = \Theta(k_0 - k)$ utilizes the Heaviside step function to introduce a strict ultraviolet momentum cutoff k_0, and v_0 represents the bare interaction strength. Formally, this separable potential is structurally homologous to the one utilized for modeling effective electron-phonon interactions in conventional superconductors.

Definition of Scattering Length and the T-Matrix

For a genuinely short-range contact potential in position space, the corresponding cutoff in momentum space must theoretically be taken to infinity, meaning $k_0 \to \infty$. We thus need to understand how to appropriately treat this divergent limit mathematically. The core physical principle is that as the cutoff k_0 goes to infinity, the bare interaction parameter v_0 must simultaneously change (renormalize) with k_0 in a precise manner in order to keep the physically observable, macroscopic scattering length a strictly constant.

To rigorously connect the mesoscopic properties of the effective potential to the universal microscopic property given by the scattering length, we introduce the **T-matrix**, denoted as $T(\mathbf{k}', \mathbf{k}, z)$, where $z \in \mathbb{C}$ is a complex energy parameter. This T-matrix satisfies the fundamental Lippmann-Schwinger integral equation:

$$T(\mathbf{k}', \mathbf{k}, z) = V(\mathbf{k}', \mathbf{k}) + \int \frac{d^3k''}{(2\pi)^3} \frac{V(\mathbf{k}', \mathbf{k}'')}{z - \frac{(k'')^2}{2m_r}} T(\mathbf{k}'', \mathbf{k}, z),$$

where m_r is the reduced mass of the colliding two-body system. For a gas of identical fermions of mass m, the reduced mass is simply $m_r = m/2$.

The formal relationship linking the T-matrix to the physical scattering length a is established by evaluating the T-matrix at zero incoming and outgoing momenta, and precisely at zero energy ($z = 0$):

$$T(\mathbf{0}, \mathbf{0}, 0) = \frac{2\pi\hbar^2}{m_r} a. \tag{2.2.3}$$

By invoking this definition, we are directly translating the physical requirement of a fixed scattering length a into a precise mathematical condition on the bare interaction strength v_0 as a function of the ultraviolet cutoff k_0. This equation elegantly bridges the two scales: $T(\mathbf{0}, \mathbf{0}, 0)$ encapsulates the mesoscopic properties depending on the specific potential approximations, while a represents the universal microscopic property of the atoms. Furthermore, the scattering length a is directly related to the standard physical scattering amplitude function $f(\mathbf{k}', \mathbf{k})$ evaluated at zero momentum:

$$a = -f(\mathbf{0}, \mathbf{0}) = \frac{m_r}{2\pi\hbar^2} T(\mathbf{0}, \mathbf{0}, 0).$$

Renormalization of the Bare Coupling Strength

We now wish to solve the integral differential equation for the T-matrix using our specific separable potential $V(\mathbf{k}, \mathbf{k}') = v_0 w_k w_{k'}$. Substituting this into the Lippmann-Schwinger equation yields:

$$T(\mathbf{k}, \mathbf{k}', z) = v_0 w_{\mathbf{k}} w_{\mathbf{k}'} + \int \frac{d^3k''}{(2\pi)^3} \frac{v_0 w_{\mathbf{k}} w_{\mathbf{k}''}}{z - \frac{(k'')^2}{2m_r}} T(\mathbf{k}'', \mathbf{k}', z).$$

To proceed, we make a highly reasonable physical ansatz for the functional form of the T-matrix. Since the potential is non-zero exclusively when the momenta reside inside the energy shell (the momentum annulus), it is mathematically sound to assume that the T-matrix shares this exact property, effectively separating the momentum dependence from the complex energy parameter z:

$$T(\mathbf{k}, \mathbf{k}', z) = w_{\mathbf{k}} w_{\mathbf{k}'} T(z).$$

Let us rigorously verify why this factorization is justified:

- If the outgoing momentum $\mathbf{k}'$ is **outside** the annulus, the cutoff function vanishes by definition ($w_{\mathbf{k}'} = 0$). Consequently, both the bare potential term and the integral term vanish identically, trivially yielding $T = 0$.
- If the incoming momentum $\mathbf{k}$ is **outside** the annulus, the first term $v_0 w_{\mathbf{k}} w_{\mathbf{k}'}$ trivially vanishes. The integral term, however, is not overtly zero at first glance. Nonetheless, we know from fundamental scattering theory that the T-matrix must be strictly symmetric under the exchange of incoming and outgoing momenta: $T(\mathbf{k}, \mathbf{k}') = T(\mathbf{k}', \mathbf{k})$. By invoking this time-reversal symmetry, we conclude that the integral term must also evaluate to 0.

Plugging our ansatz back into the integral equation and factoring out the momentum dependence gives the implicit scalar relation:

$$w_{\mathbf{k}} w_{\mathbf{k}'} T(z) = v_0 w_{\mathbf{k}} w_{\mathbf{k}'} + \int \frac{d^3k''}{(2\pi)^3} \frac{v_0 w_{\mathbf{k}} w_{\mathbf{k}''}}{z - \frac{(k'')^2}{2m_r}} w_{\mathbf{k}''} w_{\mathbf{k}'} T(z).$$

Because the term $w_{\mathbf{k}} w_{\mathbf{k}'}$ appears as a common factor in every term, we can cleanly divide it out:

$$T(z) = v_0 + v_0 \int \frac{d^3k''}{(2\pi)^3} \frac{(w_{\mathbf{k}''})^2 T(z)}{z - \frac{(k'')^2}{2m_r}}.$$

Gathering the terms containing $T(z)$ on the left-hand side, we can solve for it explicitly:

$$T(z) = \frac{v_0}{1 - v_0 \int \frac{d^3k''}{(2\pi)^3} \frac{(w_{\mathbf{k}''})^2}{z - \frac{(k'')^2}{2m_r}}}.$$

We evaluate this expression at zero energy, $z = 0$, to connect with our macroscopic scattering length boundary condition. Substituting $z = 0$, the minus sign in the denominator's integral cancels with the subtraction, allowing us to bring $2m_r$ to the numerator:

$$T(\mathbf{0}, \mathbf{0}, 0) = \frac{v_0}{1 + 2m_r v_0 \int \frac{d^3k''}{(2\pi)^3} \frac{(w_{\mathbf{k}''})^2}{(k'')^2}}.$$

Equating this result to the previously derived kinematic definition $T(\mathbf{0}, \mathbf{0}, 0) = \frac{2\pi\hbar^2}{m_r} a$, and substituting $m_r = m/2$ specifically for identical fermions, we obtain:

$$\frac{4\pi\hbar^2}{m} a = \frac{v_0}{1 + m v_0 \int \frac{d^3k}{(2\pi)^3} \frac{w_k^2}{k^2}}.$$

Remark. *Because we utilized (2.2.3) to define the scattering length, this relation strictly holds in the zero-energy limit, where the scattering amplitude becomes entirely isotropic and independent of the incoming and outgoing momenta. In this ultracold limit, the scattering length a operates as a universal quantity characterizing the low-energy physical behavior of the system, independent of the microscopic complexities of the true interaction potential.*

By isolating the inverse bare coupling strength, we can recast this equation into a highly illuminating form:

$$\frac{1}{v_0} = \frac{m}{4\pi\hbar^2 a} - \int \frac{d^3k}{(2\pi)^3} \frac{w_k^2}{2\epsilon_k}, \tag{2.2.4}$$

where we have explicitly used the free-particle kinetic energy dispersion $\epsilon_k = \hbar^2 k^2 / 2m$. Equation (2.2.4) governs the renormalization flow: it dictates exactly how the bare interaction strength v_0 must scale with the ultraviolet momentum cutoff k_0 (embedded within w_k) to maintain a fixed, physical scattering length a.

To solve this explicitly, we introduce our specific hard-sphere ultraviolet cutoff, $w_k = \Theta(k_0 - k)$. Evaluating the integral in spherical coordinates yields:

$$\int_0^{k_0} \frac{4\pi k^2 dk}{(2\pi)^3} \frac{m}{\hbar^2 k^2} = \frac{m k_0}{2\pi^2 \hbar^2}.$$

Substituting this back into our relation and isolating v_0 gives:

$$v_0 = \frac{2\pi^2 \hbar^2}{m} \frac{1}{\frac{\pi}{2a} - k_0}.$$

We must be mathematically careful when analyzing the true contact limit where the cutoff goes to infinity ($k_0 \to \infty$). If we take this limit naively, we simply get $v_0 = 0$, meaning all physical dependence on the scattering length a is disastrously lost. Instead, we take the limit $k_0 \gg 1$ under the assumption that the dimensionless product $k_0 a$ remains finite. By asymptotically expanding the equation in this limit, we find the effective regularized potential:

$$v_0 \simeq -\frac{2\pi^2 \hbar^2}{m k_0} \left(1 + \frac{\pi}{2 k_0 a}\right).$$

This asymptotic expansion clearly demonstrates that in the limiting case of a perfectly localized contact potential, the bare coupling strength must approach zero from below ($v_0 \to 0^-$) as $k_0 \to \infty$. Crucially, the physical scattering length a appears exclusively in the subleading term, successfully encoding the actual collision physics of the interacting region.

Ultimately, having a momentum cutoff in our interaction is not a theoretical hindrance but a physical reality: the ultraviolet cutoff k_0 is physically meaningful. For electrons in solid-state lattices, it is naturally determined by the inverse lattice spacing; for interacting atoms in a Fermi gas, it is intrinsically set by the strong short-distance repulsion (such as the hard core of a Lennard-Jones potential). Because the physical scattering length a is a well-defined, experimentally measurable quantity, we can robustly model the many-body physics without requiring precise knowledge of the interaction potential's extreme high-momentum microscopic details.

2.3 | Grand Canonical Hamiltonian and Ground State

To study the many-body problem, the gold standard approach is to extract thermodynamic information from the partition function $\mathcal{Z}$, defined as:

$$\mathcal{Z} = \mathrm{Tr}\left\{e^{-\beta \hat{K}}\right\}.$$

Notice that the partition function itself is not a direct physical observable in practice, as measuring it would require scaling exponentially with the system size. However, it is fundamentally crucial because its mathematical structure gives us access to measurable macroscopic observables.

To accurately describe a macroscopic piece of matter, such as a metal where the exact number of interacting electrons can dynamically fluctuate, we study the problem in the grand canonical ensemble. We redefine our working Hamiltonian by introducing the chemical potential μ:

$$\hat{K} = \hat{H}_{\mathrm{INT}} - \mu \hat{N},$$

where $\hat{N} = \sum_{\mathbf{k},\sigma} c^{\dagger}_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma}$ is the total fermionic number operator. This allows us to conveniently reabsorb the chemical potential directly into the definition of the single-particle energy dispersion:

$$\xi_k = \epsilon_k - \mu.$$

Using the standard second-quantized expression for the interaction, the full grand canonical Hamiltonian, which explicitly conserves the total momentum $\mathbf{q}$, takes the form:

$$\hat{K} = \sum_{\mathbf{k},\sigma} \xi_k c^{\dagger}_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma} + \frac{1}{2V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} \sum_{\sigma,\sigma'} V(\mathbf{k}',\mathbf{k}) c^{\dagger}_{\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma} c^{\dagger}_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\sigma'} c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\sigma'} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\sigma}.$$

For notational convenience, we will often denote the non-interacting single-particle kinetic part of $\hat{K}$ simply as $\hat{T}$.

This Hamiltonian exhibits important conservation laws associated with underlying $U(1)$ symmetries:

- **Charge Symmetry ($[\hat{K}, \hat{N}] = 0$):** The total number of particles in the system is strictly conserved. This holds mathematically because every term in the Hamiltonian preserves the particle number: the kinetic term destroys and creates exactly one particle, while the interaction term destroys two and creates two.
- **Spin Symmetry ($[\hat{K}, \hat{N}_{\uparrow} - \hat{N}_{\downarrow}] = 0$):** Also referred to as asymmetric charge symmetry, this implies that the difference between the number of spin-up and spin-down fermions is a conserved quantity. Any generic interaction process (for example, destroying one spin-up and one spin-down fermion to create a new pair of one spin-up and one spin-down) leaves this overall difference invariant.

To simplify the many-body problem, we make a crucial physical approximation: we neglect the interactions between fermions with identical spins (up-up and down-down). Due to the Pauli exclusion principle, identical fermions cannot occupy the exact same spatial position, meaning the short-distance effective interaction is overwhelmingly dominated by pairs of fermions with opposite spins ($\uparrow\downarrow$). Retaining only these anti-aligned interactions (and noting that the summation over distinguishable pairs absorbs the $1/2$ prefactor), the effective Hamiltonian becomes:

$$\hat{K} = \hat{T} + \frac{1}{V} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} V(\mathbf{k}',\mathbf{k}) c^{\dagger}_{\mathbf{k}'+\frac{\mathbf{q}}{2},\uparrow} c^{\dagger}_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\downarrow} c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\downarrow} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\uparrow}.$$

2.3.1 | Mean-Field Approximation

We now proceed in a manner conceptually similar to the treatment of the weakly interacting Bose gas (WIBG), where we successfully applied a mean-field approximation by replacing the bosonic destruction operator with a macroscopic c-number, $\langle a \rangle \simeq \sqrt{N_0}$.

Can we perform an identical substitution for individual fermions, setting $\langle c \rangle \neq 0$? The answer is strictly negative. This prohibition stems from the fundamental difference in quantum statistics: fermions inherently require a completely antisymmetric wave function. Removing or adding a single fermion physically destroys this antisymmetrization property. In other words, fermionic parity is fundamental, and the expectation value of any single fermionic operator evaluated over the exact many-body ground state must rigorously vanish:

$$\langle \psi_{GS} | c | \psi_{GS} \rangle = 0.$$

Since we cannot assign a non-zero expectation value to a single fermionic operator, we instead look at pairs of operators. We expect that $\langle cc \rangle \neq 0$ and $\langle c^\dagger c^\dagger \rangle \neq 0$. By doing so, we essentially associate the bilinear operator cc with an effective bosonic particle. This pairs-based approach allows us to mathematically describe the Cooper pair, and its macroscopic occupation will physically characterize the superfluid mode of the system.

Following the standard mean-field framework, we write the pair destruction operator cc as its macroscopic expectation value plus a small quantum fluctuation operator $\delta(cc)$:

$$cc = \langle cc \rangle + \delta(cc),$$

where we explicitly assume the fluctuations are much smaller than the mean value ($\delta(cc) \ll \langle cc \rangle$). We apply the exact same decomposition to the pair creation factor $c^\dagger c^\dagger$ and substitute this approximation into the four-fermion interaction term:

$$c^\dagger c^\dagger cc = \left(\langle c^\dagger c^\dagger \rangle + \delta(c^\dagger c^\dagger) \right) \left(\langle cc \rangle + \delta(cc) \right).$$

Expanding the product and systematically neglecting the second-order fluctuation term $\delta(c^\dagger c^\dagger)\delta(cc)$ (since it involves the product of two very small fluctuations), we rewrite the fluctuation operators back in terms of the original operators to obtain a linearized mean-field interaction:

$$c^\dagger c^\dagger cc \simeq c^\dagger c^\dagger \langle cc \rangle + \langle c^\dagger c^\dagger \rangle cc - \langle c^\dagger c^\dagger \rangle \langle cc \rangle.$$

At which specific momenta will these Cooper pairs form? We make the physical assumption that Cooper pairs will form exclusively with zero center-of-mass momentum. This implies that we only consider non-zero expectation values where the total momentum $\mathbf{q}$ of the pairing electrons is identically zero:

$$\langle c_{-\mathbf{k}+\frac{\mathbf{q}}{2},\uparrow} c_{\mathbf{k}+\frac{\mathbf{q}}{2},\downarrow} \rangle = \delta_{\mathbf{q},0} A_{\mathbf{k}},$$
$$\langle c^\dagger_{\mathbf{k}'+\frac{\mathbf{q}}{2},\uparrow} c^\dagger_{-\mathbf{k}'+\frac{\mathbf{q}}{2},\downarrow} \rangle = \delta_{\mathbf{q},0} A^*_{\mathbf{k}'},$$

where $A_{\mathbf{k}}$ and $A^*_{\mathbf{k}'}$ are momentum-dependent amplitudes.

By substituting this zero-momentum mean-field approximation back into the four-fermion interaction term of our Hamiltonian, the sum over $\mathbf{q}$ cleanly collapses due to the Kronecker delta $\delta_{\mathbf{q},0}$, yielding the effective mean-field Hamiltonian:

$$\hat{K} = \hat{T} + \frac{1}{V} \sum_{\mathbf{k},\mathbf{k}'} V(\mathbf{k}', \mathbf{k}) \left(A_{\mathbf{k}} c^\dagger_{\mathbf{k}',\uparrow} c^\dagger_{-\mathbf{k}',\downarrow} + A^*_{\mathbf{k}'} c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} - A_{\mathbf{k}} A^*_{\mathbf{k}'} \right).$$

TODO: add image of a momentum space diagram comparing a pair of fermions with perfectly opposite momenta forming a zero center of mass Cooper pair against a pair with non zero total momentum which is not allowed

To render this expression structurally clearer and physically transparent, we define the momentum-dependent superconducting **gap parameter** $\Delta_{\mathbf{k}}$ and its complex conjugate:

$$\Delta_{\mathbf{k}'} = -\frac{1}{V}\sum_{\mathbf{k}} V(\mathbf{k}',\mathbf{k})A_{\mathbf{k}}, \quad \Delta^*_{\mathbf{k}} = -\frac{1}{V}\sum_{\mathbf{k}'} V(\mathbf{k}',\mathbf{k})A^*_{\mathbf{k}'}. \tag{2.3.1}$$

(Note that these definitions are perfectly consistent because any time-reversal invariant potential satisfies the symmetry $V(\mathbf{k},\mathbf{k}') = V(\mathbf{k}',\mathbf{k})^*$.)

With these gap definitions incorporated, the Hamiltonian reduces to a fundamentally bilinear form:

$$\hat{K} = \hat{T} - \sum_{\mathbf{k}} \left[\Delta_{\mathbf{k}} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} + \Delta^*_{\mathbf{k}} c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow}\right] + \sum_{\mathbf{k}} A^*_{\mathbf{k}}\Delta_{\mathbf{k}}.$$

Notice that this expansion yields anomalous terms containing two creation operators ($c^\dagger c^\dagger$) and two annihilation operators (cc). This algebraic structure is mathematically completely analogous to the terms $a^2 + (a^\dagger)^2$ that we encountered when performing the Bogoliubov transformation for the weakly interacting Bose gas. The final term is a constant energy offset that does not depend on the fermionic operators; while usually discarded in simple normal-state models, it must be retained here as it decisively contributes to the self-consistent energy of the condensed state.

2.3.2 | Diagonalization of the Hamiltonian

To elegantly diagonalize this bilinear Hamiltonian, it is highly convenient to express it in a matrix representation using **Nambu spinors**. By leveraging the fundamental fermionic anti-commutation relations, we can rewrite the kinetic energy term for the spin-down particles as:

$$\xi_k c^\dagger_{\mathbf{k},\downarrow} c_{\mathbf{k},\downarrow} = \xi_k - \xi_k c_{\mathbf{k},\downarrow} c^\dagger_{\mathbf{k},\downarrow}\,.$$

We observe that the effective Hamiltonian contains assorted bilinear pairings of the form $c^\dagger c$, $c^\dagger c^\dagger$, cc, and $cc^\dagger$. By strategically grouping the fermionic operators (specifically, pairing the state of a spin-up particle at momentum $\mathbf{k}$ with the time-reversed state of a spin-down particle at momentum $-\mathbf{k}$) these bilinear combinations can be systematically organized into a 2×2 operator matrix structure of the form:

$$\begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} c_{\mathbf{k},\uparrow} & c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} \\ c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow} c^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix}.$$

This algebraic maneuver allows us to recast the operator $\hat{K}$ into the following compact matrix form by introducing the two-component Nambu spinor $\Psi^\dagger_{\mathbf{k}} = \begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow}\end{pmatrix}$:

$$\hat{K} = \sum_{\mathbf{k}} \begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow}\end{pmatrix} \begin{pmatrix} \xi_k & -\Delta_{\mathbf{k}} \\ -\Delta^*_{\mathbf{k}} & -\xi_k \end{pmatrix} \begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c^\dagger_{-\mathbf{k},\downarrow}\end{pmatrix} + C, \tag{2.3.2}$$

where the constant energy offset C explicitly collects all the scalar terms that do not directly multiply the fermionic operators:

$$C = \sum_{\mathbf{k}} \left(\xi_k + A^*_{\mathbf{k}}\Delta_{\mathbf{k}}\right).$$

Notice that the off-diagonal terms in the matrix representation of $\hat{K}$ are precisely proportional to the gap parameter $\Delta_{\mathbf{k}}$, which mathematically couples operators of opposite momenta and spins. This off-diagonal mixing is a direct manifestation of the fundamental pairing mechanism that underlies the BCS theory of superconductivity. The constant scalar term C is structurally important

for calculating the total thermodynamic energy, but our immediate goal is to "solve the problem"—that is, to diagonalize the Hamiltonian matrix to find the fundamental ground state and the quasiparticle excitation spectrum—without having to carry the constant term through the diagonalization algebra. It will simply be added back to the final ground-state energy at the end.

Quasiparticle Transformation

Let us define the central 2×2 Hermitian matrix $M_{\mathbf{k}}$ as:

$$M_{\mathbf{k}} = \begin{pmatrix} \xi_k & -\Delta_{\mathbf{k}} \\ -\Delta_{\mathbf{k}}^* & -\xi_k \end{pmatrix}.$$

We now show that the many-body problem of diagonalizing the quadratic form $\hat{K}$ reduces entirely to the problem of diagonalizing the independent 2×2 matrix $M_{\mathbf{k}}$ for every momentum mode $\mathbf{k}$. In strict analogy to the procedure utilized for the weakly interacting Bose gas (with the Bogoliubov transformation and quasiparticles), we seek a transformation to a new set of fermionic creation and destruction operators, $\gamma_{\mathbf{k},\sigma}$ and $\gamma_{\mathbf{k},\sigma}^\dagger$, that diagonalize $\hat{K}$. These will become our new *quasiparticle operators*. We introduce a transformation matrix $U_{\mathbf{k}}$ such that:

$$\begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c_{-\mathbf{k},\downarrow}^\dagger \end{pmatrix} = U_{\mathbf{k}} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma_{-\mathbf{k},\downarrow}^\dagger \end{pmatrix}, \quad \begin{pmatrix} c_{\mathbf{k},\uparrow}^\dagger & c_{-\mathbf{k},\downarrow} \end{pmatrix} = \begin{pmatrix} \gamma_{\mathbf{k},\uparrow}^\dagger & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} U_{\mathbf{k}}^\dagger. \tag{2.3.3}$$

Notice that we are mixing momenta and spins in this transformation, which is a direct consequence of the fundamental pairing mechanism that underlies the BCS theory (the structure of the Hamiltonian encodes this directly).

This transformation must simultaneously satisfy two critical conditions:

1. The terms appearing in the grand canonical Hamiltonian, once expressed in terms of the new operators, must form a diagonal matrix $D_{\mathbf{k}}$:

$$\begin{pmatrix} c_{\mathbf{k},\uparrow}^\dagger & c_{-\mathbf{k},\downarrow} \end{pmatrix} M_{\mathbf{k}} \begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c_{-\mathbf{k},\downarrow}^\dagger \end{pmatrix} = \begin{pmatrix} \gamma_{\mathbf{k},\uparrow}^\dagger & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} U_{\mathbf{k}}^\dagger M_{\mathbf{k}} U_{\mathbf{k}} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma_{-\mathbf{k},\downarrow}^\dagger \end{pmatrix},$$

   with $D_{\mathbf{k}} = U_{\mathbf{k}}^\dagger M_{\mathbf{k}} U_{\mathbf{k}}$ being explicitly diagonal: $U_{\mathbf{k}}$ is a unitary (as we will see in a moment) change of basis that diagonalizes the Hamiltonian.

2. The fundamental anti-commutation rules must be preserved for the new quasiparticle operators, ensuring in particular that:

$$\{\gamma_{\mathbf{k},\sigma}, \gamma_{\mathbf{k}',\sigma'}^\dagger\} = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'},$$

   with all remaining anti-commutators identically zero.

We can readily prove that requiring the transformation $U_{\mathbf{k}}$ to be unitary guarantees the preservation of these anti-commutation relations on the new operators.

Proof. Let us parameterize the inverse transformation matrix as:

$$U_{\mathbf{k}}^{-1} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

The relation linking the bare c operators to the quasiparticle γ operators is then:

$$\begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix} = U_{\mathbf{k}}^{-1} \begin{pmatrix} c_{\mathbf{k},\uparrow} \\ c^\dagger_{-\mathbf{k},\downarrow} \end{pmatrix}, \quad \begin{pmatrix} \gamma^\dagger_{\mathbf{k},\uparrow} & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} = \begin{pmatrix} c^\dagger_{\mathbf{k},\uparrow} & c_{-\mathbf{k},\downarrow} \end{pmatrix} (U_{\mathbf{k}}^{-1})^\dagger,$$

where $(U_{\mathbf{k}}^{-1})^\dagger$ is the conjugate transpose of $U_{\mathbf{k}}^{-1}$:

$$(U_{\mathbf{k}}^{-1})^\dagger = \begin{pmatrix} a_{11}^* & a_{21}^* \\ a_{12}^* & a_{22}^* \end{pmatrix}.$$

TODO: check computations from Di Sante notes (Filippo)

By explicitly substituting this into the anti-commutators and using the known algebraic relations for the bare c operators, it is straightforward to check that:

$$\{\gamma_{\mathbf{k},\uparrow}, \gamma^\dagger_{\mathbf{k}',\uparrow}\} = \left\{ \left(a_{11} c_{\mathbf{k},\uparrow} + a_{12} c^\dagger_{-\mathbf{k},\downarrow} \right), \left(a_{11}^* c^*_{\mathbf{k}',\uparrow} + a_{12}^* c^\dagger_{-\mathbf{k}',\downarrow} \right) \right\} = (|a_{11}|^2 + |a_{12}|^2)\delta_{\mathbf{k},\mathbf{k}'},$$

$$\{\gamma_{-\mathbf{k},\downarrow}, \gamma^\dagger_{-\mathbf{k}',\downarrow}\} = \left\{ \left(a_{21} c_{-\mathbf{k},\downarrow} + a_{22} c^\dagger_{\mathbf{k},\uparrow} \right), \left(a_{21}^* c^*_{-\mathbf{k}',\downarrow} + a_{22}^* c^*_{\mathbf{k}',\uparrow} \right) \right\} = (|a_{21}|^2 + |a_{22}|^2)\delta_{\mathbf{k},\mathbf{k}'},$$

$$\{\gamma_{\mathbf{k},\uparrow}, \gamma_{-\mathbf{k}',\downarrow}\} = \left\{ \left(a_{11} c_{\mathbf{k},\uparrow} + a_{12} c^\dagger_{-\mathbf{k},\downarrow} \right), \left(a_{21}^* c^*_{-\mathbf{k}',\downarrow} + a_{22}^* c^*_{\mathbf{k}',\uparrow} \right) \right\} = (a_{11} a_{21}^* + a_{12} a_{22}^*)\delta_{\mathbf{k},-\mathbf{k}'}.$$

In order to successfully satisfy the fermionic algebra, the coefficients must obey:

$$1 = |a_{11}|^2 + |a_{12}|^2,$$

$$1 = |a_{21}|^2 + |a_{22}|^2,$$

$$0 = a_{11} a_{21}^* + a_{12} a_{22}^*.$$

These three constraints are automatically and exactly satisfied by unitary matrices. This implies that $U_{\mathbf{k}}^\dagger M_{\mathbf{k}} U_{\mathbf{k}} = U_{\mathbf{k}}^{-1} M_{\mathbf{k}} U_{\mathbf{k}}$; thus, finding a unitary matrix that diagonalizes the Hermitian matrix $M_{\mathbf{k}}$ will successfully diagonalize the entire many-body Hamiltonian $\hat{K}$. □

Quasiparticle Dispersion and Coherence Factors

The eigenvalues λ of $M_{\mathbf{k}}$ are determined by its characteristic equation:

$$(\xi_k - \lambda)(-\xi_k - \lambda) - |\Delta_{\mathbf{k}}|^2 = 0,$$

from which we immediately obtain the roots $\lambda = \pm E_{\mathbf{k}}$, where:

$$E_{\mathbf{k}} = \sqrt{\xi_k^2 + |\Delta_{\mathbf{k}}|^2} > 0. \tag{2.3.4}$$

This fundamentally important quantity $E_{\mathbf{k}}$ represents the *energy dispersion of the superconducting quasiparticle excitations*. We expected two opposite sign eigenvalues because the Hamiltonian in the Nambu matrix representation artificially doubles the number of degrees of freedom to conveniently encode both electrons and holes within a single spinor: this mathematical redundancy intrinsically manifests as a built-in particle-hole symmetry in the eigenvalue spectrum.

If we let the gap parameter $\Delta_{\mathbf{k}}$ vanish, the quasiparticle dispersion reduces to the free-particle kinetic energy relative to the chemical potential, $E_{\mathbf{k}} = |\xi_k|$. In this limit, the quasiparticles are simply the original bare fermions.

The normalized eigenvector of $M_{\mathbf{k}}$ corresponding to the positive eigenvalue $E_{\mathbf{k}}$ takes the form:

$$\begin{pmatrix} u_{\mathbf{k}}^* \\ -v_{\mathbf{k}}^* \end{pmatrix}, \text{ with } |u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1.$$

Following this definition, the **coherence factors** can be expressed as:

$$u_{\mathbf{k}}^* = \frac{\Delta_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}(E_{\mathbf{k}} - \xi_k)}}, \quad v_{\mathbf{k}}^* = \frac{E_{\mathbf{k}} - \xi_k}{\sqrt{2E_{\mathbf{k}}(E_{\mathbf{k}} - \xi_k)}} = v_{\mathbf{k}},$$

such that $v_{\mathbf{k}}$ will be strictly real and positive, while $u_{\mathbf{k}}$ can be complex.

Conversely, the normalized eigenvector corresponding to the negative eigenvalue $-E_{\mathbf{k}}$ is:

$$\begin{pmatrix} v_{\mathbf{k}} \\ u_{\mathbf{k}} \end{pmatrix}.$$

Remark. *In defining these eigenvectors, a specific choice of phase factors has been made such that $v_{\mathbf{k}}^*$ is strictly real and the determinant of the transformation matrix is exactly* $\det U_{\mathbf{k}} = 1$.

Using these two orthogonal eigenvectors, one may immediately construct the explicit unitary transformation $U_{\mathbf{k}}$ that diagonalizes $M_{\mathbf{k}}$:

$$U_{\mathbf{k}} = \begin{pmatrix} u_{\mathbf{k}}^* & v_{\mathbf{k}} \\ -v_{\mathbf{k}}^* & u_{\mathbf{k}} \end{pmatrix},$$

with its corresponding inverse:

$$U_{\mathbf{k}}^{-1} = U_{\mathbf{k}}^\dagger = \begin{pmatrix} u_{\mathbf{k}} & -v_{\mathbf{k}} \\ v_{\mathbf{k}}^* & u_{\mathbf{k}}^* \end{pmatrix}.$$

Bogoliubov Transformation and Diagonalized Hamiltonian

By applying the previously derived inverse transformation, we obtain the explicit mathematical expressions for the new operators in terms of the old ones (the celebrated Bogoliubov transformation):

$$\begin{aligned} \gamma_{\mathbf{k},\uparrow} &= u_{\mathbf{k}} c_{\mathbf{k},\uparrow} - v_{\mathbf{k}} c_{-\mathbf{k},\downarrow}^\dagger, \\ \gamma_{-\mathbf{k},\downarrow}^\dagger &= v_{\mathbf{k}}^* c_{\mathbf{k},\uparrow} + u_{\mathbf{k}}^* c_{-\mathbf{k},\downarrow}^\dagger. \end{aligned} \tag{2.3.5}$$

Notice that the new annihilation operator $\gamma_{\mathbf{k},\uparrow}$ physically mixes the old creation operator $c_{-\mathbf{k},\downarrow}^\dagger$ and destruction operator $c_{\mathbf{k},\uparrow}$, specifically entangling particles with opposite spin and momentum.[2]

In these transformed variables, spin plays a non trivial role, as the same operator creates a particle with spin up and destroys a particle with spin down. This is a direct consequence of the pairing mechanism, which couples opposite spins and momenta, preserving the *parity symmetry* Z_2 of the system. The Bogoliubov transformation is a canonical unitary transformation that preserves the fundamental anti-commutation relations.

Thinking about the *Stern-Gerlach experiment*, we can see that the new quasiparticle operators $\gamma_{\mathbf{k},\sigma}$ and $\gamma_{\mathbf{k},\sigma}^\dagger$ do not correspond to any physical spin measurement along a given axis. Instead, they represent a superposition of spin-up and spin-down states, which is a direct manifestation of the entanglement between opposite spins induced by the pairing mechanism of BCS theory.

Expressed in terms of these newly defined γ operators, the grand canonical Hamiltonian assumes its fully diagonalized form:

$$\hat{K} = \sum_{\mathbf{k}} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow}^\dagger & \gamma_{-\mathbf{k},\downarrow} \end{pmatrix} D_{\mathbf{k}} \begin{pmatrix} \gamma_{\mathbf{k},\uparrow} \\ \gamma_{-\mathbf{k},\downarrow}^\dagger \end{pmatrix} + C,$$

[2] Equivalently, taking the Hermitian adjoint of the second equation yields $\gamma_{-\mathbf{k},\downarrow} = u_{\mathbf{k}} c_{-\mathbf{k},\downarrow} + v_{\mathbf{k}} c_{\mathbf{k},\uparrow}^\dagger$.

where the diagonal eigenvalue matrix $D_{\mathbf{k}} = U_{\mathbf{k}}^{\dagger} M_{\mathbf{k}} U_{\mathbf{k}}$ is given by:

$$D_{\mathbf{k}} = \begin{pmatrix} E_{\mathbf{k}} & 0 \\ 0 & -E_{\mathbf{k}} \end{pmatrix}.$$

By explicitly expanding this matrix product and once again applying the quasiparticle anti-commutation relations, we obtain:

$$\hat{K} = \sum_{\mathbf{k}} \left(E_{\mathbf{k}} \gamma_{\mathbf{k},\uparrow}^{\dagger} \gamma_{\mathbf{k},\uparrow} - E_{\mathbf{k}} \gamma_{-\mathbf{k},\downarrow} \gamma_{-\mathbf{k},\downarrow}^{\dagger} \right) + C = \sum_{\mathbf{k},\sigma} E_{\mathbf{k}} \gamma_{\mathbf{k},\sigma}^{\dagger} \gamma_{\mathbf{k},\sigma} + C', \tag{2.3.6}$$

where the newly defined constant term C' absorbs the commutation shifts, assuring that the Hamiltonian is expressed in normal-ordered form:

$$C' = C - \sum_{\mathbf{k}} E_{\mathbf{k}} = \sum_{\mathbf{k}} \left(\xi_k + A_{\mathbf{k}}^{*} \Delta_{\mathbf{k}} - E_{\mathbf{k}} \right).$$

Because the quasiparticle energy dispersion $E_{\mathbf{k}}$ is strictly positive, the operator $\sum_{\mathbf{k},\sigma} E_{\mathbf{k}} \gamma_{\mathbf{k},\sigma}^{\dagger} \gamma_{\mathbf{k},\sigma}$ is positive semi-definite. Consequently, the constant C' physically represents the lowest possible eigenvalue of $\hat{K}$, which corresponds exactly to the ground state grand potential of the superconducting system.

2.3.3 | Ground State Wave Function

The structure of the many-body wavefunction changes dramatically in the superconducting phase. A generic state can be written as a superposition of configurations with different quasiparticle occupation numbers $m_{\mathbf{k}}$ (for spin up) and $n_{\mathbf{k}}$ (for spin down):

$$|\Psi_{GS}\rangle = \sum_{m_{\mathbf{k}}, n_{\mathbf{k}}} C_{m_{\mathbf{k}}, n_{\mathbf{k}}} |m_{\mathbf{k}}, n_{\mathbf{k}}\rangle.$$

Because the newly defined γ operators obey canonical fermionic anti-commutation relations, the occupation numbers $|m_{\mathbf{k}}, n_{\mathbf{k}}\rangle$ form a vector composed strictly of 0s and 1s. The expectation value for the number of quasiparticles in a given state is thus given by the statistical weights:

$$\langle \gamma_{\mathbf{k},\uparrow}^{\dagger} \gamma_{\mathbf{k},\uparrow} \rangle \propto \sum_{\mathbf{k}} m_{\mathbf{k}} C_{m_{\mathbf{k}}, n_{\mathbf{k}}}.$$

In the grand-canonical ensemble, the thermodynamic thermal average of any physical operator $\hat{O}$ is given by the standard statistical mechanics trace formula:

$$\langle O \rangle = \frac{\mathrm{tr}\left(e^{-\beta \hat{K}} \hat{O} \right)}{\mathcal{Z}},$$

where the grand partition function is defined as:

$$\mathcal{Z} = \mathrm{tr}\left(e^{-\beta \hat{K}} \right).$$

Here, $\hat{K} = \hat{H} - \mu \hat{N}$ represents the grand canonical Hamiltonian. In the absolute zero temperature limit, $T \to 0$, the inverse temperature $\beta \to +\infty$. In this asymptotic regime, the thermal average strictly reduces to a pure quantum mechanical expectation value over the ground state, where the grand-canonical ground state $|\psi_0(\mu)\rangle$ is defined as the unique eigenstate of $\hat{K}$ corresponding to its lowest possible eigenvalue.

Based on our preceding diagonalization procedure, the energy eigenvalues $E_{\mathbf{k}}$ associated with the quasiparticles are strictly positive ($E_{\mathbf{k}} > 0$). Consequently, adding any quasiparticle inherently increases the total energy of the system. Therefore, to rigorously minimize the energy, the ground state must contain exactly zero quasiparticles. It is exactly the **vacuum of the Bogoliubov operators**:

$$\gamma_{\mathbf{k},\sigma}|\psi_0\rangle = 0 \quad \forall \mathbf{k}, \sigma. \tag{2.3.7}$$

Conversely, acting on this vacuum with quasiparticle creation operators produces elementary excitations with additive positive energies:

$$\langle 0,1|\hat{K}|0,1\rangle = E_{\mathbf{k}}, \quad \langle 0,2|\hat{K}|0,2\rangle = E_{\mathbf{k}} + E_{\mathbf{k}'}.$$

With the ground state firmly established as the quasiparticle vacuum, we can now evaluate expectation values in the original fermion basis, such as the anomalous pairing amplitude:

$$\langle c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\rangle.$$

To determine such measurable observables (since the quasiparticles are a mathematical construct, we ultimately need correlation functions of the old bare fermionic variables which can be measured experimentally), we must analyze the ground state structure explicitly in terms of the original electron operators.

By explicitly using the Bogoliubov transformation derived earlier to express $\gamma_{\mathbf{k},\uparrow}$ in terms of the bare particle operators ($\gamma_{\mathbf{k},\uparrow} = u_{\mathbf{k}} c_{\mathbf{k},\uparrow} - v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}$), this vacuum condition for the specific spin-up branch ($\sigma = \uparrow$) yields:

$$\left(u_{\mathbf{k}} c_{\mathbf{k},\uparrow} - v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}\right)|\psi_0\rangle = 0,$$

which can be rewritten as the fundamental equality:

$$u_{\mathbf{k}} \underbrace{c_{\mathbf{k},\uparrow}}_{\text{TAKE AN ELECTRON OUT}} |\psi_0\rangle = v_{\mathbf{k}} \underbrace{c^\dagger_{-\mathbf{k},\downarrow}}_{\text{TAKE AN ELECTRON IN}} |\psi_0\rangle \quad \forall \mathbf{k}. \tag{2.3.8}$$

Looking at (2.3.8), we can observe a profound physical reality regarding the nature of the BCS ground state: the action of destroying a bare fermion with momentum $\mathbf{k}$ and spin $\uparrow$ on the ground state produces the exact same resulting state (up to the scalar coherence amplitudes $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$) as creating a bare fermion with opposite momentum $-\mathbf{k}$ and opposite spin $\downarrow$.

Because the left-hand operation explicitly decreases the total particle number by 1, while the right-hand operation explicitly increases the total particle number by 1, for these two operations to equate, **the ground state itself must inherently be a coherent quantum superposition of states with different total particle numbers**.

Consequently, we draw three major theoretical conclusions:

1. The macroscopic BCS ground state $|\psi_0\rangle$ **does not possess a definite number of fermions**.
2. This macroscopic indefiniteness implies **Spontaneous Symmetry Breaking**.
3. The specific continuous symmetry being spontaneously broken is the global $U(1)$ phase symmetry, which in the normal phase classically dictates the strict conservation of the total particle number $\hat{N}_\uparrow + \hat{N}_\downarrow$.

Remark. *It is crucial to clarify the exact nature of this symmetry breaking. The global $U(1)$ gauge symmetry associated with the total particle number $\hat{N} = \hat{N}_\uparrow + \hat{N}_\downarrow$ is spontaneously broken by the macroscopic phase coherence of the condensate. However, as previously established, the Hamiltonian still precisely commutes with the "asymmetric charge" operator governing the spin difference $\hat{N}_\uparrow - \hat{N}_\downarrow$. This means the $U(1)$ spin symmetry remains strictly conserved and unbroken in the superconducting phase. Note also that we are referring here to a global gauge symmetry; local gauge symmetries (like electromagnetism) cannot be spontaneously broken in the strict sense, as established by Elitzur's theorem.*

Quasiparticle Operators and Parity Symmetry

Let us now elaborate on the critical relationship between the Bogoliubov quasiparticle operators $\gamma_{\mathbf{k},\sigma}$ and the parity symmetry of the fermionic system. In your previous notes, there was a slight conceptual misunderstanding: the γ operators do not preserve the parity eigenvalue identically; rather, they definitively flip it.

Because $\gamma_{\mathbf{k},\sigma}$ is defined as a linear combination of a single-particle annihilation operator and a single-particle creation operator, applying it inherently changes the total fermion number by ± 1. Therefore, it rigorously changes the fermion number parity of a wave function from even to odd, or from odd to even. However, the crucial point is that $\gamma_{\mathbf{k},\sigma}$ keeps the parity of a state **well-defined**: parity remains a perfectly good quantum number when acting on a state of definite parity.

Specifically, if we apply $\gamma_{\mathbf{k},\sigma}$ to a physical state $|\psi(P = +1)\rangle$ that possesses a well-defined even parity ($P = +1$), we will always obtain a new state $|\varphi(P = -1)\rangle$ with a well-defined odd parity ($P = -1$). We never obtain an undefined superposition of states with different parities (such as an unphysical mixture of $P = -1$ and $P = +1$):

$$\gamma_{\mathbf{k},\sigma}|\psi(P = +1)\rangle = |\varphi(P = -1)\rangle \neq \alpha|\varphi(P = -1)\rangle + \beta|\varphi'(P = +1)\rangle.$$

This holds fundamentally because the Bogoliubov transformation only mixes operators that either destroy one fermion or create one fermion (both of which flip the parity in the exact same manner), without introducing any constant c-number term in the linear combination, which would instead mix different parities and destroy the quantum number.

More formally, the fermion number parity operator $\hat{P}$ is mathematically defined as the product over all possible momenta $\mathbf{k}$ and spins σ:

$$\hat{P} = \prod_{\mathbf{k},\sigma} (1 - 2\hat{n}_{\mathbf{k},\sigma}),$$

where $\hat{n}_{\mathbf{k},\sigma} = c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma}$ is the standard number operator for the mode $(\mathbf{k}, \sigma)$. If a mode is occupied ($\langle \hat{n}_{\mathbf{k},\sigma} \rangle = 1$), the corresponding factor in the product yields $1 - 2(1) = -1$; if the mode is empty ($\langle \hat{n}_{\mathbf{k},\sigma} \rangle = 0$), the factor yields $+1$. Thus, $\hat{P}$ assigns a factor of (-1) for every single particle present in the many-body state. Since $\gamma_{\mathbf{k},\sigma}$ changes the number of particles by ± 1, it strictly does not commute with the parity operator:

$$[\gamma_{\mathbf{k},\sigma}, \hat{P}] \neq 0.$$

(In fact, the anticommutator formally vanishes, $\{\gamma_{\mathbf{k},\sigma}, \hat{P}\} = 0$, confirming that $\gamma_{\mathbf{k},\sigma}$ cleanly flips the parity eigenvalue while perfectly maintaining it as a well-defined observable.)

Example. To explicitly verify that applying $\gamma_{\mathbf{k},\uparrow}$ produces a superposition of states with vastly different particle numbers N_p that nevertheless share the exact same parity P, let us correctly

compute the action on a two-particle state. Consider the state (which clearly has even parity, $P = +1$):

$$|\psi\rangle = c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{p},\downarrow} |0\rangle.$$

Assuming for simplicity that the momenta are distinct ($-\mathbf{k} \neq \mathbf{p}$), let us apply $\gamma_{\mathbf{k},\uparrow} = u_{\mathbf{k}} c_{\mathbf{k},\uparrow} - v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}$ to $|\psi\rangle$:

$$\begin{aligned}\gamma_{\mathbf{k},\uparrow}|\psi\rangle &= \left(u_{\mathbf{k}} c_{\mathbf{k},\uparrow} - v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow}\right)\left(c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{p},\downarrow}\right)|0\rangle \\ &= u_{\mathbf{k}} c_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{p},\downarrow}|0\rangle - v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{p},\downarrow}|0\rangle.\end{aligned}$$

In the first term, we utilize the fundamental fermionic anti-commutation relation $c_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\uparrow} = 1 - \hat{n}_{\mathbf{k},\uparrow}$. Since $\hat{n}_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{p},\downarrow}|0\rangle = 0$ (the state has no particle at $\mathbf{k},\uparrow$), the first term elegantly simplifies to $u_{\mathbf{k}} c^\dagger_{\mathbf{p},\downarrow}|0\rangle$. We therefore obtain a clean quantum superposition of a one-particle state and a three-particle state:

$$\gamma_{\mathbf{k},\uparrow}|\psi\rangle = \underbrace{u_{\mathbf{k}} c^\dagger_{\mathbf{p},\downarrow}|0\rangle}_{\substack{N_p=1 \\ P=(-1)^1=-1}} - \underbrace{v_{\mathbf{k}} c^\dagger_{-\mathbf{k},\downarrow} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{p},\downarrow}|0\rangle}_{\substack{N_p=3 \\ P=(-1)^3=-1}}.$$

Although the resulting state is a complex superposition of states with differing particle numbers ($N_p = 1$ and $N_p = 3$), **both structural components possess an odd parity ($P = -1$)**. Consequently, applying $\gamma_{\mathbf{k},\uparrow}$ cleanly changes the overall parity of $|\psi\rangle$ from $+1$ to -1, without ever creating an unphysical superposition of $+1$ and -1 parity states.

BCS Ground State Wavefunction Ansatz

This entangled behavior is structurally very similar to what we previously found for the correlated ground state of a dilute interacting Bose gas. In that bosonic case, we determined that the ground state wave function was heavily populated by pairs of opposite momenta and could be expressed using an exponential ansatz:

$$|\psi_0\rangle = A e^{B a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}} = A + \left(B a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}\right) + \frac{A}{2}\left(B a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}\right)^2 + \dots$$

with A and B representing specific coherent coefficients that we determined.

However, in the present scenario, we are strictly dealing with fermions. Fermions are governed by the Pauli exclusion principle, which mathematically enforces that the square of any fermionic creation operator identically vanishes ($c^\dagger c^\dagger = 0$). Consequently, all terms in the previous coherent expansion proportional to $\left(a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}\right)^n$ with $n > 1$ must vanish exactly for fermions. Indeed sometimes we called them bosons a bit improperly, since they just behave like bosons in some limits: they still have to satisfy the Pauli principle and the anti-commutation relations, in fact:

$$a^\dagger_{\mathbf{k}} a^\dagger_{\mathbf{k}} |0\rangle = |2\rangle\,, \text{ but } (c^\dagger_{\mathbf{k}} c^\dagger_{-\mathbf{k}})(c^\dagger_{\mathbf{k}} c^\dagger_{-\mathbf{k}})|0\rangle = 0.$$

We call them **hard-core bosons**, and hey respect the modified commutation relations:

$$[c_{\mathbf{k},\uparrow} c_{-\mathbf{k},\downarrow}, c^\dagger_{\mathbf{k}',\uparrow} c^\dagger_{-\mathbf{k}',\downarrow}] = \delta_{\mathbf{k},\mathbf{k}'} P_{\mathbf{k}},$$

where the last operator $P_{\mathbf{k}}$ is the projector onto the subspace ensuring only one pair of fermions can occupy the same momentum mode $\mathbf{k}$.

Truncating the series to respect the Pauli principle, we thus try the following linear pairing ansatz for the wavefunction of a single momentum mode:

$$\left(A + B c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle,$$

where $|0\rangle$ represents the absolute vacuum of the original bare fermionic operators $c_{\mathbf{k},\sigma}$ (meaning a state containing zero particles). To determine the unknown coefficients A and B, we strictly require that this proposed ansatz satisfies the defining vacuum condition (2.3.8).

Evaluating the left-hand side of the condition by acting with the destruction operator yields:

$$u_{\mathbf{k}} c_{\mathbf{k},\uparrow} \left(A + B c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} \right) |0\rangle = B u_{\mathbf{k}} c_{\mathbf{k},\uparrow} c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} |0\rangle$$
$$= B u_{\mathbf{k}} c^{\dagger}_{-\mathbf{k},\downarrow} |0\rangle.$$

Evaluating the right-hand side of the condition by acting with the creation operator yields:

$$v_{\mathbf{k}} c^{\dagger}_{-\mathbf{k},\downarrow} \left(A + B c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} \right) |0\rangle = A v_{\mathbf{k}} c^{\dagger}_{-\mathbf{k},\downarrow} |0\rangle.$$

By comparing these two results, we immediately see that the proposed state successfully verifies the vacuum condition if and only if $B = v_{\mathbf{k}}$ and $A = u_{\mathbf{k}}$. Because the different momentum modes $\mathbf{k}$ are decoupled in the diagonalized Hamiltonian, this structural requirement must be independently satisfied for every momentum pair $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ across the entire system. Taking the tensor product over all possible momentum states yields the complete many-body ground state:

$$|\psi_0(\mu)\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} \right) |0\rangle. \tag{2.3.9}$$

The state described by (2.3.9) is the celebrated BCS wave function. It was first proposed historically by Bardeen, Cooper, and Schrieffer in 1957 as a brilliant variational wave function to approximate the ground state of a metallic superconductor. The macroscopic state describes a coherent superposition where each momentum pair state $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ is either empty with probability amplitude $u_{\mathbf{k}}$, or occupied by a Cooper pair with probability amplitude $v_{\mathbf{k}}$.

Since the coherence factors $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are momentum-dependent, the BCS wave function is highly non-trivial state that cannot be factorized into a simple product of single-particle states. Anyway, the BCS wave function is a pure state, and not a mixed state. It is a superposition of different particle number states, but it is still a single quantum state, with even parity (since it is a superposition of states with an even number of particles).

Remark. *It is a strong assertion that the BCS wave function is a good approximation of the true ground state of a superconductor, since this wave function is not an eigenstate of the particle number operator (where the original hamiltonian is) and does not have entanglement between different momentum modes. However, it is a very good approximation, and it captures the essential physics of superconductivity.*

Normalization of the Ground State

Finally, one can easily check that this constructed BCS ground state is properly normalized to unity. Because the creation and annihilation operators for distinct momentum modes completely commute (pairs of fermions behave as hard-core bosons that commute for $\mathbf{k} \neq \mathbf{k}'$), the grand inner product factorizes into an independent product over each individual momentum state evaluated over the bare vacuum $|0\rangle$:

$$\langle \psi_0 | \psi_0 \rangle = \langle 0 | \left[\prod_{\mathbf{k}} (u^*_{\mathbf{k}} + v^*_{\mathbf{k}} c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow}) \right] \left[\prod_{\mathbf{k}'} \left(u_{\mathbf{k}'} + v_{\mathbf{k}'} c^{\dagger}_{\mathbf{k}',\uparrow} c^{\dagger}_{-\mathbf{k}',\downarrow} \right) \right] |0\rangle$$
$$= \prod_{\mathbf{k}} \langle 0 | \left(u^*_{\mathbf{k}} + v^*_{\mathbf{k}} c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \right) \left(u_{\mathbf{k}} + v_{\mathbf{k}} c^{\dagger}_{\mathbf{k},\uparrow} c^{\dagger}_{-\mathbf{k},\downarrow} \right) |0\rangle.$$

Expanding the product for a single momentum mode $\mathbf{k}$ yields four terms. The cross terms that change the particle number evaluate to zero because the vacuum expectation value of an unequal number of creation and annihilation operators vanishes identically ($\langle 0|cc|0\rangle = 0$ and $\langle 0|c^\dagger c^\dagger|0\rangle = 0$). We are left strictly with the diagonal terms:

$$\begin{aligned}\langle\psi_0|\psi_0\rangle &= \prod_{\mathbf{k}} \left(|u_{\mathbf{k}}|^2 \langle 0|0\rangle + |v_{\mathbf{k}}|^2 \langle 0|c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}|0\rangle \right) \\ &= \prod_{\mathbf{k}} \left(|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 \langle 0|c_{-\mathbf{k},\downarrow}(1 - c^\dagger_{\mathbf{k},\uparrow} c_{\mathbf{k},\uparrow}) c^\dagger_{-\mathbf{k},\downarrow}|0\rangle \right).\end{aligned}$$

Applying the fermionic anti-commutation relations systematically pushes the destruction operators to the right until they annihilate the vacuum. The sequential operation $c_{\mathbf{k},\uparrow} c^\dagger_{\mathbf{k},\uparrow}|0\rangle = |0\rangle$ and $c_{-\mathbf{k},\downarrow} c^\dagger_{-\mathbf{k},\downarrow}|0\rangle = |0\rangle$ leaves only the scalar coefficients:

$$\langle\psi_0|\psi_0\rangle = \prod_{\mathbf{k}} \left(|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 \right) = 1,$$

which is physically guaranteed to equal 1 exactly thanks to the fundamental normalization condition on the coherence factors, $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$, that we rigorously enforced to ensure the Bogoliubov transformation remained unitary.

Remark. ***(A Note on the Density Matrix and the Zero-Temperature Limit)***

Throughout this derivation, we have explicitly utilized the parameter μ, which stems from the fact that we are solving the many-body problem via the partition function in the grand canonical ensemble. In this formalism, the grand partition function is defined as:

$$\mathcal{Z} = Tr\left\{ e^{-\beta \hat{K}} \right\},$$

where $\hat{K} = \hat{H} - \mu \hat{N}$. It is important to remember that the operator $e^{-\beta \hat{K}}$ (or equivalently the normalized thermal state $\hat{\rho} = e^{-\beta \hat{K}}/\mathcal{Z}$) is ***always a density matrix****, and never a single pure state vector at any finite temperature. Therefore, when we discuss a single, pure "ground state" wave function $|\Psi_{GS}\rangle$, we are strictly referring to the zero-temperature limit ($\beta \to \infty$, i.e., $T \to 0$). In this asymptotic limit, only the lowest-energy eigenstate survives, and the thermal operator strictly projects onto the ground-state projector (up to a ground-state energy factor $e^{-\beta C'}$, which completely cancels out in normalized thermal averages):*

$$|\Psi_{GS}\rangle\langle\Psi_{GS}| = \lim_{\substack{\beta \to \infty \\ (T \to 0)}} e^{-\beta \hat{K}}.$$

This limit rigorously justifies our theoretical approach of writing down a pure state $|\Psi_0(\mu)\rangle$ to represent the macroscopic system at absolute zero temperature.

2.4 | Spectrum of the BCS Hamiltonian

2.4.1 | Gap Equation at Zero Temperature

The elementary excitations over the many-body ground state $|\psi_0\rangle$ are created by the Bogoliubov quasiparticle operators $\gamma^\dagger_{\mathbf{k},\sigma}$. These states naturally form the eigenvectors of the diagonalized grand canonical Hamiltonian:

$$\hat{K}\gamma^\dagger_{\mathbf{k},\sigma}|\psi_0\rangle = (E_\mathbf{k} + C')\gamma^\dagger_{\mathbf{k},\sigma}|\psi_0\rangle,$$

with the quasiparticle energy dispersion given by $E_\mathbf{k} = \sqrt{\xi_k^2 + |\Delta_\mathbf{k}|^2}$. We are considering single quasiparticle excitations, which are the lowest-lying excited states of the system. The energy of a single quasiparticle excitation is therefore $E_\mathbf{k}$ above the ground state energy C'.

Consider the physical implications of this dispersion relation: whenever the momentum-dependent parameter $\Delta_\mathbf{k} \neq 0$, the excitation energy satisfies $E_\mathbf{k} > 0$ for all momenta $\mathbf{k}$. Consequently, there exists a strictly positive energy gap between the superconducting ground state and the lowest-lying first excited state (which is determined by the global minimum of $E_\mathbf{k}$). For this fundamental reason, the function $\Delta_\mathbf{k}$ is commonly referred to as the *gap function*.

Recall that the gap function $\Delta_\mathbf{k}$ was originally defined self-consistently through the expectation value of the pair destruction operator, $A_\mathbf{k} = \langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow}\rangle$, which looks like a self-consistency condition for the gap function: we are trying to find a solution for $\Delta_\mathbf{k}$ that is consistent with the expectation value of the pair operator in the BCS ground state. The gap function is therefore determined by the microscopic details of the interaction potential $V(\mathbf{k},\mathbf{k}')$ and the underlying Fermi surface. Let us explicitly compute this thermodynamic average at absolute zero temperature ($T = 0$), meaning we evaluate the expectation value strictly over the BCS ground state. Using the inverse Bogoliubov transformations, we express the bare particle operators in terms of the quasiparticle operators:

$$A_\mathbf{k} = \langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow}\rangle = \left\langle \left(-v_\mathbf{k}\gamma^\dagger_{\mathbf{k},\uparrow} + u^*_\mathbf{k}\gamma_{-\mathbf{k},\downarrow}\right)\left(v_\mathbf{k}\gamma^\dagger_{-\mathbf{k},\downarrow} + u^*_\mathbf{k}\gamma_{\mathbf{k},\uparrow}\right)\right\rangle = u^*_\mathbf{k} v_\mathbf{k},$$

where we have used the fact that $\gamma_{\mathbf{k},\sigma}|\psi_0\rangle = 0$. One can readily verify from the explicit definitions of the coherence factors $u_\mathbf{k}$ and $v_\mathbf{k}$ that their product yields:

$$u^*_\mathbf{k} v_\mathbf{k} = \frac{\Delta_\mathbf{k}}{2E_\mathbf{k}}.$$

Substituting this relation back into the definition of the gap parameter, we obtain the discrete self-consistency equation:

$$\Delta_\mathbf{k} = -\frac{1}{V}\sum_{\mathbf{k}'} V(\mathbf{k},\mathbf{k}')\frac{\Delta_{\mathbf{k}'}}{2E_{\mathbf{k}'}}.$$

Taking the thermodynamic limit, the momentum sum transitions into an integral, yielding the celebrated BCS gap equation:

$$\Delta_\mathbf{k} = -\int \frac{\mathrm{d}^3\mathbf{k}'}{(2\pi)^3}\frac{V(\mathbf{k},\mathbf{k}')}{2E_{\mathbf{k}'}}\Delta_{\mathbf{k}'}. \tag{2.4.1}$$

This is a non-linear integral equation for the gap function $\Delta_\mathbf{k}$. Because $E_{\mathbf{k}'}$ itself depends on $\Delta_{\mathbf{k}'}$, it generally requires a numerical solution. However, in the physically relevant limit of weak interactions (which accurately models several conventional metallic superconductors), the gap equation can be solved analytically. This condition is realted to the many-body gap, which is essentially the energy cost to insert a single quasiparticle into the system. We are still considering momenta in the vicinity of the Fermi surface, where the gap function is non-zero. The energy cost to insert a single quasiparticle is therefore the minimum of the dispersion $\min_k E_\mathbf{k} = |\Delta_\mathbf{k}|$.

Weak Coupling Solution for BCS Model of Interaction

Let us first consider the effective phonon-mediated potential characteristic of a conventional superconductor. We approximate it as a separable, attractive well:

$$V(\mathbf{k},\mathbf{k}') = -g w_k w_{k'},$$

with a positive, constant coupling strength $g = g_{e,ph}^2 > 0$. We are implying that adding a quasiparticle to the system costs energy, and this cost is captured by this very rough mean-field approximation of the potential. The momentum dependence is captured by the step function $w_k = \Theta(\hbar\omega_D - |\xi_k|)$, which restricts the interaction to a narrow energy shell of width $2\hbar\omega_D$ around the Fermi surface. Here, the kinetic energy relative to the chemical potential is $\xi_k = \epsilon_k - \mu$. Note that the chemical potential μ is approximately equal to the non-interacting Fermi energy E_F (the exact equality is slightly broken by the interaction, but $\mu \simeq E_F$ holds to leading order).

To solve the integral gap equation (2.4.1), we proceed with a piecewise ansatz for the gap function that mirrors the symmetry of the potential:

$$\Delta_{\mathbf{k}} = \Delta_k = \Delta w_k.$$

Substituting this ansatz and the explicit form of the interaction potential into (2.4.1), we obtain:

$$\Delta w_k = g \int \frac{\mathrm{d}^3\mathbf{k}'}{(2\pi)^3} \frac{w_k w_{k'} \Delta w_{k'}}{2E_{\mathbf{k}'}}.$$

Factoring out Δw_k (assuming a non-trivial solution for the many body factor, $\Delta \neq 0$, and remembering that the dependence on Δ is kept inside the energy denominator), the self-consistency condition simplifies to:

$$\frac{1}{g} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{\Theta(\hbar\omega_D - |\xi_k|)}{2E_k}.$$

In the relevant region of integration specified by the Heaviside step function Θ, the quasiparticle dispersion is:

$$E_{\mathbf{k}} = E_k = \sqrt{\xi_k^2 + |\Delta|^2}.$$

The minimum of this dispersion is $\min_k E_k = |\Delta|$, which is reached exactly at the Fermi surface where $\xi_k = 0$ (corresponding to $k = \sqrt{2m\mu}/\hbar \simeq k_F$). Notice that the gap equation only dictates the modulus of the gap $|\Delta|$, leaving the global phase undetermined. Without loss of generality, we can therefore assume Δ to be strictly real and positive.

To evaluate the momentum integral, we transition to an energy integration for convenience by introducing the single-particle density of states $N(\xi)$ *per spin component*[3] and per unit volume:

$$N(\xi) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \delta(\xi - \xi_k) = \frac{m^{3/2}}{\sqrt{2}\pi^2\hbar^3} \sqrt{\xi + \mu}, \tag{2.4.2}$$

where the last equality is obtained by explicitly evaluating the delta function integral in spherical coordinates. Physically, the density of states $N(\xi)$ counts the number of single-particle states per unit energy interval and per unit volume. Using this definition, the gap equation transforms into:

$$\frac{1}{g} = \int_{-\mu}^{+\infty} d\xi \frac{N(\xi)}{2E_\xi} \Theta(\hbar\omega_D - |\xi|),$$

[3] We are ignoring the spin degeneracy in this discussion.

where $E_\xi = \sqrt{\xi^2 + |\Delta|^2}$, and the integral is still counting the volume of the phase space (in energy, the density of states takes the role of a weight function). Because the Debye energy is typically orders of magnitude smaller than the Fermi energy ($\hbar\omega_D \ll E_F \approx \mu$), the lower limit of integration $-\mu$ can be safely replaced by the step function's cutoff $-\hbar\omega_D$:

$$\frac{1}{g} = \int_{-\hbar\omega_D}^{\hbar\omega_D} d\xi \frac{N(\xi)}{2E_\xi}.$$

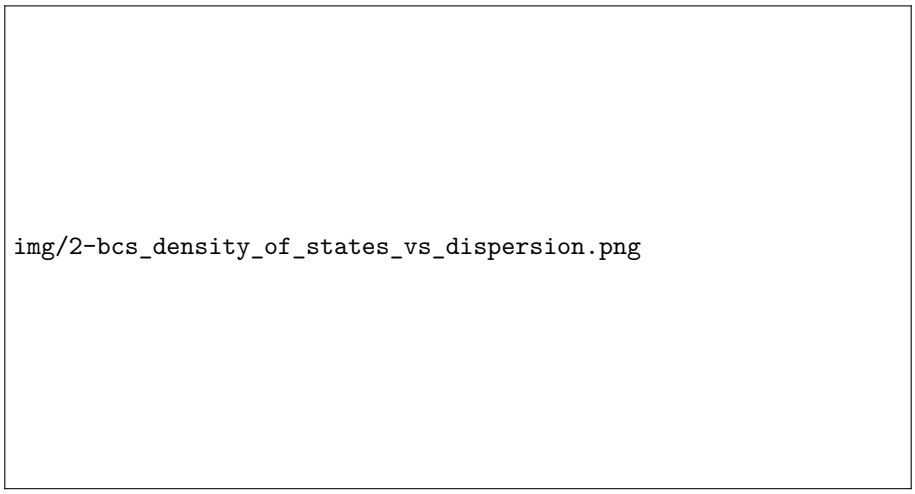


Figure 2.7: Plot of the single-particle density of states $N(\xi)$ and the inverse of the quasiparticle dispersion $1/E_\xi$, where $E_\xi = \sqrt{\xi^2 + |\Delta|^2}$. The shaded region visually represents the narrow integration shell defined by the Debye energy cutoff $\pm\hbar\omega_D$. Within this small window around the Fermi surface ($\xi = 0$), $1/E_\xi$ is strongly peaked, whereas the density of states varies negligibly and can be robustly approximated by its constant value at the Fermi level, $N_0 = N(0)$.

We must now carefully analyze the weak-coupling limit, where $g \to 0$. For the equation to hold, the right-hand side integral must correspondingly diverge towards $+\infty$. This logarithmic divergence is only possible if the gap parameter itself vanishes, $\Delta \to 0$, ensuring the integrand $1/E_\xi \to \infty$ near the Fermi surface ($\xi = 0$). Under these weak-coupling conditions, the function $1/E_\xi$ is extremely sharply peaked at $\xi = 0$ and varies rapidly over an energy scale defined by Δ (the peak depends on the value of Δ). Conversely, over this exact same narrow energy shell, the density of states $N(\xi)$ is nearly constant and can be reliably approximated by its value precisely at the Fermi level, $N_0 \equiv N(\xi = 0)$. Notice that when we say $\Delta, g \to 0$, we mean that both parameters approach zero but never actually reach it. The limit is taken in such a way that the ratio Δ/g remains finite and well-defined.

Pulling the constant N_0 out of the integral, we evaluate the remaining expression analytically:

$$\frac{1}{g} = \frac{N_0}{2} \int_{-\hbar\omega_D}^{\hbar\omega_D} \frac{\mathrm{d}\xi}{\sqrt{\xi^2 + \Delta^2}} = N_0 \int_0^{\hbar\omega_D} \frac{1}{\Delta\sqrt{(\frac{\xi}{\Delta})^2 + 1}} \mathrm{d}\xi = N_0 \mathrm{arcsinh}\left(\frac{\hbar\omega_D}{\Delta}\right).$$

Inverting this relation yields:

$$\frac{\hbar\omega_D}{\Delta} = \sinh\left(\frac{1}{N_0 g}\right) = \frac{e^{\frac{1}{N_0 g}} - e^{-\frac{1}{N_0 g}}}{2} \sim \frac{1}{2}\exp\left(\frac{1}{N_0 g}\right),$$

where the final asymptotic approximation relies on the behavior of $\sinh(x) \sim e^x/2$ for large arguments $x = 1/(N_0 g)$. Solving for the physical gap parameter, we obtain the seminal weak-coupling BCS result:

$$\Delta = 2\hbar\omega_D e^{-\frac{1}{N_0 g}}. \tag{2.4.3}$$

Remark. *The exponential form of (2.4.3) carries profound physical implications. At $T = 0$, even an infinitesimally small attractive interaction ($g > 0$) is sufficient to open a finite (albeit exponentially suppressed) energy gap Δ at the Fermi surface. Furthermore, the function $e^{-1/(N_0 g)}$ exhibits an essential singularity at $g = 0$. This signifies that the superconducting state cannot be reached via standard perturbative expansions starting from the non-interacting Fermi gas; the Cooper instability is a fundamentally non-perturbative phenomenon. The dimensionless parameter gN_0 acts as the effective coupling strength, which is typically quite small in conventional phonon-mediated superconductors.*

The fact that now we have the Debye frequency ω_D in the prefactor of the gap equation makes this a real many-body effect. The Debye frequency is a material property that is determined by the phonon spectrum of the solid. It is not a property of the electrons themselves, but rather of the lattice vibrations, stiffness, and the speed of sound in the material (everything is then related to the mass of the ions). The fact that the gap depends on this frequency shows that the superconducting state is a collective effect that arises from the interaction between electrons and the lattice vibrations.

Weak Coupling Solution for Ultracold Fermi Gases

We now shift our focus to ultracold Fermi gases, where the effective interaction is characterized by a short-range contact potential. In momentum space, this is modeled as $V(\mathbf{k}, \mathbf{k}') = v_0 w_k w_{k'}$, with the cutoff function $w_k = \Theta(k_0 - k)$. We are physically interested in the limit where the ultraviolet cutoff momentum is taken to infinity, $k_0 \to \infty$. To ensure physical observables remain well-defined, the bare coupling strength v_0 must be simultaneously rescaled (renormalized) in order to hold the low-energy s-wave scattering length a constant.

TODO: add image of omega as cutoff function

Applying the same piecewise ansatz for the gap function, $\Delta_{\mathbf{k}} = \Delta_k = \Delta w_k$, the generalized gap equation (2.4.1) becomes:

$$\Delta w_k = -v_0 \int \frac{\mathrm{d}^3\mathbf{k}'}{(2\pi)^3} \frac{w_k w_{k'} \Delta w_{k'}}{2E_{k'}}.$$

From now on we repeat the same computation done for the superconducing case. Dividing by Δw_k yields the bare relation:

$$-\frac{1}{v_0} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{w_k^2}{2E_k}.$$

To eliminate the divergent bare coupling v_0 in favor of the physical scattering length a, we formally add and subtract an identical free-particle integral containing the bare kinetic energy $\epsilon_k = \hbar^2 k^2/(2m)$:

$$-\frac{1}{v_0} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \left(\frac{1}{2E_k} - \frac{1}{2\epsilon_k} \right) w_k^2 + \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{w_k^2}{2\epsilon_k}.$$

We recognize that the bare coupling and the last integral term are exactly related by the Lippmann-Schwinger equation for the T-matrix at zero energy, which defines the running of v_0 with the scattering length a:

$$-\frac{m}{4\pi\hbar^2 a} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \left(\frac{1}{2E_k} - \frac{1}{2\epsilon_k} \right) w_k^2.$$

Because the integrand in the parenthesis now scales as k^{-4} for large momenta, the integral is explicitly ultraviolet convergent. We can therefore safely remove the artificial regularization by taking the limit $k_0 \to \infty$, yielding the fully regularized gap equation:

$$-\frac{m}{4\pi\hbar^2 a} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\left(\frac{1}{2E_k} - \frac{1}{2\epsilon_k}\right). \tag{2.4.4}$$

This convergent integral can be evaluated analytically in terms of complete elliptic integrals. Let us consider the specific regime of a weak attractive interaction, characterized by a small and negative scattering length $a \to 0^-$. In this BCS-like limit, the chemical potential approaches the non-interacting Fermi energy ($\mu \simeq E_F$) and the gap is small ($\Delta \ll \mu$). The regularized integration yields the explicit asymptotic solution:

$$\Delta = \frac{8E_F}{e^2} e^{\frac{\pi}{2k_F a}}. \tag{2.4.5}$$

This profound result explicitly dictates how the pairing gap becomes exponentially suppressed in the limit $k_F a \to 0^-$. Physically, adding particles to the system costs energy, and this cost is captured by this very rough mean-field approximation of the potential. The gap is exponentially small in the weak-coupling limit, but it is non-zero for any finite attractive interaction, indicating that the system is unstable.

It is structurally insightful to compare this ultracold gas result with the solid-state BCS expression (2.4.3). Recall that the 3D density of states at the Fermi surface is $N_0 = (mk_F)/(2\pi^2\hbar^2)$. If we identify the effective BCS coupling parameter with the first-order Born approximation of the scattering amplitude, $g = -4\pi\hbar^2 a/m$, the exponential dependence $e^{\pi/(2k_F a)}$ strictly coincides with the BCS term $e^{-1/(N_0 g)}$. This mapping perfectly reproduces the exponent; however, retaining the full $1/(2\epsilon_k)$ counter-term in the integral regularization—rather than relying strictly on the lowest-order Born approximation—is mathematically critical for capturing the exact $8/e^2$ numerical prefactor governing the overall energy scale of the pairing gap.

Comparison with Spectrum of the Weak Interacting Bose Gas

TODO: complete, graph and compare

In the case of a weakly interacting Bose gas, the spectrum of elementary excitations is determined by the Bogoliubov dispersion relation:...

[drawing of "BCS wavefunction" as a product of pairs of fermions in momentum space, with arrows indicating the pairing of opposite momenta and spins]

In order to create an excitation, we have to add momenta to an electron inside a cooper pair, but this cannot be done continuously because the two electrons are bound together. The energy cost to break a pair is the gap Δ. In contrast, in a weakly interacting Bose gas, the Bogoliubov spectrum is gapless, and excitations can be created with arbitrarily small energy by adding small momenta to the condensate. This fundamental difference in the excitation spectrum leads to distinct physical properties between fermionic superfluids and bosonic condensates.

the cooper pairs can be broke releasing energy or absorbing energy interacting with the phonons (we have ignored them for a while but they are still in the description).

transition to the landau criterion...

2.4.2 | Elementary Excitations and Landau Criterion

All the excited states of the grand-canonical Hamiltonian $\hat{K}$ are systematically obtained by acting with the quasiparticle creation operator $\gamma^\dagger_{\mathbf{k},\sigma}$ on the many-body ground state $|\psi_0(\mu)\rangle$. The energy dispersion of these elementary excitations in a BCS superconductor or superfluid is thus given by $E_k = \sqrt{\xi_k^2 + |\Delta|^2}$.

To fully appreciate the physical consequences of the Bogoliubov quasiparticle spectrum derived from the zero-temperature gap equation, it is highly instructive to compare the dispersion profile $\varepsilon = E_{\mathbf{k}}$ as a function of the momentum magnitude $k = |\mathbf{k}|$ for a Weakly Interacting Bose Gas (WIBG) against that of a BCS superconductor:

- **Weakly Interacting Bose Gas (WIBG):** Via the bosonic Bogoliubov transformation, the elementary excitations are created by the bosonic quasiparticle operators $\alpha^\dagger_{\mathbf{k}}$ and follow the gapless dispersion relation:

  $$E^{\text{WIBG}}_{\mathbf{k}} = \sqrt{(\epsilon_{\mathbf{k}} + nU_0)^2 - (nU_0)^2} = \sqrt{\epsilon^2_{\mathbf{k}} + 2nU_0\epsilon_{\mathbf{k}}},$$

  which vanishes linearly as $E^{\text{WIBG}}_{\mathbf{k}} \simeq c\hbar k$ in the long-wavelength limit ($k \to 0$, representing the phononic Goldstone mode with sound velocity $c = \sqrt{nU_0/m}$) and smoothly crosses over to the quadratic free-particle behavior $\epsilon_{\mathbf{k}} + nU_0 \sim \frac{\hbar^2 k^2}{2m}$ at large momenta.

- **BCS Superconductor:** By contrast, in a BCS superconductor the elementary excitations are created by the fermionic Bogoliubov quasiparticle operators $\gamma^\dagger_{\mathbf{k},\sigma}$ and are governed by the strictly gapped dispersion:

  $$E^{\text{BCS}}_{\mathbf{k}} = \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}, \quad \text{with} \quad \xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu.$$

  At $k = 0$, the excitation energy starts at a finite value. Taking into account the momentum profile of the gap around the active energy shell, the dispersion exhibits a characteristic non-monotonic shape that reaches its absolute minimum Δ at a finite momentum $k_{\min} \neq 0$ (located precisely at the Fermi surface $k_{\min} \simeq k_F$, where $\xi_{\mathbf{k}} = 0$ and the phonon-mediated attractive interaction within the Debye window $\hbar\omega_D$ is active), before growing quadratically at large momenta ($k \gg k_{\min}$). In order to create a single-particle excitation, one must impart momentum to an electron bound within a Cooper pair; this process cannot occur continuously from zero energy because the pair is bound together, requiring a minimum threshold energy of Δ (or 2Δ to break a pair into two quasiparticles) via the absorption or emission of lattice phonons.

By applying **Landau's criterion for superfluidity** to this specific dispersion relation $E_{\mathbf{k}}$, we look geometrically for the minimum of the ratio $E_k/(\hbar k)$. Because the excitation spectrum is strictly gapped around $k_{\min} \neq 0$, the straight line starting from the origin and tangent to the minimum of the dispersion curve possesses a strictly positive slope:

$$v_c = \min_{\mathbf{k}} \frac{E_{\mathbf{k}}}{\hbar k} \simeq \frac{\Delta}{\hbar k_F} > 0,$$

which holds in the physically relevant regime where $\Delta \ll E_F$. For flow velocities below v_c, no elementary excitations can be kinematically nucleated, and the system supports dissipationless flow without friction. In a charged system, this directly translates into the complete vanishing of electrical resistivity ($\rho = 0$) below the critical temperature T_c, contrasting sharply with the normal

TODO: add image of a comparison plot between the elementary excitation spectrum of a weakly interacting Bose gas and a BCS superconductor showing the linear bounding slope from the origin for the Landau critical velocity

metallic state where resistivity remains finite and typically scales as T^3 (or T^5 in the standard Bloch-Grüneisen electron-phonon regime) at low temperatures before dropping abruptly to zero at the superconducting transition. We thus arrive at the striking physical conclusion that BCS theory combined with the Landau kinematic criterion rigorously predicts that an attractively interacting Fermi gas fundamentally behaves as a superfluid (if neutral) or a superconductor (if charged) at $T = 0$: *superconductors are superfluids of Cooper pairs.*

Comparison of Ground-State Wave Functions and the Role of Quantum Statistics

To truly appreciate what it means for a superconductor to behave as a superfluid of Cooper pairs, we must address a subtle and seemingly counterintuitive question regarding the quantum statistics of the underlying pairs. Let us compare the explicit algebraic structure of the ground-state wave function of a BCS superconductor with that of a Weakly Interacting Bose Gas:

1. **BCS Ground-State Wave Function:** Introducing the Cooper-pair creation operator $D^\dagger_{\mathbf{k}} = c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}$, which creates a pair of fermions with opposite momenta and opposite spins $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$, the variational BCS ground state reads:

$$|\Psi_{\mathrm{BCS}}\rangle = \prod_{\mathbf{k}} \left(A_{\mathbf{k}} + B_{\mathbf{k}} D^\dagger_{\mathbf{k}} \right) |0\rangle, \quad \text{where} \quad \left(D^\dagger_{\mathbf{k}}\right)^2 = 0, \tag{2.4.6}$$

   with variational coherence coefficients $A_{\mathbf{k}} = u_{\mathbf{k}}$ and $B_{\mathbf{k}} = v_{\mathbf{k}}$ normalized such that $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$. Crucially, because the constituent particles are fermions obeying canonical anti-commutation relations $\{c^\dagger_{\mathbf{k},\sigma}, c^\dagger_{\mathbf{k}',\sigma'}\} = 0$, the pair creation operator is strictly **nilpotent**:

$$\left(D^\dagger_{\mathbf{k}}\right)^2 = c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} = -\left(c^\dagger_{\mathbf{k},\uparrow}\right)^2 \left(c^\dagger_{-\mathbf{k},\downarrow}\right)^2 = 0.$$

   Physically, the Pauli exclusion principle strictly forbids placing two identical fermions—and hence two Cooper pairs with the exact same relative momentum $\mathbf{k}$—into the same single-particle quantum state.

2. **Weakly Interacting Bose Gas (WIBG) Ground-State Wave Function:** By contrast, recall how the ground state of a Weakly Interacting Bose Gas is constructed in terms of the elementary bosonic creation operators $a^\dagger_{\mathbf{k}}$. Its coherent-like structure over the momentum modes is summarized schematically as $|\Psi_{\mathrm{WIBG}}\rangle = \prod_{\mathbf{k}} e^{\alpha a^\dagger_{\mathbf{k}}} |0\rangle$, which encapsulates the exact Bogoliubov ground-state wave function combining the coherent state for the $\mathbf{k} = \mathbf{0}$ condensate mode with the exponential of bosonic pair creation operators for $\mathbf{k} \neq \mathbf{0}$:

$$|\Psi_{\mathrm{WIBG}}\rangle = \frac{1}{\mathcal{N}} e^{\sqrt{N_0} a^\dagger_{\mathbf{0}}} \exp\left(-\frac{1}{2} \sum_{\mathbf{k}\neq\mathbf{0}} \frac{v_{\mathbf{k}}}{u_{\mathbf{k}}} a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}} \right) |0\rangle = \frac{1}{\mathcal{N}} e^{\sqrt{N_0} a^\dagger_{\mathbf{0}}} \prod_{\mathbf{k}\neq\mathbf{0}} \exp\left(-\frac{1}{2} \frac{v_{\mathbf{k}}}{u_{\mathbf{k}}} a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}} \right) |0\rangle.$$

Comparing the operators generating the two ground states highlights a fundamental algebraic distinction between Cooper pairs and true elementary bosons:

$$\left(D^\dagger_{\mathbf{k}}\right)^2 = 0 \quad \longleftrightarrow \quad a^2_{\mathbf{k}} \neq 0 \quad \left(\text{and } (a^\dagger_{\mathbf{k}})^2 \neq 0\right).$$

In a Bose gas, the operators $a^\dagger_{\mathbf{k}}$ (as well as the bosonic pair operators $a^\dagger_{\mathbf{k}} a^\dagger_{-\mathbf{k}}$) can be applied an arbitrary number of times n to the vacuum without vanishing, generating all higher-order terms

$\frac{1}{n!}(a_{\mathbf{k}}^\dagger)^n$ in the Taylor expansion of the exponential. In a BCS superconductor, instead, the hard-core fermionic constraint $(D_{\mathbf{k}}^\dagger)^2 = 0$ truncates the exponential series $\exp\left(\frac{v_{\mathbf{k}}}{u_{\mathbf{k}}} D_{\mathbf{k}}^\dagger\right) = 1 + \frac{v_{\mathbf{k}}}{u_{\mathbf{k}}} D_{\mathbf{k}}^\dagger$ at first order, meaning that a given $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ mode can accommodate at most one Cooper pair.

Given that $(D_{\mathbf{k}}^\dagger)^2 = 0$ prevents multiple occupation of the same relative momentum state, a natural question arises: *How can we legitimately invoke Landau's criterion—which was originally derived for a Bose gas—to conclude that a BCS superconductor is a superfluid of Cooper pairs?* The resolution of this apparent paradox lies in a fundamental physical principle: **Landau's criterion does not care about quantum statistics; it is based exclusively on the excitation spectrum and energetics**. To see why rigorously, let us recall how Landau's criterion is constructed:

- **Single-Excitation Energetics:** In the reference frame moving with the fluid, the system starts in its many-body ground state $|\Psi_0\rangle$ with energy $\mathcal{E}_0$. To determine the onset of dissipation (friction), Landau's kinematic argument simply asks whether it is energetically favorable to create a *single extra elementary excitation* of momentum $\hbar\mathbf{k}$ and energy $E_{\mathbf{k}}$, which shifts the energy in the laboratory frame by $\Delta E = E_{\mathbf{k}} + \hbar\mathbf{k}\cdot\mathbf{v}$.
- **Indistinguishability of Statistics for a Single Quasiparticle:** In the Weakly Interacting Bose Gas, creating a single elementary excitation above the ground state corresponds to acting once with the bosonic Bogoliubov operator $\alpha_{\mathbf{k}}^\dagger|\Psi_{\mathrm{WIBG}}\rangle$. In the BCS superconductor, creating a single elementary excitation corresponds to acting once with the fermionic Bogoliubov quasiparticle operator $\gamma_{\mathbf{k},\sigma}^\dagger|\Psi_{\mathrm{BCS}}\rangle$. As long as we consider the creation of a **single excitation** (one quasiparticle), it is physically and mathematically impossible to distinguish whether that single excitation obeys bosonic or fermionic statistics.
- **When Would Statistics Matter?** Quantum statistics (Fermi-Dirac versus Bose-Einstein, or Pauli blocking) only manifests itself when we attempt to populate *multiple* particles or excitations in the exact same quantum state. If the stability criterion for superfluidity had required the simultaneous creation of a macroscopic multiparticle state in a single mode, statistics would have played a decisive role. However, because the threshold for dissipation is governed entirely by the kinematic cost of nucleating the *first* single elementary excitation, Landau's criterion depends exclusively on the single-quasiparticle dispersion relation $E_{\mathbf{k}}$ (purely energetics), remaining completely blind to the underlying quantum statistics.

Momentum Distribution Function

The average number of fermions with momentum $\mathbf{k}$ and spin σ in the ground state $|\psi_0\rangle$ (at $T = 0$) is defined by the expectation value of the bare number operator:

$$n_{\mathbf{k},\sigma} = \langle c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma}\rangle.$$

Experimentally, this quantity can be probed directly in ultracold atomic gases via time-of-flight (TOF) expansion measurements: when the trapping potential is abruptly switched off, the gas expands freely, and in the ballistic regime the final spatial positions of the independent atoms map directly onto their initial momenta, rendering the imaged spatial density profile directly proportional to the momentum distribution $n_{\mathbf{k},\sigma}$. For an ideal non-interacting Fermi gas at $T = 0$, this distribution is a sharp Heaviside step function $\Theta(k_F - k)$; in the interacting BCS state, we expect to recover this exact step behavior everywhere outside the narrow energy shell surrounding the Fermi surface, within which the emergent many-body pairing correlations will smear the distribution.

To evaluate $n_{\mathbf{k},\sigma}$ analytically, we utilize the inverse Bogoliubov transformations, which explicitly dictate the relations between the bare particle operators $c_{\mathbf{k},\sigma}, c^\dagger_{\mathbf{k},\sigma}$ and the Bogoliubov quasiparticle operators $\gamma_{\mathbf{k},\sigma}, \gamma^\dagger_{\mathbf{k},\sigma}$:

$$c_{\mathbf{k},\uparrow} = u^*_{\mathbf{k}}\gamma_{\mathbf{k},\uparrow} + v_{\mathbf{k}}\gamma^\dagger_{-\mathbf{k},\downarrow}, \quad c_{\mathbf{k},\downarrow} = u^*_{\mathbf{k}}\gamma_{\mathbf{k},\downarrow} - v_{\mathbf{k}}\gamma^\dagger_{-\mathbf{k},\uparrow}.$$

Taking the Hermitian conjugate yields the corresponding creation operators:

$$c^\dagger_{\mathbf{k},\uparrow} = u_{\mathbf{k}}\gamma^\dagger_{\mathbf{k},\uparrow} + v^*_{\mathbf{k}}\gamma_{-\mathbf{k},\downarrow}, \quad c^\dagger_{\mathbf{k},\downarrow} = u_{\mathbf{k}}\gamma^\dagger_{\mathbf{k},\downarrow} - v^*_{\mathbf{k}}\gamma_{-\mathbf{k},\uparrow}.$$

Because the BCS ground state is defined as the quasiparticle vacuum ($\gamma_{\mathbf{k},\sigma}|\psi_0\rangle = 0$), only the terms involving anti-normal-ordered quasiparticle operators $\gamma\gamma^\dagger$ contribute to the expectation value. We thus obtain:

$$\langle c^\dagger_{\mathbf{k},\uparrow}c_{\mathbf{k},\uparrow}\rangle = \langle v^*_{\mathbf{k}}v_{\mathbf{k}}\gamma_{-\mathbf{k},\downarrow}\gamma^\dagger_{-\mathbf{k},\downarrow}\rangle = |v_{\mathbf{k}}|^2, \quad \langle c^\dagger_{\mathbf{k},\downarrow}c_{\mathbf{k},\downarrow}\rangle = \langle(-v^*_{\mathbf{k}})(-v_{\mathbf{k}})\gamma_{-\mathbf{k},\uparrow}\gamma^\dagger_{-\mathbf{k},\uparrow}\rangle = |v_{\mathbf{k}}|^2.$$

Recalling the explicit analytical expression for the coherence factor $v_{\mathbf{k}}$, one obtains the ground-state momentum distribution:

$$n_{\mathbf{k},\sigma} = |v_{\mathbf{k}}|^2 = \frac{1}{2}\left(1 - \frac{\xi_k}{E_k}\right). \tag{2.4.7}$$

TODO: add image of a plot showing the momentum distribution function as a function of momentum displaying a smeared Fermi step near the Fermi momentum compared to the sharp step of a non interacting Fermi gas

In the weak-coupling regime where $\Delta \ll E_F$, the pairing gap is much smaller than the relative kinetic energy $|\xi_k|$, except within a very narrow momentum window of width $\delta k \sim (m\Delta)/(\hbar^2 k_F)$ centered directly around the Fermi momentum k_F. Outside this specific window, the dispersion smoothly approximates the absolute value of the kinetic energy, $E_k = \sqrt{\xi_k^2 + |\Delta|^2} \approx |\xi_k|$. Consequently, the distribution sharply approaches $n_{\mathbf{k},\sigma} \simeq 1$ for states deep inside the Fermi sea ($\xi_k < 0$), or $n_{\mathbf{k},\sigma} \simeq 0$ for states far above it ($\xi_k > 0$). One thus naturally recovers a momentum distribution that essentially coincides with the standard Heaviside step distribution of an ideal non-interacting Fermi gas at $T = 0$, with the crucial modification of a continuous smearing confined to a small momentum window around k_F whose width is governed directly by the magnitude of the pairing gap Δ.

2.4.3 | Gap Equation at Finite Temperature

At absolute zero temperature ($T = 0$), the system resides in a pure quantum state—the BCS ground state $|\psi_0\rangle$—and the anomalous Cooper-pair amplitude $\langle c_{-\mathbf{k},\downarrow}c_{\mathbf{k},\uparrow}\rangle$ entering the gap function $\Delta_{\mathbf{k}}$ was evaluated using the vacuum property $\gamma_{\mathbf{k},\sigma}|\psi_0\rangle = 0$, which implies $\langle\gamma^\dagger_{\mathbf{k},\sigma}\gamma_{\mathbf{k},\sigma}\rangle = 0$.

At any finite temperature ($T > 0$), we can no longer evaluate expectation values over a single pure wave function because thermal fluctuations excite the system into a **statistical mixed state**. Consequently, pure ground-state averages must be rigorously replaced by thermal averages computed via the grand-canonical density matrix:

$$\langle\hat{O}\rangle = \frac{\mathrm{Tr}\left(e^{-\beta\hat{K}}\hat{O}\right)}{\mathcal{Z}}, \quad \text{with} \quad \mathcal{Z} = \mathrm{Tr}\left(e^{-\beta\hat{K}}\right).$$

To characterize this mixed state for $T > 0$, recall that our diagonalized grand-canonical Hamiltonian operator $\hat{K} = \hat{H} - \mu\hat{N}$ takes the form:

$$\hat{K} = \sum_{\mathbf{k},\sigma} E_{\mathbf{k}}\gamma^\dagger_{\mathbf{k},\sigma}\gamma_{\mathbf{k},\sigma} + C',$$

where the scalar constant offset C' identically cancels out between the numerator and denominator in thermal averages, leaving the effective operator $\hat{K}' = \sum_{\mathbf{k},\sigma} E_{\mathbf{k}}\gamma^\dagger_{\mathbf{k},\sigma}\gamma_{\mathbf{k},\sigma}$.

Remark. *Notice that $\hat{K}'$ formally describes a gas of* ***independent, non-interacting fermionic quasiparticles****, coinciding structurally with the Hamiltonian of an ideal Fermi gas, with the critical distinction that the bare single-particle dispersion $\xi_k = \epsilon_k - \mu$ is replaced by the superconducting quasiparticle dispersion $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$.*

With this direct structural identification, we conclude immediately that in the thermal mixed state at $T > 0$, the average occupation of the fermionic quasiparticle modes is governed by the standard **Fermi-Dirac distribution**:

$$\langle \gamma_{\mathbf{k},\sigma}^\dagger \gamma_{\mathbf{k},\sigma} \rangle = f(E_{\mathbf{k}}) = \frac{1}{e^{\beta E_{\mathbf{k}}} + 1},$$

where $\beta = 1/(k_B T)$, and $f(E_{\mathbf{k}})$ is independent of the spin index σ because the dispersion $E_{\mathbf{k}}$ is spin-independent. Notice that this relation holds in any spatial dimension; at any finite temperature $T > 0$, every quasiparticle state of arbitrary energy $E_{\mathbf{k}}$ acquires a small but strictly non-zero thermal occupation.

Evaluation of the Finite-Temperature Anomalous Pairing Average

We now re-evaluate the thermal expectation value of the anomalous Cooper-pair operator at finite temperature by expressing the bare fermionic operators in terms of the Bogoliubov quasiparticle operators via the inverse Bogoliubov transformation:

$$\langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \rangle = \left\langle \left(-v_{\mathbf{k}} \gamma_{\mathbf{k},\uparrow}^\dagger + u_{\mathbf{k}}^* \gamma_{-\mathbf{k},\downarrow} \right) \left(u_{\mathbf{k}}^* \gamma_{\mathbf{k},\uparrow} + v_{\mathbf{k}} \gamma_{-\mathbf{k},\downarrow}^\dagger \right) \right\rangle.$$

(Note that since the global phase can be chosen such that $\Delta_{\mathbf{k}}$ and the coherence factors $u_{\mathbf{k}}, v_{\mathbf{k}}$ are real, the expectation values $\langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \rangle = A_{\mathbf{k}}$ and $\langle c_{\mathbf{k},\uparrow}^\dagger c_{-\mathbf{k},\downarrow}^\dagger \rangle = A_{\mathbf{k}}^*$ coincide.) Expanding the product, the off-diagonal cross terms $\langle \gamma_{\mathbf{k},\uparrow}^\dagger \gamma_{-\mathbf{k},\downarrow}^\dagger \rangle$ and $\langle \gamma_{-\mathbf{k},\downarrow} \gamma_{\mathbf{k},\uparrow} \rangle$ vanish identically because the diagonal thermal density matrix $e^{-\beta \hat{K}'}$ strictly conserves the quasiparticle number. Using the canonical fermionic anticommutation relation $\langle \gamma_{-\mathbf{k},\downarrow} \gamma_{-\mathbf{k},\downarrow}^\dagger \rangle = 1 - \langle \gamma_{-\mathbf{k},\downarrow}^\dagger \gamma_{-\mathbf{k},\downarrow} \rangle = 1 - f(E_{\mathbf{k}})$, we obtain step by step:

$$\begin{aligned}
\langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \rangle &= -v_{\mathbf{k}} u_{\mathbf{k}}^* \langle \gamma_{\mathbf{k},\uparrow}^\dagger \gamma_{\mathbf{k},\uparrow} \rangle + u_{\mathbf{k}}^* v_{\mathbf{k}} \langle \gamma_{-\mathbf{k},\downarrow} \gamma_{-\mathbf{k},\downarrow}^\dagger \rangle \\
&= -v_{\mathbf{k}} u_{\mathbf{k}}^* f(E_{\mathbf{k}}) + u_{\mathbf{k}}^* v_{\mathbf{k}} [1 - f(E_{\mathbf{k}})] \\
&= u_{\mathbf{k}}^* v_{\mathbf{k}} [1 - 2 f(E_{\mathbf{k}})] \\
&= u_{\mathbf{k}}^* v_{\mathbf{k}} \left(1 - \frac{2}{e^{\beta E_{\mathbf{k}}} + 1} \right) = u_{\mathbf{k}}^* v_{\mathbf{k}} \left(\frac{e^{\beta E_{\mathbf{k}}} - 1}{e^{\beta E_{\mathbf{k}}} + 1} \right) \\
&= u_{\mathbf{k}}^* v_{\mathbf{k}} \left(\frac{e^{\beta E_{\mathbf{k}}/2} - e^{-\beta E_{\mathbf{k}}/2}}{e^{\beta E_{\mathbf{k}}/2} + e^{-\beta E_{\mathbf{k}}/2}} \right) \frac{e^{\beta E_{\mathbf{k}}/2}}{e^{\beta E_{\mathbf{k}}/2}} = u_{\mathbf{k}}^* v_{\mathbf{k}} \tanh \left(\frac{\beta E_{\mathbf{k}}}{2} \right).
\end{aligned}$$

TODO: add image of a plot showing the profile of the thermal hyperbolic tangent factor as a function of inverse temperature

Let us verify the physical consistency of this thermal factor across the two opposite temperature limits:

- **Zero-Temperature Limit ($T \to 0, \beta \to \infty$):** In this asymptotic regime, $\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) \xrightarrow{\beta \to \infty} 1$, which smoothly recovers the exact zero-temperature BCS ground-state expectation value $\langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \rangle \xrightarrow{T \to 0} u_{\mathbf{k}}^* v_{\mathbf{k}} = \frac{\Delta_{\mathbf{k}}}{2E_{\mathbf{k}}}$.
- **High-Temperature Limit ($T \to \infty, \beta \to 0$):** Conversely, at extreme temperatures, $\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) \xrightarrow{\beta \to 0} 0$, causing the pairing amplitude to vanish: $\langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \rangle \xrightarrow{T \to \infty} 0$. Physically, when thermal fluctuations $k_B T$ far exceed the pairing gap protecting the Cooper pairs,

all quasiparticle states become equally populated and destructive interference completely wipes out the pairing amplitude (in fact, the superconducting phase and the mean-field description break down well before $\beta \to 0$, as the gap already vanishes continuously at the critical temperature $T = T_c$).

Self-Consistent BCS Gap Equation at Finite Temperature and Low-Temperature Expansion

By inserting the finite-temperature pairing amplitude $\langle c_{-\mathbf{k},\downarrow} c_{\mathbf{k},\uparrow} \rangle = \frac{\Delta_{\mathbf{k}}}{2E_{\mathbf{k}}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)$ back into the fundamental self-consistent definition of the gap parameter and taking the thermodynamic continuum limit, the zero-temperature BCS gap equation is modified solely by the insertion of the hyperbolic tangent factor inside the momentum integral:

$$\Delta_{\mathbf{k}} = -\int \frac{\mathrm{d}^3\mathbf{k}'}{(2\pi)^3} \frac{V(\mathbf{k},\mathbf{k}')}{2E_{\mathbf{k}'}} \tanh\left(\frac{\beta E_{\mathbf{k}'}}{2}\right) \Delta_{\mathbf{k}'}. \tag{2.4.8}$$

Specializing to the case of a conventional superconductor governed by the separable BCS model interaction $V_{\mathbf{k},\mathbf{k}'} = -g w_{\mathbf{k}} w_{\mathbf{k}'}$ with $\Delta_{\mathbf{k}} = \Delta w_{\mathbf{k}}$ and $w_{\mathbf{k}} = \Theta(\hbar\omega_D - |\xi_{\mathbf{k}}|)$, we pass from momentum integration to energy integration over $\xi = \epsilon_{\mathbf{k}} - \mu$ via the single-particle density of states $N(\xi)$, where $E_\xi = \sqrt{\xi^2 + \Delta^2}$:

$$\frac{1}{g} = \int_{-\hbar\omega_D}^{\hbar\omega_D} \mathrm{d}\xi \frac{N(\xi)}{2E_\xi} \tanh\left(\frac{\beta E_\xi}{2}\right).$$

Close to the Fermi energy E_F (an approximation that holds robustly in the weak-coupling regime where $\Delta \ll E_F$ and $T \ll T_c \ll T_F$), the density of states can be approximated as constant across the narrow Debye shell, $N(\xi) \simeq N_0 = N(0)$, and by exploiting the even symmetry of the integrand ($\xi \leftrightarrow -\xi$) we obtain:

$$\frac{1}{g} = N_0 \int_0^{\hbar\omega_D} \mathrm{d}\xi \frac{1}{E_\xi} \tanh\left(\frac{\beta E_\xi}{2}\right).$$

Unlike the $T = 0$ case—where the integral reduced to $\int_0^{\hbar\omega_D} \frac{\mathrm{d}\xi}{\sqrt{\xi^2+\Delta^2}} = \operatorname{arcsinh}\left(\frac{\hbar\omega_D}{\Delta}\right)$ and could be inverted analytically—the presence of the thermal factor $\tanh\left(\frac{\beta\sqrt{\xi^2+\Delta^2}}{2}\right)$ prevents a closed-form analytical solution for arbitrary T, requiring numerical evaluation in general. However, we can derive an explicit analytical solution in the **low-temperature limit** ($k_B T \ll \Delta_0$, i.e., large $\beta \to \infty$):

- **Heuristic Dimensional Estimate:** To illustrate the structure of the subleading thermal integral from a rough dimensional-analysis perspective, one might initially consider a schematic algebraic expansion of the form:

$$\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right) \simeq 1 - \left(\frac{\beta E_{\mathbf{k}}}{2}\right)^{-1} + \cdots \implies \frac{1}{gN_0} = \int_0^{\hbar\omega_D} \mathrm{d}\xi \left[\frac{1}{\sqrt{\xi^2+\Delta^2}} - \frac{2k_B T}{\xi^2+\Delta^2}\right] + \ldots$$

- **Exact Asymptotic Expansion for $\beta E_\xi \gg 1$:** Rigorously, however, for large arguments $x = \frac{\beta E_\xi}{2} \to \infty$, the exact asymptotic expansion of the hyperbolic tangent is governed by an exponential tail rather than a power law: $\tanh(x) = \frac{1-e^{-2x}}{1+e^{-2x}} \simeq 1 - 2e^{-2x} = 1 - 2e^{-\beta E_\xi}$. Substituting this exact expansion into the gap equation yields:

$$\frac{1}{gN_0} = \int_0^{\hbar\omega_D} \frac{\mathrm{d}\xi}{\sqrt{\xi^2+\Delta(T)^2}} - 2\int_0^{\hbar\omega_D} \mathrm{d}\xi \frac{e^{-\beta\sqrt{\xi^2+\Delta(T)^2}}}{\sqrt{\xi^2+\Delta(T)^2}}.$$

Using the zero-temperature relation $\Delta_0 = 2\hbar\omega_D e^{-1/(gN_0)}$, the first integral evaluates to $\operatorname{arcsinh}\left(\frac{\hbar\omega_D}{\Delta(T)}\right) \simeq \ln\left(\frac{2\hbar\omega_D}{\Delta(T)}\right) = \frac{1}{gN_0} + \ln\left(\frac{\Delta_0}{\Delta(T)}\right)$. In the second integral, because the integrand is exponentially suppressed for $\xi \gg \sqrt{\Delta_0 k_B T}$, we can safely extend the upper integration limit to ∞ and expand the dispersion as $E_\xi \simeq \Delta_0 + \frac{\xi^2}{2\Delta_0}$ in the exponent while setting $E_\xi \simeq \Delta_0$ in the denominator. This reduces the thermal correction to a standard Gaussian integral:

$$2\frac{e^{-\beta\Delta_0}}{\Delta_0}\int_0^\infty \mathrm{d}\xi\, e^{-\frac{\beta\xi^2}{2\Delta_0}} = \left(\frac{2\pi k_B T}{\Delta_0}\right)^{1/2} e^{-\frac{\Delta_0}{k_B T}}.$$

Equating the terms and expanding the logarithm for small thermal deviations, $\ln\left(\frac{\Delta(T)}{\Delta_0}\right) \simeq \frac{\Delta(T)-\Delta_0}{\Delta_0}$, we arrive at the celebrated low-temperature expansion of the BCS gap:

$$\Delta(T) = \Delta_0\left[1 - \left(\frac{2\pi k_B T}{\Delta_0}\right)^{\frac{1}{2}} e^{-\frac{\Delta_0}{k_B T}}\right]. \tag{2.4.9}$$

Just as the zero-temperature gap $\Delta_0 \propto e^{-1/(gN_0)}$ is non-perturbative in the coupling constant g, the leading finite-temperature correction to the gap in Eq. (2.4.9) is strictly **non-perturbative in temperature**, being governed by the Boltzmann activation factor $e^{-\Delta_0/(k_B T)}$ rather than a power law in T. Because the quasiparticle excitation spectrum is gapped, thermal excitations are exponentially rare at $k_B T \ll \Delta_0$, leaving the Cooper-pair condensate and $\Delta(T)$ virtually unchanged near $T = 0$ (which manifests as an exponentially flat plateau in the profile of $\Delta(T)$ at low temperatures).

TODO: add image of a schematic plot showing the typical behavior of the superconducting gap order parameter as a function of temperature with a flat plateau at low temperature and a steep square root drop at the critical temperature

Overall, the full temperature profile of $\Delta(T)$ exhibits the universal behavior of an order parameter in a second-order phase transition: it remains flat near Δ_0 at low temperatures, decreases monotonically as thermal excitations proliferate, and vanishes continuously at a critical temperature T_c. Very close to the critical phase transition, in the Ginzburg-Landau regime where $(T_c - T)/T_c \ll 1$, the gap closes with a characteristic mean-field square-root critical exponent:

$$\Delta(T) = 3.06 k_B T_c \left(1 - \frac{T}{T_c}\right)^{\frac{1}{2}},$$

where the universal numerical prefactor stems from the Riemann zeta function:

$$\pi\left(\frac{8}{7\zeta(3)}\right)^{1/2} \simeq 3.06.$$

In a completely analogous manner, utilizing the zero-range contact interaction regularized by the scattering length a, the renormalized finite-temperature gap equation for an ultracold attractive Fermi gas reads:

$$-\frac{m}{4\pi\hbar^2 a} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\left[\frac{\tanh(\beta E_k/2)}{2E_k} - \frac{1}{2\epsilon_k}\right]. \tag{2.4.10}$$

Both the low-temperature asymptotic expansion (2.4.9) and the near-T_c square-root scaling hold identically for the ultracold Fermi gas as well, provided one substitutes the corresponding zero-temperature gap $\Delta_0 = \frac{8E_F}{e^2} e^{\frac{\pi}{2k_F a}}$ characteristic of the contact potential.

2.4.4 | Critical Temperature

Gap Equation at the Critical Temperature and Analytical Integration

Let us now determine the critical temperature T_c at which the system ceases to be a superconductor (or a superfluid of Cooper pairs) and reverts to a normal metallic state. As the temperature approaches the critical temperature from below ($T \to T_c^-$), the superconducting order parameter vanishes continuously, $\Delta(T \to T_c) \to 0$. Consequently, the quasiparticle dispersion seamlessly reduces to the absolute value of the normal-state free-fermion kinetic energy measured relative to the chemical potential μ:

$$\Delta(T \to T_c) \to 0 \implies E_\xi = \sqrt{\xi^2 + \Delta^2} \xrightarrow{T \to T_c} |\xi|.$$

Substituting $E_\xi = \xi$ (for $\xi \geq 0$) and the critical inverse temperature $\beta = \beta_c = (k_B T_c)^{-1}$ into the finite-temperature gap equation, we obtain a linear integral condition that directly relates $T_c = T_c(g, N_0)$ to the microscopic parameters of the theory:

$$\frac{1}{N_0 g} = \int_0^{\hbar\omega_D} \mathrm{d}\xi \frac{1}{\xi} \tanh\left(\frac{\beta_c \xi}{2}\right).$$

To evaluate this integral analytically, we perform the dimensionless change of variables:

$$x = \frac{\beta_c \xi}{2} \implies \mathrm{d}\xi = \frac{2}{\beta_c} \mathrm{d}x,$$

where the upper integration limit $\xi = \hbar\omega_D$ maps directly to $x = \frac{\hbar\omega_D \beta_c}{2} = \frac{\hbar\omega_D}{2k_B T_c}$. Integrating by parts by choosing $\frac{1}{x}$ as the factor to integrate (with primitive $\ln(x)$) and $\tanh(x)$ as the factor to differentiate (with derivative $\frac{1}{(\cosh(x))^2}$), we find:

$$\begin{aligned}
\frac{1}{N_0 g} = \int_0^{\hbar\omega_D} \mathrm{d}\xi \frac{1}{\xi} \tanh\left(\frac{\beta_c \xi}{2}\right) &= \int_0^{\frac{\hbar\omega_D \beta_c}{2}} \mathrm{d}x \frac{\tanh(x)}{x} \\
&= \left[\ln(x) \tanh(x)\right]_0^{\frac{\hbar\omega_D \beta_c}{2}} - \int_0^{\frac{\hbar\omega_D \beta_c}{2}} \mathrm{d}x \frac{\ln(x)}{(\cosh(x))^2} \\
&= \ln\left(\frac{\hbar\omega_D}{2k_B T_c}\right) \tanh\left(\frac{\hbar\omega_D}{2k_B T_c}\right) - \int_0^{\frac{\hbar\omega_D}{2k_B T_c}} \mathrm{d}x \frac{\ln(x)}{(\cosh(x))^2},
\end{aligned}$$

where the lower boundary term at $x \to 0^+$ vanishes identically because $\tanh(x) \simeq x$ goes to zero much faster than $\ln(x)$ diverges ($\lim_{x \to 0^+} x \ln(x) = 0$).

We now invoke our physical weak-coupling assumption: since the zero-temperature Cooper-pair gap $\Delta_0 = 2\hbar\omega_D e^{-\frac{1}{gN_0}}$ is exponentially smaller than the Debye cutoff $\hbar\omega_D$, the critical thermal energy $k_B T_c$ (which is expected on physical grounds to be of the same order of magnitude as Δ_0) must likewise satisfy $k_B T_c \ll \hbar\omega_D$. In this asymptotic regime, the upper integration limit diverges:

$$k_B T_c \ll \hbar\omega_D \implies \frac{\hbar\omega_D}{2} \beta_c = \frac{\hbar\omega_D}{2k_B T_c} \to \infty \implies \tanh\left(\frac{\hbar\omega_D \beta_c}{2}\right) \xrightarrow{k_B T_c \ll \hbar\omega_D} \tanh(\infty) = 1.$$

Furthermore, because the integrand $\frac{\ln(x)}{(\cosh(x))^2}$ decays exponentially at large x, we can safely extend the upper limit of the remaining integral to $+\infty$ and utilize the standard definite integral:

$$-\int_0^\infty \mathrm{d}x \frac{\ln(x)}{(\cosh(x))^2} = \ln\left(\frac{4e^\gamma}{\pi}\right),$$

where $\gamma \approx 0.577216$ is the Euler-Mascheroni constant. Combining the two logarithmic terms into the logarithm of a product yields:

$$\frac{1}{N_0 g} \simeq \ln\left(\frac{\hbar\omega_D}{2k_B T_c}\right) + \ln\left(\frac{4e^\gamma}{\pi}\right) = \ln\left(\frac{2\hbar\omega_D e^\gamma}{\pi k_B T_c}\right).$$

Universal BCS Ratio, Isotope Effect, and Density of States Dependence

Exponentiating both sides of this logarithmic relation and recalling that $\Delta_0 = 2\hbar\omega_D e^{-\frac{1}{gN_0}}$, we can solve explicitly for the critical temperature:

$$k_B T_c = \frac{2\hbar\omega_D e^{\gamma}}{\pi} e^{-\frac{1}{gN_0}} = \frac{e^{\gamma}}{\pi}\Delta_0,$$

which confirms that the thermal energy $k_B T_c$ at the transition is indeed of the exact same order of magnitude as the energy gap of a Cooper pair. Numerically, $\frac{e^{\gamma}}{\pi} \approx \frac{1.781}{3.1416} \approx \frac{1}{1.764}$, meaning $\frac{\Delta_0}{k_B T_c} = \frac{\pi}{e^{\gamma}} \approx 1.764$. Multiplying by two to reference the minimum pair-breaking threshold energy $2\Delta_0$ required to excite two quasiparticles out of the condensate, we arrive at the celebrated universal BCS ratio:

$$\frac{2\Delta_0}{k_B T_c} = \frac{2\pi}{e^{\gamma}} \approx 3.53. \tag{2.4.11}$$

Furthermore, for ultracold Fermi gases governed by a zero-range contact potential, the renormalized finite-temperature gap equation identically reduces at T_c to:

$$-\frac{m}{4\pi\hbar^2 a} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\left[\frac{\tanh(\beta_c\xi_k/2)}{2\xi_k} - \frac{1}{2\epsilon_k}\right].$$

By evaluating this integral in the weak-coupling BCS limit where $k_F a \to 0^-$, one obtains an analytic expression for T_c, proving that the exact same universal ratio $2\Delta_0/(k_B T_c) \approx 3.53$ remains valid in this entirely different physical system, utilizing the appropriate contact-potential zero-temperature gap Δ_0.

This full-fledged many-body result yields three major physical and experimental predictions:

1. **Universal Ratio:** The ratio $\frac{2\Delta_0}{k_B T_c} = \frac{2\pi}{e^{\gamma}} \approx 3.53$ is strictly *universal*. All microscopic material parameters (g, N_0, ω_D) cancel out completely, confirming that the critical temperature is set directly by the zero-temperature pairing gap Δ_0, independent of the specific details of the attractive interaction provided the system remains in the weak-coupling regime. Experimental measurements across numerous conventional weak-coupling superconductors have spectacularly confirmed this prediction.

2. **Isotope Effect on T_c:** The critical temperature T_c (identically to Δ_0) is directly proportional to the Debye frequency ω_D. Because the lattice vibration frequencies of crystal ions of mass M scale as $\omega_D \propto M^{-1/2}$, the many-body formula rigorously predicts that the critical temperature scales as:
$$T_c \propto \frac{1}{\sqrt{M}}.$$
This proportionality explains the experimentally observed isotope effect: replacing the constituent lattice ions with a heavier or lighter equivalent isotope leaves the electronic band structure completely untouched while modifying the phonon frequencies ω_D, thereby shifting T_c in direct agreement with experiment. Historically, this finding provided one of the earliest direct indications that lattice phonons play the central role in mediating conventional superconductivity.

3. **Dependence on the Density of States N_0 at the Fermi Energy:** Because N_0 appears inside the non-perturbative exponent $e^{-\frac{1}{gN_0}}$, the critical temperature T_c is exponentially sensitive to the single-particle density of electronic states at the Fermi level. Both in theory and in experiment, N_0 can be actively tuned by shifting the chemical potential μ (for instance,

by varying the carrier density via chemical doping or electrostatic gating) across a crystal lattice energy band $E_{\mathbf{k}}$. For non-relativistic fermions on a lattice, the single-particle band dispersion $E_{\mathbf{k}}$ is typically quadratic at very small k near the band bottom, steepens in the middle of the band, and flattens out again near the band edges. Since the density of states $N(\mu)$ counts the number of k-states per unit energy window $\mathrm{d}E$ and is inversely proportional to the dispersion slope $|\partial E_{\mathbf{k}}/\partial k|$, a flat dispersion (such as near a band extremum in a tight-binding band) packs many states into a narrow energy window, yielding a high density of states N_0, whereas a steep slope yields a low density of states (conversely, in a 3D free-fermion continuum, $N(\xi) \propto \sqrt{\xi+\mu}$ grows monotonically with carrier filling). Selecting different carrier fillings alters N_0 and consequently modifies T_c exponentially via $e^{-1/(gN_0)}$.

TODO: add image of a single particle lattice band dispersion curve versus momentum illustrating how shifting the chemical potential across flat and steep regions tunes the density of states at the Fermi level

2.4.5 | Low Temperature Expansion for the Specific Heat

Excitation Energy at Low Temperature

While measuring T_c provides a primary test of BCS theory, an even more direct experimental probe of the gapped superconducting excitation spectrum is the **specific heat capacity at constant volume**, $C_V = \frac{\partial E}{\partial T}\big|_V$. Let us investigate the thermodynamic internal energy difference $\delta E(T) = E(T) - E(0)$ between the total energy of the superconducting system at a finite temperature T and its absolute zero ground state. At $T = 0$, all Bogoliubov quasiparticle modes are empty ($\langle \gamma^\dagger_{\mathbf{k},\sigma}\gamma_{\mathbf{k},\sigma}\rangle = 0$) and the system sits in its ground state with constant energy $E(0)$. At $T > 0$, thermally excited independent quasiparticles increase the internal energy by $\delta E(T)$. Because these elementary excitations are fermionic, their thermal occupation is governed by the Fermi-Dirac distribution function $f(E_k)$ (with the quasiparticle chemical potential set to zero, since their average number is not an independently conserved thermodynamic variable, but rather fluctuates dynamically as a function of temperature). Summing over both spin species (which introduces a factor of 2) and passing from momentum integration to energy integration via the density of states $N(\xi)$, one has:

$$\delta E(T) = 2V \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} E_k f(E_k) = 2V \int_{-\mu}^{+\infty} \mathrm{d}\xi\, N(\xi) E_\xi f(E_\xi).$$

To evaluate this integral analytically, we focus specifically on the extreme **low-temperature limit**, where the thermal energy is drastically smaller than the pairing gap, $k_B T \ll \Delta(T) \simeq \Delta_0 \equiv \Delta$. In this strongly suppressed regime, where $E_\xi = \sqrt{\xi^2+\Delta^2} \geq \Delta \gg k_B T$, the Fermi-Dirac distribution effectively reduces to the classical Maxwell-Boltzmann exponential tail:

$$f(E_\xi) = \frac{1}{e^{\beta E_\xi}+1} \simeq e^{-\beta E_\xi}.$$

Because the exponential factor $e^{-\beta E_\xi}$ rapidly suppresses the integrand outside a very narrow energy window $|\xi| \lesssim \sqrt{\Delta k_B T} \ll \Delta \ll E_F \simeq \mu$, we can perform three systematic approximations:

1. Safely extend the integration limits from $[-\mu, +\infty)$ to $(-\infty, +\infty)$;
2. Approximate the density of states by its constant value evaluated at the Fermi level, $N(\xi) \simeq N_0$, and set $E_\xi \simeq \Delta$ in the prefactor outside the exponential function;
3. Retain the quadratic Taylor expansion of the quasiparticle dispersion $E_\xi = \sqrt{\xi^2+\Delta^2} \simeq \Delta\left(1+\frac{\xi^2}{2\Delta^2}\right) = \Delta + \frac{\xi^2}{2\Delta}$ inside the exponential, which transforms the remaining integral into a standard Gaussian integral.

This yields the asymptotic excitation energy density:

$$\frac{\delta E(T)}{V} \simeq 2N_0\Delta e^{-\beta\Delta}\int_{-\infty}^{+\infty} \mathrm{d}\xi\, e^{-\frac{\beta\xi^2}{2\Delta}} = 2N_0\Delta^2 e^{-\beta\Delta}\sqrt{\frac{2\pi}{\beta\Delta}}.$$

Temperature Derivative and Exponential Suppression of the Specific Heat

The specific heat capacity per unit volume at constant volume, C_V/V, is obtained by differentiating this internal energy density with respect to temperature T (or equivalently using $-k_B\beta^2\frac{\partial}{\partial\beta}$, noting that $\beta = \frac{1}{k_BT}$ implies $\frac{\partial}{\partial T}e^{-\frac{\Delta}{k_BT}} = \frac{\Delta}{k_BT^2}e^{-\frac{\Delta}{k_BT}}$):

$$\begin{aligned}\frac{C_V}{V} &= \frac{1}{V}\frac{\partial E(T)}{\partial T} = \frac{1}{V}\frac{\partial \delta E(T)}{\partial T} = -\frac{k_B\beta^2}{V}\frac{\partial \delta E(\beta)}{\partial \beta}\\ &= 2N_0\Delta^2\left[\frac{\Delta}{k_BT^2}e^{-\beta\Delta}\sqrt{\frac{2\pi k_BT}{\Delta}} + e^{-\beta\Delta}\sqrt{\frac{2\pi k_B}{\Delta}}\left(\frac{1}{2\sqrt{T}}\right)\right] + \text{sub-leading terms.}\end{aligned}$$

In the strict low-temperature limit $k_BT \ll \Delta$ ($T \to 0$), only the first term inside the brackets is relevant, as it scales as $T^{-2}\cdot T^{1/2} = T^{-3/2}$ and strictly dominates over the second term (which scales as $T^{-1/2}$). Furthermore, the neglected sub-leading terms include contributions containing the temperature derivative of the gap itself, $\frac{\partial\Delta(T)}{\partial T}$; because the low-temperature expansion of $\Delta(T)$ exhibits an exponentially flat plateau near Δ_0, its temperature derivative is exponentially suppressed and completely non-dominant. Consistently setting $\Delta = \Delta_0$, we obtain the final leading-order expression for the low-temperature specific heat of a BCS superconductor:

$$\frac{C_V}{V} = 2k_BN_0\sqrt{2\pi}\Delta_0\left(\frac{\Delta_0}{k_BT}\right)^{\frac{3}{2}} e^{-\frac{\Delta_0}{k_BT}}. \tag{2.4.12}$$

TODO: add image of a plot comparing the low temperature specific heat versus temperature for a BCS superconductor exhibiting exponential suppression and a normal Fermi gas exhibiting linear behavior

We clearly observe that the specific heat vanishes exponentially as $C_V \propto T^{-3/2}e^{-\Delta_0/(k_BT)}$ when $T \to 0$. This dramatic exponential suppression is a direct macroscopic consequence of the finite energy gap Δ_0 in the quasiparticle dispersion spectrum, which effectively freezes out thermal excitations at low temperatures. This stands in sharp contrast to the characteristic linear behavior of an ideal normal Fermi gas (or a Landau Fermi liquid).

It is highly instructive to apply this exact same mathematical procedure to calculate the specific heat of an ideal, gapless Fermi gas to explicitly highlight this contrast. In the normal state, the excitation energy relative to the chemical potential is simply $|\xi_k|$, yielding:

$$\begin{aligned}\delta E(T) &= 2V\int\frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}|\xi_k|f(|\xi_k|) = 2V\int_{-\mu}^{+\infty}\mathrm{d}\xi\, N(\xi)|\xi|f(|\xi|)\\ &\simeq 4N_0V\int_0^{+\infty}\mathrm{d}\xi\,\xi f(\xi)\\ &= 4N_0Vk_B^2T^2\int_0^{+\infty}\mathrm{d}x\frac{x}{e^x+1} = 4N_0Vk_B^2T^2\frac{\pi^2}{12}.\end{aligned}$$

Differentiating this normal-state excitation energy with respect to temperature gives the classic linear specific heat capacity:

$$\frac{C_V}{V} = 2N_0k_B^2T\frac{\pi^2}{3} = 2\frac{mk_F}{2\pi^2\hbar^2}k_B^2T\frac{\pi^2}{3} = \frac{mk_F}{3\hbar^2}k_B^2T.$$

Remark. ***Why Does the Maxwell-Boltzmann Approximation $f(E_\xi) \simeq e^{-\beta E_\xi}$ Hold at Low Temperature?*** *In standard statistical mechanics, the quantum Fermi-Dirac distribution*

approaches the classical Maxwell-Boltzmann distribution $e^{-\beta E}$ *in the* high-temperature *(or low-density) limit. It might therefore seem counterintuitive that we are permitted to replace* $f(E_\xi)$ *with* $e^{-\beta E_\xi}$ *in the exact opposite limit of* very low temperature *(*$k_B T \ll \Delta_0$*).*

The mathematical and physical justification lies entirely in the presence of the strictly positive ***energy gap*** $\Delta > 0$ *in the Bogoliubov quasiparticle spectrum:*

$$f(E_\xi) = \frac{1}{e^{\frac{E_\xi}{k_B T}} + 1}, \quad \textit{with} \quad E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2} \geq \Delta.$$

Because the excitation energy $E_{\mathbf{k}}$ *is bounded from below by* Δ*, at low temperatures where* $\frac{\Delta}{k_B T} \gg 1$*, the exponent in the Fermi-Dirac denominator is guaranteed to be* extremely large for every single momentum mode $\mathbf{k}$*:*

$$\frac{\Delta}{k_B T} \gg 1 \implies \frac{E_{\mathbf{k}}}{k_B T} \geq \frac{\Delta}{k_B T} \gg 1 \implies e^{\frac{E_\xi}{k_B T}} \gg 1 \implies f(E_\xi) = \frac{1}{e^{\frac{E_\xi}{k_B T}} + 1} \simeq e^{-\frac{E_\xi}{k_B T}} = e^{-\beta E_\xi}.$$

By contrast, in a ***gapless*** *system such as an ideal normal Fermi gas, the excitation energy relative to the chemical potential is* $|\xi_{\mathbf{k}}| = |\epsilon_{\mathbf{k}} - \mu|$*, which vanishes continuously at the Fermi surface (*$\xi_{\mathbf{k}} \to 0$ *as* $k \to k_F$*). Consequently, no matter how small the temperature* T *becomes, there are always low-energy states within the thermal window* $|\xi_{\mathbf{k}}| \lesssim k_B T$ *for which the ratio* $\frac{|\xi_{\mathbf{k}}|}{k_B T} \lesssim 1$*. In that gapless window, the* $+1$ *in the Fermi-Dirac denominator cannot be neglected, and those precise states are responsible for generating the linear specific heat* $C_V \propto T$*. It is exclusively the presence of the finite superconducting gap* Δ *that enforces* $\beta E_{\mathbf{k}} \gg 1$ *across the entire Brillouin zone as* $T \to 0$*, rigorously justifying the Maxwell-Boltzmann tail and causing the exponential suppression of the specific heat.*

2.5 | Spontaneous Symmetry Breaking and Collective Excitations

2.5.1 | $O(2)$ Model

The $O(2)$ model is a simple model of spontaneous symmetry breaking, where the continuous symmetry is broken to a discrete subgroup. In the context of superconductivity, this model helps to understand the emergence of a condensate and the associated Goldstone mode.

In antiferromagnets, the spins are ordered in a way that creates a pattern of alternating magnetic moments, leading to a breaking of the continuous rotational symmetry. This results in the appearance of magnons, which are the quantized spin waves.

The $U(1)$ gauge group is associated with the phase of the complex scalar field. In the context of superconductivity, this gauge group is responsible for the electromagnetic interactions and the emergence of the Meissner effect.

We consider a complex scalar field $\varphi(x)$ and the lagrangian density $\mathcal{L} = \frac{1}{2}\partial_\mu\varphi^*\partial^\mu\varphi - V(\varphi)$, where $V(\varphi)$ is a potential that has a minimum at some non-zero value of φ:

$$V(\varphi) = -|\mu|^2|\varphi|^2 + \lambda|\varphi|^4.$$

The potential has a "Mexican hat" shape, with a circle of minima at $|\varphi| = \sqrt{\frac{|\mu|^2}{2\lambda}}$:

TODO: add image of a mexican hat potential

$$4\lambda|\varphi|^2 - 2|\mu|^2 = 0 \implies |\varphi| = \sqrt{\frac{|\mu|^2}{2\lambda}}.$$

The lagrangian is invariant under global $U(1)$ phase rotations of the field $\varphi \to e^{i\theta}\varphi$.

When particles condense into the ground state, they choose a specific phase, breaking the continuous symmetry. In the circle of minima, the system can choose any point, but once a point is chosen, the symmetry is broken. So we have a portion of particles condensing into the ground state, costituing the lowest energy **manifold**, and are called gapless modes: their spectrum is gapless, meaning thtey can be excited with arbitrarily low energy. These gapless modes are the Goldstone modes, massive excitations that arise due to the spontaneous breaking of continuous symmetries. In the case of superconductivity, these Goldstone modes correspond to phase fluctuations of the order parameter.

TODO: check comments

So initially we had a U(1) symmetric system, but after the condensation, we have a discrete symmetry. The Goldstone modes are associated with the broken continuous symmetry and are responsible for the low-energy excitations in the system.

Example (Ising Field Theory).

$\varphi \in \mathbb{R}$

The Ising model is a simple model of ferromagnetism, where the spins can take on two values, up or down. The ising field theory describes the behavior of the system near the critical point, where the symmetry is spontaneously broken. In this model, the order parameter is a scalar field that takes on a non-zero value in the ordered phase, breaking the Z_2 symmetry of the system.

It can be seen as a limiting case of the $O(2)$ model, where the continuous symmetry is broken to a discrete subgroup: the phase space of the order parameter is reduced from a circle to two

points and the potential is a section of the Mexican hat potential. The Ising model is a simple example of spontaneous symmetry breaking, where the system chooses one of the two possible ground states, breaking the symmetry of the system.

TODO: add image of a double well potential as a section of the mexican hat potential

There are no gapless modes in the Ising model, as the excitations are massive and require a finite amount of energy to be excited. The Ising model is a simple example of spontaneous symmetry breaking, where the system chooses one of the two possible ground states, breaking the symmetry of the system.

TODO: check

Example ($\mathbb{Z}_3$ Field Theory).

$\varphi \in \mathbb{R}^+$

The $\mathbb{Z}_3$ model is a simple model of ferromagnetism, where the spins can take on three values, up, down or zero. The $\mathbb{Z}_3$ field theory describes the behavior of the system near the critical point, where the symmetry is spontaneously broken. In this model, the order parameter is a complex scalar field that takes on a non-zero value in the ordered phase, breaking the $\mathbb{Z}_3$ symmetry of the system. It can be seen as a limiting case of the $O(2)$ model, where the continuous symmetry is broken to a discrete subgroup: the phase space of the order parameter is reduced from a circle to three points.

2.5.2 | The $U(1)$ Gauge Group and Phase Rotation

The original many-body Hamiltonian H for the interacting Fermi gas is fundamentally invariant under global $U(1)$ phase transformations. This continuous symmetry corresponds to the strict conservation of the total particle number $\hat{N} = \sum_{\mathbf{k},\sigma} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma}$. We can define the unitary phase rotation operator as $\hat{U}(\varphi) = e^{i\varphi\hat{N}}$, under which the Hamiltonian remains invariant: $\hat{U}^\dagger(\varphi) H \hat{U}(\varphi) = H$.

Applying this unitary transformation to the bare fermionic operators yields a phase shift:

$$\hat{U}^\dagger(\varphi) c_{\mathbf{k},\sigma} \hat{U}(\varphi) = e^{i\varphi} c_{\mathbf{k},\sigma} \implies \hat{U}^\dagger(\varphi) c^\dagger_{\mathbf{k},\sigma} \hat{U}(\varphi) = e^{-i\varphi} c^\dagger_{\mathbf{k},\sigma}.$$

However, the hallmark of the superconducting state is the non-vanishing anomalous average $\langle\psi_0|c_{\mathbf{k},\uparrow}c_{-\mathbf{k},\downarrow}|\psi_0\rangle \neq 0$, which immediately implies that the macroscopic BCS ground state $|\psi_0\rangle$ is *not* invariant under this $U(1)$ transformation. We can see this explicitly by applying the transformation operator to the BCS ground state wavefunction:

$$\begin{aligned}|\psi_0(\varphi)\rangle = \hat{U}(\varphi)|\psi_0\rangle &= \hat{U}(\varphi) \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle \\ &= \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} e^{i\varphi\hat{N}} c^\dagger_{\mathbf{k},\uparrow} e^{-i\varphi\hat{N}} e^{i\varphi\hat{N}} c^\dagger_{-\mathbf{k},\downarrow} e^{-i\varphi\hat{N}} e^{i\varphi\hat{N}}\right)|0\rangle.\end{aligned}$$

Since $e^{i\varphi\hat{N}}|0\rangle = |0\rangle$ and the creation operators pick up a phase $e^{i\varphi}$ when commuted with the generator, we obtain an entirely new state rotated by a global phase 2φ:

$$|\psi_0(\varphi)\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + e^{2i\varphi} v_{\mathbf{k}} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow}\right)|0\rangle.$$

Clearly, $|\psi_0(\varphi)\rangle \neq |\psi_0\rangle$ for any arbitrary phase φ that is not a multiple of π. Despite this, the energy of this rotated state remains identical to the original ground state:

$$\langle\psi_0(\varphi)|H|\psi_0(\varphi)\rangle = \langle\psi_0|\hat{U}^\dagger(\varphi) H \hat{U}(\varphi)|\psi_0\rangle = \langle\psi_0|H|\psi_0\rangle.$$

This mathematically demonstrates Spontaneous Symmetry Breaking (SSB): the ground state possesses a lower symmetry than the Hamiltonian that governs it.

2.5.3 | Goldstone Modes in Ultracold Fermi Gases

According to Goldstone's theorem, the spontaneous breaking of a continuous global symmetry must be accompanied by the emergence of gapless collective excitations, known as Goldstone modes. In a conventional charged superconductor, the long-range Coulomb interaction combined with the Anderson-Higgs mechanism pushes these phase fluctuations up to the plasma frequency, effectively gapping them out. However, ultracold Fermi gases are composed of strictly neutral atoms; therefore, there is no Higgs mechanism, and the gapless Goldstone modes are unequivocally present.

These collective phase fluctuations are not captured by the effective BCS mean-field Hamiltonian $\hat{H}_{\text{eff}} = \sum_{\mathbf{k}} E_{\mathbf{k}} \gamma_{\mathbf{k}}^{\dagger} \gamma_{\mathbf{k}} + C'$, which only describes the gapped, single-particle pair-breaking excitations. To properly describe the Goldstone modes, one must theoretically go beyond the mean-field approximation to account for the collective dynamics of the pairs.

The dispersion relation of the Goldstone mode in the long-wavelength limit ($k \to 0$) is linear, representing gapless acoustic sound waves (often called Bogoliubov-Anderson sound):

$$\omega_{\text{Gold}}(k) = ck.$$

The sound velocity c for this mode in a Fermi gas is extremely large, explicitly given by $c = v_F/\sqrt{3}$, where v_F is the Fermi velocity. Because this collective velocity is so high, it requires a substantial amount of energy to excite these modes at any finite momentum. Consequently, they are exceedingly difficult to thermally excite, even compared to the gapped pair-breaking excitations when the pairing gap Δ is small.

Why Goldstone Modes?

The charge Q^a is associated with the number of particles conserved by the $U(1)$ symmetry and it can be written as the integral of the time component of the Noether current J_0^a:

$$Q^a = \int \mathrm{d}^3x \, J_0^a(x), \quad \partial_\mu J_a^\mu = 0.$$

If φ rotations are broken, then the vacuum is not invariant under the action of Q^a:

$$\langle 0 | \, [\mathcal{O}, Q^a] \, | 0 \rangle \neq 0,$$

where $\mathcal{O}$ is a generic operator: this expression indicates that the vacuum is not invariant under the symmetry transformation generated by Q^a. This implies that the vacuum is not invariant under the symmetry transformation generated by Q^a:

$$\begin{aligned} Q^a \, |0\rangle &\neq \alpha \, |0\rangle \, , \\ Q^a \, |0\rangle &= \alpha \, |0\rangle + \beta \, |\pi_p\rangle \, , \end{aligned}$$

where $|\pi_p\rangle$ is a single-particle state of the Goldstone boson. This means that the vacuum is not invariant under the symmetry transformation generated by Q^a, and there must exist a massless excitation (the Goldstone boson) associated with this broken symmetry.

The current J_a^μ is associated with the broken symmetry, and:

$$\langle 0 | \, J^\mu(?) \, | \pi_p \rangle = iFp^\mu e^{-ipx},$$

which implies current conservation:

$$p_\mu p^\mu = 0 \implies p^2 = 0 \implies m_\pi = 0,$$

such that the Goldstone boson is massless, and the dispersion relation is linear in momentum.

Goldstone Modes in BCS Superconductors

The grand-canonical BCS Hamiltonian is invariant under global $U(1)$ phase rotations, which correspond to the conservation of the total particle number:

$$\hat{K} = \sum_{\mathbf{k},\sigma} \xi_{\mathbf{k}} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma} - g \sum_{\mathbf{k},\mathbf{k}'} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} c_{-\mathbf{k}',\downarrow} c_{\mathbf{k}',\uparrow}.$$

If we apply the $U(1)$ transformation to the fermionic operators, we find that the Hamiltonian remains invariant:

$$\begin{cases} c_{\mathbf{k},\sigma} \to e^{i\varphi} c_{\mathbf{k},\sigma}, \\ \hat{K} \to \hat{K}. \end{cases}$$

Since the $U(1)$ symmetry is global, φ is a constant and the Hamiltonian commutes with the total particle number operator $\hat{N} = \hat{N}_\uparrow + \hat{N}_\downarrow = \sum_{\mathbf{k},\sigma} c^\dagger_{\mathbf{k},\sigma} c_{\mathbf{k},\sigma}$:

$$[\hat{K}, \hat{N}] = 0.$$

Now considering the BCS wavefunction, we see that it is not invariant under the $U(1)$ transformation:

$$|\psi_{BCS}\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} \right) |0\rangle \to \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + e^{2i\varphi} v_{\mathbf{k}} c^\dagger_{\mathbf{k},\uparrow} c^\dagger_{-\mathbf{k},\downarrow} \right) |0\rangle ,$$

where $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are the BCS coherence factors. This indicates that the BCS ground state spontaneously breaks the global $U(1)$ symmetry, since the ground state is not invariant under the transformation, but only under a $\mathbb{Z}_2$ transformation:

$$\varphi' = n\pi, \quad n \in \mathbb{Z}.$$

This spontaneous symmetry breaking leads to the emergence of a gapless Goldstone mode, which corresponds to the phase fluctuations of the order parameter.

Thermodynamics and Specific Heat Competition

To determine the thermodynamic relevance of the Goldstone modes, we compare their contribution to the specific heat capacity at constant volume (C_{Gold}) against the contribution from the gapped pair-breaking excitations (C_{pair}, the excitations breaking Cooper pairs). Our goal is to understand the temperature regime where the gapless Goldstone modes dominate the thermodynamics of the system.

The specific heat of the acoustic Goldstone modes is computed using the exact same standard Debye formalism utilized for phonons or a weakly interacting Bose gas, yielding a T^3 dependence:

$$\frac{C_{\text{Gold}}}{V} = \frac{2\pi^2}{15} k_B \left(\frac{k_B T}{\hbar c} \right)^3 .$$

Conversely, the specific heat stemming from the pair-breaking excitations is exponentially suppressed due to the presence of the energy gap $E_{\mathbf{k}} \simeq \sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2}$:

$$\frac{C_{\text{pair}}}{V} = 2k_B N_0 \sqrt{2\pi}|\Delta| \left(\frac{|\Delta|}{k_B T}\right)^{3/2} e^{-\frac{|\Delta|}{k_B T}}.$$

To identify the temperature regime where the collective modes dominate, we equate the two specific heats, $C_{\text{Gold}} = C_{\text{pair}}$:

$$\frac{\pi^2}{15}\left(\frac{k_B T}{\hbar c}\right)^3 = N_0\sqrt{2\pi}|\Delta|\left(\frac{|\Delta|}{k_B T}\right)^{3/2} e^{-\frac{|\Delta|}{k_B T}}, \quad \text{now } \tilde{\Delta} = \frac{\Delta}{k_B T}, \text{ and } \tilde{c} = \frac{c}{k_B T},$$

$$\frac{\pi^2}{15}\left(\frac{k_B T}{\hbar c}\right)^3 = N_0\sqrt{2\pi}|\Delta|\left(\frac{|\Delta|}{k_B T}\right)^{3/2} e^{-\frac{|\Delta|}{k_B T}}$$

We substitute the Fermi surface density of states $N_0 = mk_F/(2\pi^2\hbar^2)$ and use the low-temperature limit for the gap, $\Delta \simeq \Delta_0 \approx 1.76 k_B T_c \equiv \alpha$. Defining the reduced temperature $\tilde{T} = T/T_c$, we can expand the left-hand side of the equation in terms of T_c/T_F and $\tilde{T}$:

TODO: check and complete computation

$$\frac{C_{\text{Gold}}}{V} = \frac{2\pi^2}{15}k_B\left(\frac{k_B T}{\hbar c}\right)^3 = \frac{2\pi^2}{15}k_B\left(\frac{k_B T_c}{\hbar c}\right)^3\tilde{T}^3 = \frac{2\pi^2}{15}k_B\left(\frac{k_B T_c}{\hbar v_F/\sqrt{3}}\right)^3\tilde{T}^3 = \frac{2\pi^2}{15}k_B\left(\frac{\sqrt{3}k_B T_c}{\hbar v_F}\right)^3\tilde{T}^3,$$

since $v_F = \hbar k_F/m$, we can write $\hbar v_F = \hbar^2 k_F/m$ and $c = v_F/\sqrt{3}$. In the end we have:

TODO: check and complete computation

$$\tilde{T}^{9/2}\left(\frac{T_c}{T_F}\right)^2 \frac{\sqrt{6}\pi^{7/2}}{20\alpha^{5/2}} = e^{-\frac{\alpha}{\tilde{T}}} \implies \tilde{T}^{9/2}\left(\frac{T_c}{T_F}\right)^2 1.6 \simeq e^{-\frac{1.76}{\tilde{T}}}. \tag{2.5.1}$$

In standard weak-coupling superfluids, the critical temperature is orders of magnitude smaller than the Fermi temperature, $T_c \ll T_F$. Using the BCS relation $T_c/T_F = (8e^{\gamma}/\pi e^2)e^{-\pi/(2k_F a)} \ll 1$, a typical experimental parameter for a weak-coupling Fermi gas is $T_c/T_F \sim 10^{-3}$.

By numerically plotting the two sides of Eq. (2.5.1) as functions $f(\tilde{T})$ (the power-law Goldstone contribution scaled by the small prefactor) and $g(\tilde{T})$ (the exponential pair-breaking contribution), we observe that the gapped single-particle excitations completely dominate the thermodynamics for $\tilde{T} \gtrsim 0.1$. The curves intersect at a severely suppressed critical threshold of $T \simeq 0.11 T_c$.

TODO: add image of plots showing the theoretical specific heat contributions as a function of temperature intersecting at a highly suppressed temperature ratio, noticing that y axis is really zoomed in to show the intersection

This implies that the gapless Goldstone mode only becomes the dominant thermodynamic excitation for extremely low temperatures, $T \lesssim 0.1 T_c$. However, at such minuscule temperatures, the absolute magnitude of the specific heat is astronomically small ($C_V \simeq 0$). Therefore, in practice, the Goldstone contribution is entirely masked and generally not experimentally observable in weakly interacting Fermi gases. The collective Goldstone mode only becomes thermodynamically relevant and macroscopically observable if the interaction strength is tuned away from the weak-coupling limit (for example, toward unitarity via Feshbach resonances), which significantly raises the ratio T_c/T_F and consequently pushes the intersection to higher, measurable temperatures.

3 | Monte carlo and boh

[...]

3.1 | The Ising Model

lattice, spins, hamiltonian, external coupling term, internal coupling term (interaction), energy configuations

TODO: arrows on a lattice

The partition function is defined by

...

We are now interested to study strongly correlated systems, where the interaction term is not negligible. In particular, are interested in the **critical behavior** of the system, which is characterized by a phase transition (second order) at a critical temperature T_c. The critical behavior is characterized by the divergence of the correlation length ξ and the susceptibility χ as $T \to T_c$. We are not interested in first order phase transitions, which are characterized by a discontinuity in the first derivative of the free energy (latent heat) and a finite correlation length at the transition. We will be interested in second and infinite order phase transitions.

> **Remark (Lattice QCD).** *The natural way to regularize QFT is to put it on a lattice, which is the basis of lattice QCD. The Ising model is a toy model for lattice QCD, where the spins are replaced by quarks and the interaction term is replaced by the gauge field. The Ising model is a classical model, while lattice QCD is a quantum model. The lattice naturally introduces a cutoff, which is the lattice spacing. The continuum limit is obtained by taking the lattice spacing to zero, which corresponds to taking the cutoff to infinity.*

There is spontaneous symmetry breaking in the Ising model, which is characterized by a non-zero magnetization below the critical temperature. The hamiltonian is invariant under a global $\mathbb{Z}_2$ symmetry (if we neglect the external field), which is spontaneously broken below the critical temperature. The order parameter is the magnetization, which is zero above the critical temperature and non-zero below the critical temperature, where the system chooses one of the two degenerate ground states. The critical exponents characterize the behavior of the order parameter and the correlation length near the critical temperature.

There is no local order parameter in the Ising model, only global information about the system.[1]

Using a topological description of the system, we can arrive to a lattice gauge theory, which is a quantum model. The Ising model is a classical model, while the lattice gauge theory is a quantum model. The lattice gauge theory is a generalization of the Ising model, where the spins are replaced by qubits and the interaction term is replaced by a gauge field. (Kitaev's toric code is a lattice gauge theory, which is a quantum model that exhibits topological order.) The topological order is characterized by a ground state degeneracy that depends on the topology of the system, and by the presence of anyons, which are quasiparticles that obey fractional statistics. The topological order is a global property of the system, which is not captured by local order parameters.

Integer quantum Hall effect, topological response of the system: robust to perturbations... topological information is preserved under continuous deformations of the system. The topological information is encoded in the ground state wavefunction, which is a global property of the system. The topological information is robust to perturbations, which is a consequence of the gap in the

[1] An order parameter is an observable that is zero in one phase and non-zero in another phase, which characterizes the phase transition. A local order parameter is an order parameter that is defined at each point in space, while a global order parameter is an order parameter that is defined for the whole system. The magnetization is a global order parameter, and if we have to build a local one we have to seek for a finite supported operator (it does not need information from the whole system to be defined) that is non-zero in one phase and zero in another phase.

spectrum of the system. Topological quantum computing is a way to encode information in the topological properties of the system, which is robust to perturbations.

We are going tostudy the Ising model in different regimes, theoretically and numerically. after studying the classical Ising model, we will study its quantized version, and in order to do that we will need to use and understand different methods (mean field theory, renormalization group, Monte Carlo simulations, etc.) to study the Ising model in different regimes. We will also study the Ising model in different dimensions, and we will see that the critical behavior is different in different dimensions. We will study the duality of the Ising model, which is a powerful tool to study the critical behavior of the system.

3.1.1 | 1D Ising Model

We are going to study the free energy and its discontinuities, given by the partition function, which is defined in . TODO: add label

So the free energy is given by

$$F = -\lim_{N\to\infty} \frac{k_B T}{N} \log \mathcal{Z},$$

where $k_B = 1.38 \times 10^{-23}\,\mathrm{J/K}$ is the Boltzmann constant, T is the temperature, N is the number of spins, and $\mathcal{Z}$ is the partition function. The free energy is a thermodynamic potential that characterizes the equilibrium properties of the system. The discontinuities in the free energy correspond to phase transitions, which are characterized by a change in the order parameter.

Transfer Matrix Method

We can rewrite the partition function in terms of a transfer matrix, which is a matrix that encodes the interactions between neighboring spins:

$$\begin{aligned}\mathcal{Z} &= \sum_{\sigma_N=\pm1} \sum_{\sigma_{N-1}=\pm1} \cdots \sum_{\sigma_1=\pm1} e^{-\beta H(\sigma_1,\sigma_2,\ldots,\sigma_N)} \\ &= \sum_{\sigma_N=\pm1} \sum_{\sigma_{N-1}=\pm1} \cdots \sum_{\sigma_1=\pm1} T_{\sigma_N,\sigma_{N-1}} T_{\sigma_{N-1},\sigma_{N-2}} \cdots T_{\sigma_2,\sigma_1},\end{aligned}$$

where T is the 2×2 transfer matrix, which is defined by

$$T_{\sigma_i,\sigma_{i-1}} = e^{-\beta H(\sigma_i,\sigma_{i-1})}, \quad \begin{pmatrix} e^{\beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J} \end{pmatrix}.$$

Using this formalism, we can write the partition function as a product of transfer matrices and then take the trace to obtain

$$\mathcal{Z} = \mathrm{Tr}(T^N).$$

So the whole problem reduces to finding the eigenvalues of the transfer matrix T. The eigenvalues of T are given by its diagonalization, which is a standard linear algebra problem. We can compute them as follows:

$$\begin{aligned}\det(T - \lambda I) = 0, \\ (e^{\beta J} - \lambda)^2 - (e^{-\beta J})^2 = 0,\end{aligned}$$

which implies that the eigenvalues are

$$\begin{cases} \lambda_+ = e^{\beta J} + e^{-\beta J} = 2\cosh(\beta J), \\ \lambda_- = e^{\beta J} - e^{-\beta J} = 2\sinh(\beta J). \end{cases}$$

So in the end we have that the partition function is given by

$$\mathcal{Z} = \lambda_+^N + \lambda_-^N,$$

such that the free energy is given by

$$F = -k_B T \lim_{N\to\infty} \frac{1}{N} \log\left(\lambda_+^N + \lambda_-^N\right) = -k_B T \log \lambda_+ = -k_B T \log(2\cosh(\beta J)),$$

since the term λ_-^N is negligible in the thermodynamic limit $N \to \infty$. The free energy is a smooth function of the temperature, in the form of a logarithm of a positive and continuous function, so it is continuous and differentiable. Therefore, there is no phase transition in the 1D Ising model, which is consistent with the fact that there is no spontaneous symmetry breaking in one dimension.

Correlation Functions

If we consider the connected correlation function between spins, we can compute it using the transfer matrix formalism. A simple computation shows that

$$\langle \sigma_1 \sigma_N \rangle = \left(\frac{\lambda_-}{\lambda_+}\right)^N = (\tanh(\beta J))^N \sim e^{-N/\xi},$$

where ξ is the correlation length, which is given by

$$\xi = -\frac{1}{\log(\tanh(\beta J))}.$$

The correlation length is finite for any finite temperature, which implies that there is no long-range order in the 1D Ising model. The correlation length diverges only in the limit $T \to 0$, which corresponds to the ground state of the system. Therefore, there is no phase transition in the 1D Ising model, which is consistent with the fact that there is no spontaneous symmetry breaking in one dimension.

So in the 1D Ising model, the correlation length is finite for any finite temperature: the system does not exhibit a phase transition, and there it remains in a paramagnetic phase for any finite temperature. The system only exhibits a phase transition in the limit $T \to 0$, where the correlation length diverges and the system becomes ordered.

TODO: add plot of correlation length vs distance for different temperatures (log log)

3.1.2 | 2D Ising Model

TODO: image of ising on different lattices

Different definitions of the 2D Ising model: square lattice, triangular lattice, Kagome lattice, etc. The 2D Ising model exhibits a phase transition at a finite temperature T_c, which is characterized by a divergence of the correlation length and the susceptibility. But does the lattice matter? If we imagine the spacing J of the lattices to be the same, then the critical temperature T_c is different for different lattices, but we wont go into that.

Instead, let us imagine the triangular lattice, given two different spins in the bottom angles of the triangle, we can see that the spin in the top angle is *frustrated*, since it cannot satisfy both interactions with the bottom spins. This is an example of **frustration**, which is a phenomenon that occurs in systems with competing interactions: the system cannot satisfy all interactions simultaneously, which leads to a degenerate ground state and a complex energy landscape. Frustration can lead to interesting phenomena, such as spin glasses, which are disordered magnetic systems with a complex energy landscape and slow dynamics. For now we are sticking to the square lattice, which is the simplest case, but we will see that on a triangular lattice the connection with QFT is more direct...

Exact Onsager Solution

The 2D Ising model on a square lattice was solved exactly by Lars Onsager in 1944, using the transfer matrix method and the theory of Toeplitz determinants. The exact solution shows that the 2D Ising model exhibits a phase transition at a finite temperature T_c, which is characterized by a divergence of the correlation length and the susceptibility. It is a second order phase transition, which is characterized by a continuous change in the order parameter and a divergence of the correlation length.

TODO: add drawing of correlation length vs temperature

Below the critical temperature, the system exhibits spontaneous symmetry breaking, which is characterized by a non-zero magnetization. The order parameter is the magnetization, which is zero above the critical temperature and non-zero below the critical temperature, where the system chooses one of the two degenerate ground states.

Peierl's Argument

Imagine to put a domain wall in the system, which is a line that separates two regions with different magnetization: in 1D imagine a line that separates two regions with different magnetization, one of all $\uparrow$ and the other of all $\downarrow$. The energy cost of creating a domain wall is proportional to its length, which is given by $E \sim J\ell$, where ℓ is the length of the domain wall. The entropy associated with the domain wall is proportional to its length, which is given by $S \sim k_B\ell$. So the domain wall gives a contribution to the free energy and one to the entropy:

TODO: wrong computations, check

$$\Delta F = \Delta E - T\Delta S \sim J\ell - k_B T\ell = \ell(J - k_B T).$$

As we take the TD limit in 1D we will have

$$\Delta F \sim \ell(J - k_B T) \to -\infty,$$

which implies that the domain wall is favorable and the system will be always disordered: no phase transition and always in a paramagnetic phase.

If we try to promote the argument to 2D, we will have an energy cost of creating a domain wall that is proportional to its length, which is given by $E \sim J\ell$, where ℓ is the length of the domain wall. The entropy associated with the domain wall is proportional to its area, which is given by $S \sim k_B\ell^2$. So the domain wall gives a contribution to the free energy and one to the entropy:

$$\Delta F = \Delta E - T\Delta S \sim J\ell - k_B T\ell^2 = \ell(J - k_B T\ell).$$

As we take the TD limit in 2D we will have a region in which the domain wall is not favorable, which implies that the system can be ordered and exhibit a phase transition at a finite temperature T_c. The Peierl's argument shows that the 2D Ising model exhibits a phase transition at a finite temperature T_c, which is characterized by a divergence of the correlation length and the susceptibility. It is a second order phase transition, which is characterized by a continuous change in the order parameter and a divergence of the correlation length.

TODO: check computations of the whole subsubsection, add drawing of domain wall in 1D and 2D

3.2 | Motivation for Monte Carlo Methods

Speaking about the ising model, we have seen that the 1D case is exactly solvable, while the 2D case is exactly solvable only in the absence of an external field. In general, the Ising model is not exactly solvable in higher dimensions or in the presence of an external field. Therefore, we need to use numerical methods to study the Ising model in these cases.

Even if the ising model is ideal and simple, its universal behavior is shared by many other systems, and therefore it is a good model to study the critical behavior of strongly correlated systems. The famous bootstrap program was developed with the aim of solving the 3D Ising model, which is not exactly solvable.

In the ising model, studying its partition function we can notice that the number of configurations grows exponentially with the number of spins

$$\mathcal{Z} = \sum_{\{\sigma_i\}} e^{-\beta H(\{\sigma_i\})} \sim 2^N,$$

where N is the number of spins. Even if we don't consider the simulation of the system but only the computation of the partition function, we can see that the number of configurations reaches astronomical numbers for a relatively small number of spins. For example, for $N = 100$ spins, the number of configurations is $2^{100} \sim 10^{30}$, which is larger than the number of atoms in the observable universe, orders of magnitude more data than what we can store in a computer. Therefore, we need to use numerical methods to study the Ising model in these cases.

Speaking about $P \neq NP$ problems, we can see that the Ising model is a NP problem, since we can verify a solution in polynomial time, but we cannot find a solution in polynomial time. The Ising model is a combinatorial optimization problem, which is a class of problems that are hard to solve but easy to verify.

An attempt to address these problems was made by Metropolis et al. in 1953, who developed the Metropolis algorithm, which is a Monte Carlo method that allows to sample configurations of the system according to the Boltzmann distribution. The Metropolis algorithm is a Markov chain Monte Carlo method, which means that it generates a sequence of configurations that are correlated with each other, but the correlation decays exponentially with the number of steps.

It was a revolution in how peaple conceaved mathematics: machine started to be used to solve problems that were not solvable analytically, and the Monte Carlo method was a way to use random numbers to solve problems that were not solvable deterministically.

computing partition function in low temperature limit... organizing principle... probability theory...

3.2.1 | Sampling a Probability Distribution

Mean and variance, finite number of samples

good estimator of the mean is the average

a good estimator of the variance is the "error", final formula... we can trust the error iff the A_i are uncorrelated.

3.2.2 | Hit or Miss: Evaluating the Area of a Circle

[...]

TODO: draw unit circle inscribed in a square, with random points inside the square, some in the circle and some outside the circle.

Estimator for the Average

[...]

Confidence Interval for the Average

computation, check professor and notes

the distribution may not be gaussian, but the central limit theorem guarantees that the distribution of the average is gaussian, and therefore we can use the standard deviation to compute the confidence interval for the average.

independence of observations is eady to achive in laboratory experiments, but in numerical simulations it is not easy to achieve, since the configurations are correlated with each other. The problem of how to generate uncorrelated configurations is a central problem in Monte Carlo simulations, and it is addressed by using Markov chain Monte Carlo methods, which generate a sequence of configurations that are correlated with each other, but the correlation decays exponentially with the number of steps.

3.2.3 | Direct Sampling

integral...

probability distribution...

sampled function...

simpson method... error of this methods scales as a power of the number of chunks in which we divide the integration domain, and the exponent depends on the dimensionality...

integrals and sums are basically the same thing in this context, and we need to compute the partition function, which is a sum over an exponential number of configurations.

for the case of the circle...

this was an example of direct or homogeneus sampling, where we sample the whole domain of integration uniformly.

Failure of homogeneous sampling

[...]

the choice of the sampling distribution is crucial for the efficiency of the Monte Carlo method, and it is a central problem in Monte Carlo simulations. The choice of the sampling distribution is a trade-off between the ease of sampling and the efficiency of the method. The ideal sampling distribution is the one that is proportional to the integrand, which minimizes the variance of the estimator, but it may be difficult to sample from this distribution.

we can choose a pd which is similar to the integrand, but less explosive, and we can use importance sampling to correct for the difference between the sampling distribution and the integrand...

A smart choice of the sampling distribution is one that makes the sample function as smooth as possible, which minimizes the variance of the estimator. This means we want to transfer as much of the complexity of the integrand to the sampling distribution, which is a trade-off between the ease of sampling and the efficiency of the method.

Motivation for Markov Chain Monte Carlo

We can imagine to substitute the distribution with program sampling a sequence of configurations that are correlated with each other, by letting him move to the points in which the function is larger, and therefore the sampling distribution is similar to the integrand. This is the idea behind Markov chain Monte Carlo methods, but the correlation between the configurations is a problem, since it introduces a bias in the estimator.

We need to find a way to select a chain which is representative of the distribution, and we need to find a way to decorrelate the configurations, which is a central problem in Monte Carlo simulations. The problem of how to generate uncorrelated configurations is addressed by using Markov chain Monte Carlo methods, which generate a sequence of configurations that are correlated with each other, but the correlation decays exponentially with the number of steps.

3.3 | Markov Chain Monte Carlo

One possibility to sample a probability distribution is to engineer a chain of events spanning the L-dimensional configuration target space: ...

[...]

We need to find an algorithm that implements these ideas in a practical way, and we need to find a way to decorrelate the configurations, which is a central problem in Monte Carlo simulations.

3.3.1 | Transition Matrix

[...]

rewriting the probability as products of transition matrices does not guarantee that each step have memory only of the previous step, since the transition matrix may depend on the whole history of the chain. We need to find a way to construct a transition matrix that depends only on the current configuration, which is a central problem in Markov chain Monte Carlo methods.

Appendices

A | Notation and Conventions

B | Computations and further developments